The Visible, The Hidden and The Invisible

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I would like to thank jury members, my collaborators, and my family.

Abstract

This manuscript synthesizes my research work from 2015 to 2025. My research work focuses on three problems. *(P1)* Shape perception of an object. *(P2)* Shape servoing of an object. *(P3)* Augmentation of an object. The object is mainly deformable and visible or hidden. *P1* and *P2* form solutions for robotic manipulation tasks. *P1* and *P3* form solutions for human manipulation tasks. *P1*, *P2* and *P3* form solutions for human-robot co-manipulation tasks. The solutions have potential to impact manufacturing and surgical applications.

Introduction

Context. This research work investigates fundamental problems in computer vision and vision-based control. Particularly it focuses on deformable object shape perception and control. The research work addresses objects that are visible, hidden, or even invisible to standard sensors, exploring challenges in their 3D shape understanding and manipulation. Key areas of investigation include shape perception from single 2D and 3D images, shape servoing using 2D and 3D visual feedback, and the perception and augmentation of hidden anatomical structures.

Applications. The solutions developed in this research work have significant potential impact across various domains. In manufacturing, they can enable the robotic remanufacturing of deformable industrial products. In the medical field, particularly in computer-assisted and robot-assisted laparoscopic surgery, they can improve surgical guidance for tasks such as tumor localization and resection.

Goal. The goal of this research work is to advance the state of the art in deformable object understanding and manipulation. This involves creating robust and real-time methods for perceiving the shape of visible, hidden, and invisible objects, developing effective shape servoing techniques for robotic manipulation, and designing intuitive augmentation strategies to enhance human perception of the hidden world, especially in critical medical applications. The research work seeks to bridge the gap between theoretical advancements in computer vision and robotics and their practical application in real-world scenarios.

Outline. The rest of the manuscript presents the research work in three parts and ends with a conclusion.

Part I focuses on "The Visible", detailing shape perception from single 2D and 3D images using methods like Shape-from-Template, and subsequently addresses shape servoing with 2D and 3D visual feedback.

Part II then explains "The Hidden", concentrating on shape perception through 3D-to-2D deformable and rigid registration in medical applications like laparoscopic liver surgery, followed by augmentation techniques for hidden anatomy to enhance depth perception.

Part III introduces "The Invisible", presenting the research project proposal which aims at shape perception and augmentation during tumor resection in liver laparoscopy, outlining its objectives and methodology involving advanced registration, visualization, and robotic control.

"Conclusion" overviews my overall research work and recalls research perspectives.

"Appendices" provide my curriculum vitae, significant achievements in research, publications, supervision, and professional activities, respectively.

Part I – The Visible

If an object can be observed by a camera, then it is considered visible.

Objectives

I set two objectives.

Objective-1. Shape perception of a visible object from a single image. The image is either 2D (*i.e.*, RGB) or 3D (*i.e.*, RGBD).

Objective-2. Shape servoing of a visible object using visual feedback. The visual feedback is provided by either a 2D camera or a 3D camera.

Outline

Part I explores my works addressing these objectives in a general context.

Chapter 1 focuses on **Objective-1**. Section 1.1 overviews *(i)* Shape-from-Template and *(ii)* sphere pose estimation problem from a single 2D image. It also highlights, respectively, my contributions [C1, C2] and [C3]. Section 1.2 presents the template-based deformable shape perception problem from a single 3D image. Subsequently, it demonstrates my contributions [C4].

Chapter 2 focuses on **Objective-2**. Section 2.1 recalls the shape servoing problem using a 2D camera. It subsequently details my contributions [C5–C7]. Section 2.2 recalls the shape servoing problem using a 3D camera. It then elaborates my contributions [C4].

Contributions

[C1] M. Shetab-Bushehri et al. “ROBUSfT: Robust Real-Time Shape-from-Template, a C++ Library”. In: *Image and Vision Computing*. 2024.

[C2] E. Ozgur and A. Bartoli. “Particle-SfT: a Provably-Convergent, Fast Shape-from-Template Algorithm”. In: *International Journal of Computer Vision*. 2017.

[C3] E. Ozgur et al. “Reconstructing Spheres by Fitting Planes”. In: *British Machine Vision Conference*. 2024.

[C4] M. Shetab-Bushehri et al. “Lattice-Based Shape Tracking and Servoing of Elastic Objects”. In: *IEEE Transactions on Robotics*. 2024.

[C5] M. Aranda et al. “Monocular Visual Shape Tracking and Servoing for Isometrically Deforming Objects”. In: *IEEE/RSJ International Conference on Intelligent Robots and Systems*. 2020.

[C6] M. Shetab-Bushehri et al. “As-Rigid-As-Possible Shape Servoing”. In: *IEEE Robotics and Automation Letters*. 2022.

[C7] V. Giraud et al. “Optimal Shape Servoing with Task-focused Convergence Constraints”. In: *IEEE/RSJ International Conference on Intelligent Robots and Systems*. 2022.

Chapter 1

Shape perception of the visible

1.1 Shape perception from a single 2D image

Seeing a 3D object's shape from a 2D image is a challenge. This is because a 2D image is a flattened view. It loses depth information. Many different 3D shapes can therefore create the same 2D image. Figure 1 illustrates a paper's different shapes with almost identical projections.

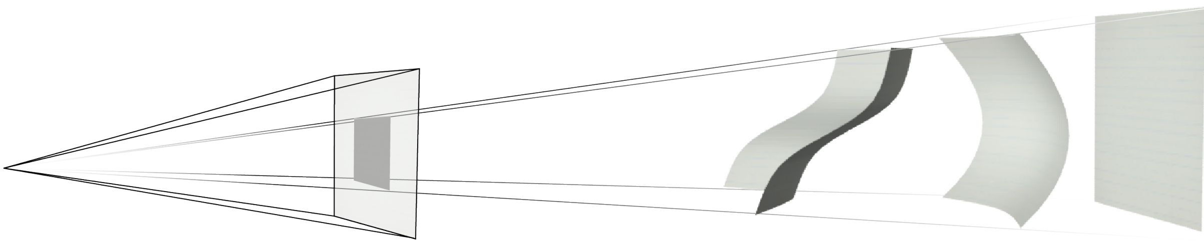

Figure 1. A paper in different postures with very similar projections.

1.1.1 Shape-from-Template

I use Shape-from-Template (SfT) to find a deformable object's 3D shape. The next subsection overviews briefly SfT problem. The subsequent subsection presents my contributions to SfT.

1.1.1.1 Problem overview

Problem statement. Infer the deformed 3D shape of a deformable object, given a 2D input image, the camera intrinsics, and the object's template. The template includes:

1. **3D shape at rest.** A representation of the object's shape when it is not deformed, often obtained using techniques like Structure-from-Motion (SfM).
2. **Texture map.** An image that captures the object's surface appearance, used in establishing correspondences between the template and the input image.
3. **Deformation law.** A mathematical model that describes how the object is allowed to deform, such as isometric and non-isometric (*e.g.*, elastic, conformal) deformation.

Working principle. SfT relies on two parts: *(i) registration* and *(ii) shape inference.* The *registration* establishes correspondences between the points in the input image and the texture map. The *shape inference* uses these correspondences to infer a deformed 3D shape by constraining the possible 3D shapes with the deformation law. Figure 2 shows the interactions between SfT components.

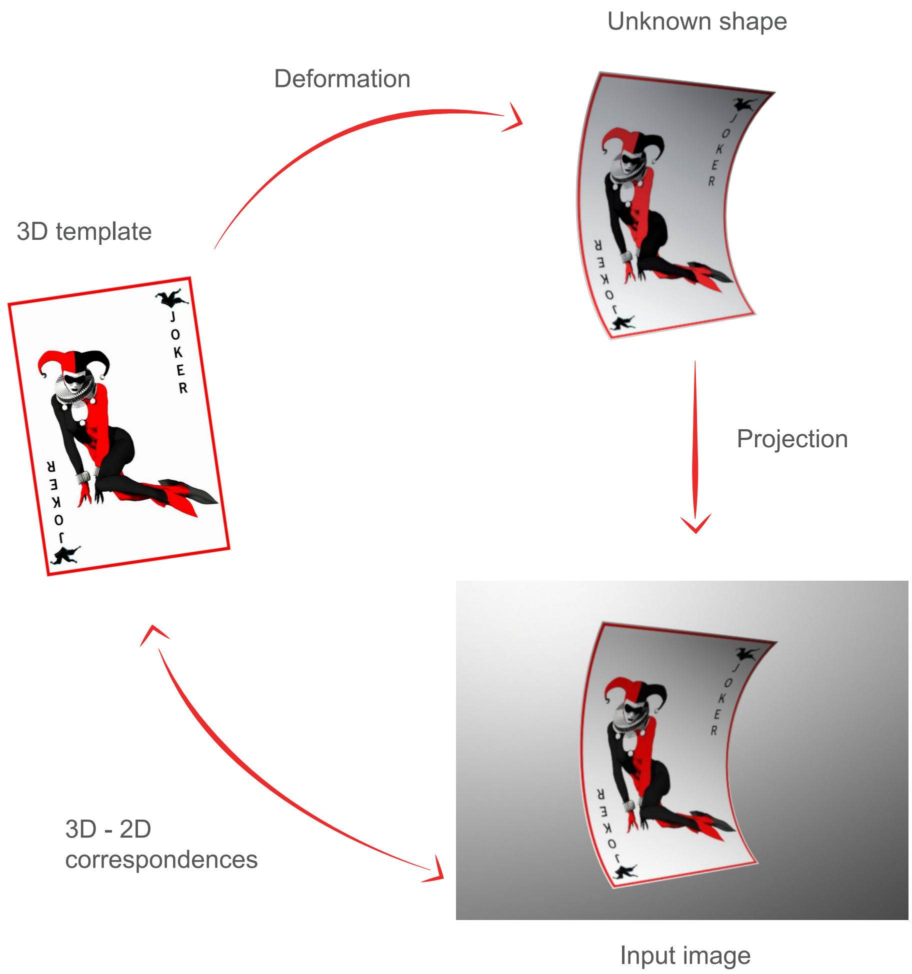

Figure 2. Interactions between SfT components.

Uniqueness of the solution. SfT yields a unique solution, if the deformation is isometric (*i.e.*, preserves geodesic distances) under perspective imaging. If the deformation is non-isometric, then SfT requires additional constraints for a unique solution. These constraints can include shading information, boundary conditions, and so on.

Advantages. Two important advantages can be listed as follows. *(i)* SfT can handle large deformations. *(ii)* SfT can infer the occluded parts of the object using the template.

Challenges. Three important challenges can be listed as follows. *(i)* SfT's *registration* part. It can easily introduce wrong correspondences under occlusions and complex deformations. *(ii)* SfT's real-time implementability. SfT with *registration* and *shape inference* parts is a computationally-demanding pipeline. *(iii)* SfT's template obtainability. The template may not always be fully obtainable. This happens in some medical applications.

State-of-the-art. SfT methods can be categorized into three groups:

1. **Shape inference methods (G1).** They focus on inferring the 3D shape given correspondences. Thus they require other algorithms for registration. This can make them computationally less efficient and slow. They are generally wide-baseline. Subsequently, they do not require an initial guess close to the solution.
2. **Integrated methods (G2).** They solve both registration and shape inference simultaneously. They are faster than G1 methods. However, they are usually short-baseline. Thus they require an initial guess close to the solution. They are also prone to failure in cases of occlusions or large deformations.
3. **DNN-based methods (G3).** They use deep neural networks for both registration and shape inference. They are wide-baseline and fast. However, they are object-specific. They require extensive training data for each new object. This makes them less practical.

1.1.1.2 Contribution to Shape-from-Template

I present `ROBUSfT` [C1]. It is built on `Particle-SfT` [C2]. `ROBUSfT` is a complete SfT pipeline for tracking isometrically deforming thin-shell objects with matchable appearance. First, `ROBUSfT`'s notable contributions with respect to the state-of-the-art are listed. Afterward, `ROBUSfT`'s methodology, results and limitations are highlighted.

Contributions. `ROBUSfT`'s four notable contributions are as follows.

1. `ROBUSfT` **is real-time.** It uses a CPU-GPU architecture that allows it to track up to 30 fps using 640 x 480 images on off-the-shelf hardware. It does not require initialization and implements tracking-by-detection.
2. `ROBUSfT` **is wide-baseline.** It is robust to occlusions, objects being out of the field of view, large deformations and fast motions.
3. `ROBUSfT` **introduces a novel mismatch removal algorithm.** This improves the registration step. The registration step can handle efficiently deforming scenes and a large percentage of mismatches. It is also very fast, reaching 200 fps.
4. `ROBUSfT` **is available as an open-source C++ library.** The code, a tutorial, a video for experiments can be found at https://github.com/mrshetab/ROBUSfT.

Existing SfT methods all have one or several limitations, including not covering the whole pipeline, not being wide-baseline, being limited to a specific texture or geometry, requiring training for a new object, being slow, and lacking public code access. Table 1.1 summarizes this information.

Methodology. `ROBUSfT`'s pipeline is shown in figure 3. `ROBUSfT` solves the problem of tracking the 3D shape of deforming objects. The pipeline is divided into offline and online sections. The offline section focuses on the creation of the template. The online section involves four steps. These steps are *(i)* keypoint extraction and matching, *(ii)* mismatch removal, *(iii)* warp estimation, and *(iv)* 3D shape inference. The first step's input is the image grabbed from the camera. Keypoints are extracted and matched with those previously extracted from the template's texturemap. The mismatch removal algorithm is then applied to detect and eliminate mismatches. The list of correct matches is used to estimate a warp between the template's texturemap and the image. This warp transforms

Group	Method	Registration	Real Time	Wide Baseline	General Geometry	No Training	Code Access
G1	[SfT1]	×	–	✓	✓	✓	×
	[SfT2]	×	×	✓	✓	✓	✓
	[SfT3]	×	–	✓	✓	✓	✓
	[SfT4]	×	×	✓	✓	✓	×
	[SfT5]	×	✓	✓	✓	✓	✓
	[SfT6]	×	✓	✓	✓	✓	×
G2	[SfT7]	✓	✓	×	✓	✓	×
	[SfT8]	✓	✓	×	✓	✓	×
	[SfT9]	✓	✓	×	✓	✓	×
	[SfT10]	✓	✓	×	✓	✓	×
G3	[SfT11]	✓	×	✓	×	×	×
	[SfT12]	✓	✓	✓	×	×	×
	[SfT13]	✓	✓	✓	✓	×	×
	[SfT14]	✓	✓	✓	×	×	×
	[SfT15]	✓	✓	✓	×	✓	×
	`ROBUSfT`	✓	✓	✓	✓	✓	✓

Table 1.1: Comparison of the existing SfT methods and `ROBUSfT`. `ROBUSfT` outperforms the state-of-the-art methods. Symbols ✓, × and – imply yes, no, and not available, respectively.

the registered mesh of the template to the image space, which is used as input for the 3D shape inference algorithm. This process is repeated for each image, allowing for tracking-by-detection. The whole pipeline is made real-time and robust by seamlessly integrating both novel and state-of-the-art algorithms. You can read more about the details here.

Results. An evaluation of `ROBUSfT` on real data sets is demonstrated in figure 4. The data sets include images of Spiderman poster, a chopping mat, and a t-shirt. The first column shows the texturemaps of the templates. Four images are shown from each data set. The inferred 3D shape with the estimated 3D coordinates of the estimated correct matches (red particles) as well as their ground truth (green particles) are shown below each image. The 2D projections of the 3D inferred shapes are also overlaid on the images. For each image, the median Euclidean distance between the estimated 3D coordinates of the estimated correct matches and their ground truth is given below the inferred shape.

Limitations. Two limitations of `ROBUSfT` are as follows. *(i)* `ROBUSfT` models only thin-shell objects and the surface of volumetric objects. As future work, `ROBUSfT` shall handle volumetric objects. To do so, it will exploit the inner volume deformation constraints. These should improve its accuracy compared to the current surface model. *(ii)* `ROBUSfT` requires textured object surfaces to establish correspondences, even though it is able to cope with reasonably low numbers of correspondences. As future work, `ROBUSfT` shall use the object's silhouette (*i.e.*, the occluding contour) to complement the correspondences. These limitations also apply to the most of existing SfT methods.

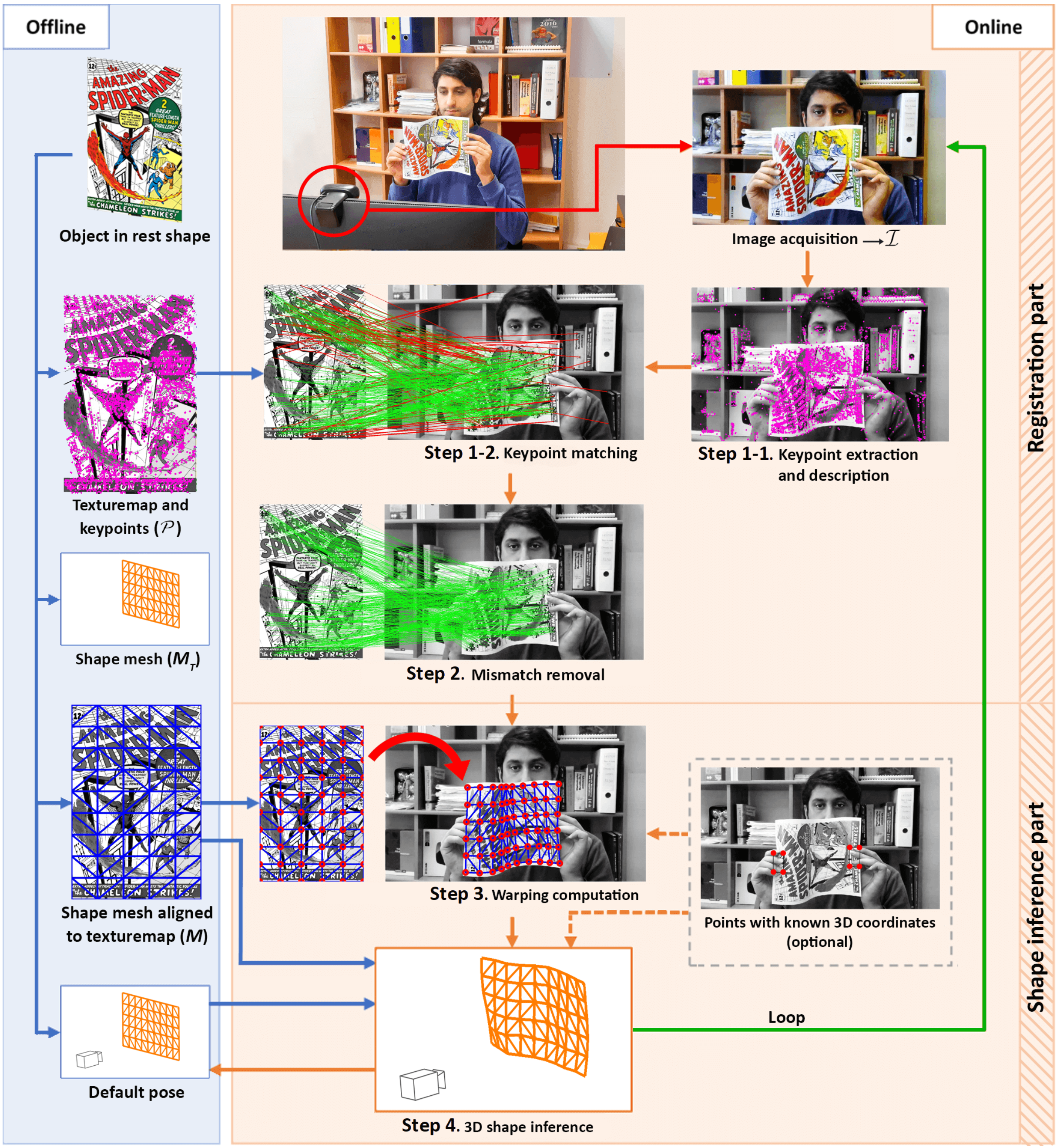

Figure 3. `ROBUSfT` pipeline.

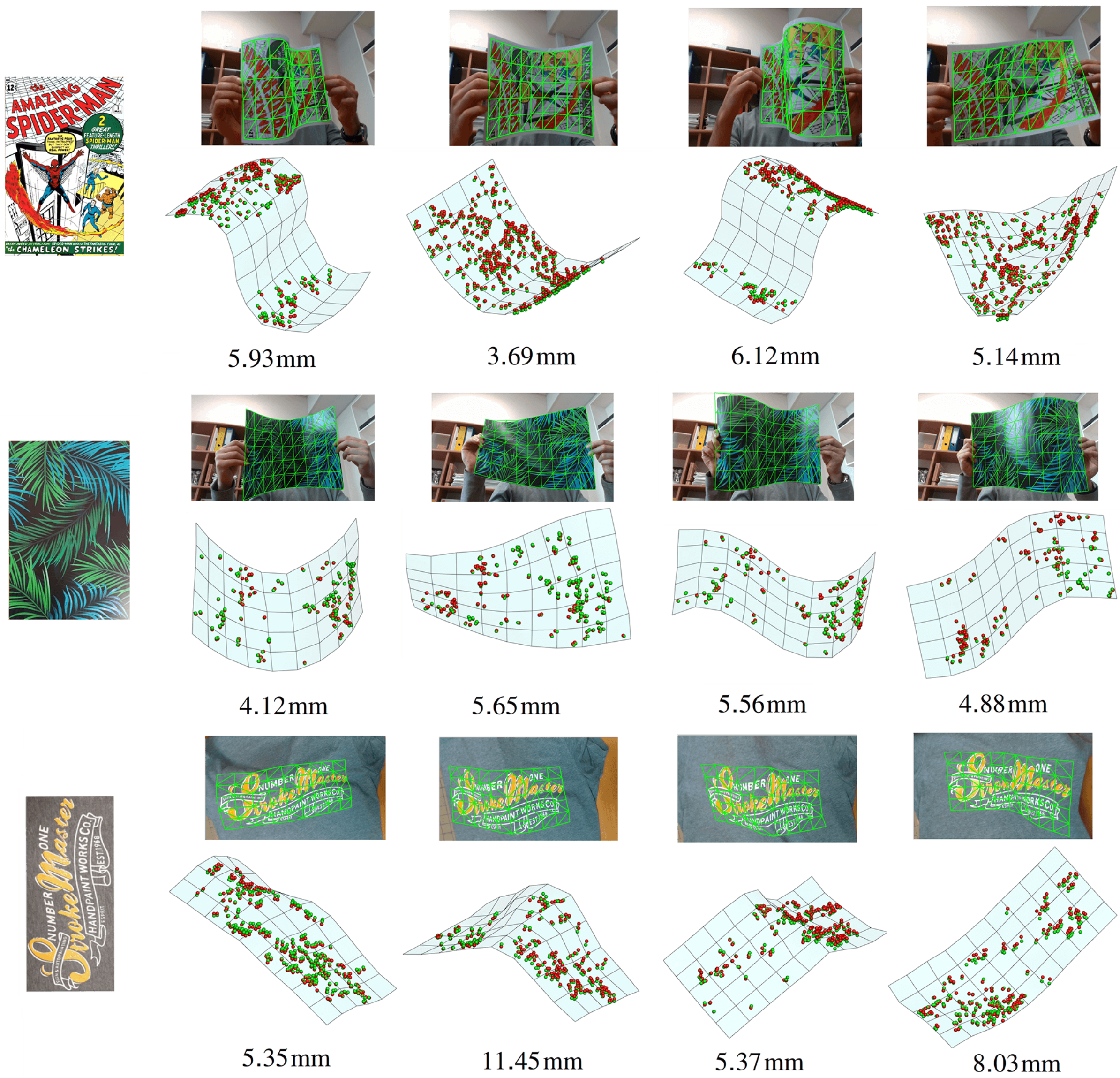

Figure 4. `ROBUSfT`'s performance on some real data sets. Click to watch.

References for SfT methods

[SfT1] Marc Salzmann et al. "Closed-form Solution to Non-rigid 3D Surface Registration". In: *European Conference on Computer Vision*. 2008.

[SfT2] Florence Brunet, Adrien Bartoli, and Richard I Hartley. "Monocular Template-based 3D Surface Reconstruction: Convex Inextensible and Nonconvex Isometric Methods". In: *IEEE Conference on Computer Vision and Pattern Recognition*. 2014.

[SfT3] Adrien Bartoli et al. "Shape-from-Template". In: *IEEE Transactions on Pattern Analysis and Machine Intelligence*. 2015.

[SfT4] Erol Özgür and Adrien Bartoli. "Particle-SfT: A Provably-convergent, Fast Shape-from-Template Algorithm". In: *International Journal of Computer Vision*. 2017.

[SfT5] Mohammad Famouri, Adrien Bartoli, and Zahra Azimifar. "Fast Shape-from-Template using Local Features". In: *Machine Vision and Applications*. 2018.

[SfT6] Miguel Aranda et al. "Monocular Visual Shape Tracking and Servoing for Isometrically Deforming Objects". In: *IEEE/RSJ International Conference on Intelligent Robots and Systems*. 2020.

[SfT7] Johan Östlund et al. "Laplacian Meshes for Monocular 3D Shape Recovery". In: *European Conference on Computer Vision*. 2012.

[SfT8] Dat Tien Ngo, Johan Östlund, and Pascal Fua. "Template-based Monocular 3D Shape Recovery using Laplacian Meshes". In: *IEEE Transactions on Pattern Analysis and Machine Intelligence*. 2015.

[SfT9] Toby Collins and Adrien Bartoli. "Realtime Shape-from-Template: System and Applications". In: *IEEE International Symposium on Mixed and Augmented Reality*. 2015.

[SfT10] Toby Collins et al. "Robust, Real-time, Dense and Deformable 3D Organ Tracking in Laparoscopic Videos". In: *International Conference on Medical Image Computing and Computer-Assisted Intervention*. 2016.

[SfT11] Albert Pumarola et al. "Geometry-aware Network for Non-rigid Shape Prediction from a Single View". In: *IEEE Conference on Computer Vision and Pattern Recognition*. 2018.

[SfT12] Victor Golyanik et al. "HDM-Net: Monocular Non-rigid 3D Reconstruction with Learned Deformation Model". In: *International Conference on 3D Vision*. 2018.

[SfT13] Diego Fuentes-Jimenez et al. "Deep Shape-from-Template: Single-image Quasi-isometric Deformable Registration and Reconstruction". In: *Image and Vision Computing*. 2022.

[SfT14] Shohei Shimada et al. "IsMo-GAN: Adversarial Learning for Monocular Non-rigid 3D Reconstruction". In: *IEEE Conference on Computer Vision and Pattern Recognition Workshops*. 2019.

[SfT15] Diego Fuentes-Jimenez et al. "Texture-generic Deep Shape-from-Template". In: *IEEE Access*. 2021.

1.1.2 Sphere pose estimation

Sphere pose estimation (SPE) forms a practical tool for applications requiring the pose of sphere-shaped objects, including visual servoing of robots. The next subsection overviews briefly the SPE problem. The subsequent subsection presents my contribution.

1.1.2.1 Problem overview

Problem statement. Find the sphere's 3D center point, given its radius and its occluding contour in a single image modelled by the calibrated perspective camera model.

State-of-the-art. A sphere's occluding contour generally appears as an ellipse and existing methods use ellipse fitting. Most methods thus require $\geqslant 5$ contour points. Only a few methods can deal with the minimal case of 3 points. However they all share two shortcomings. *(i)* They fail for non-elliptic occluding contours, parabola and hyperbola. *(ii)* They use the point-to-ellipse distance, whose computation is not closed-form.

1.1.2.2 Contribution to sphere pose estimation

I present `SpherO` [C3]. `SpherO` is a novel simple and convex method which solves the sphere pose estimation problem. First, `SpherO`'s notable contributions with respect to the state-of-the-art are listed. Afterward, `SpherO`'s methodology and results are highlighted.

Contributions. `SpherO`'s four notable contributions are as follows.

1. **`SpherO` handles both minimal and redundant cases.** It can thus serve as minimal method in the inner loop of RANSAC-like robust methods and as final refinement method given the inlier set.
2. **`SpherO` uses geometric point-to-plane distance.** It entirely avoids the point-to-conic distance required by existing methods. It thus provides a quick and sound way of determining the inlier set within RANSAC.
3. **`SpherO` deals with the occluding contours of all proper conics.** It does not matter if the sphere is seen as an ellipse, a parabola or a hyperbola. It seamlessly handles all in the same setting.
4. **`SpherO` provides a one-parameter family of spheres when the sphere's radius is unknown.** It uniquely reconstructs the sphere otherwise.

There exist only three methods which use 3 points or more [SPE1–SPE3]. Methods [SPE1] and [SPE2] are not robust; they minimize the squared algebraic point-to-ellipse image distance. Methods in [SPE3] minimize the geometric distance from a cone tangent to the sphere (method `Toth`); it provides a RANSAC-based robust solution specifically relying on the ellipse to form the consensus set (method `Robust-Toth`). It is shown in [SPE3] that `Toth` outperforms the two methods [SPE1, SPE2] and other existing methods requiring at least 5 points [SPE4–SPE6], and is thus the state of the art. Table 1.2 summarizes the characteristics of the proposed `SpherO` and `Robust-SpherO` in comparison to `Toth` and `Robust-Toth`. `SpherO` is simpler to derive and faster. `Robust-SpherO` is the first robust method to handle any conical occluding contours, as `Robust-Toth` is restricted to ellipses.

Table 1.2: Comparison of $\geq$ 3-point sphere pose estimation methods.

	Handles all image conics	Fits	Geometric costs
`Toth` [SPE3]	✓	cone	secant
`Robust-Toth` [SPE3]	only ellipse	ellipse cone	point-to-ellipse distance* secant
`SpherO`	✓	plane	point-to-plane distance
`Robust-SpherO`	✓	plane	point-to-plane distance

**numerically approximated by algebraic distance*

Methodology. `SpherO` uses spherical normalization, which places an image point on the camera-centred unit sphere, and an observation formalized in proposition 1 and illustrated in figure 5.

Proposition 1. *A sphere's spherically-normalized occluding contour is a sphero-circle.*

A sphero-conic is a curve formed by intersecting a quadric cone with the unit sphere whose center is the vertex of the cone. If the cone is circular then the sphero-conic is a circle also known as 'sphero-circle'. Using this result with the camera center taken as vertex and the sphere's projection lines taken as circular cone proves proposition 1.

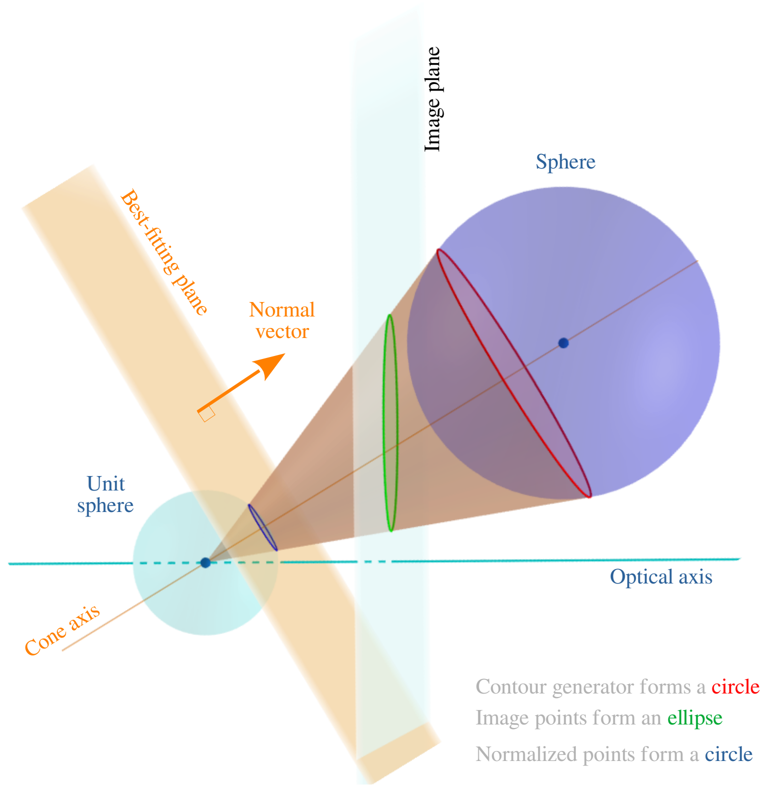

Figure 5. The geometric observation subtending the proposed `SpherO` method. Tilting the camera's image plane (in light blue) towards the cone axis leads to the sphere's occluding contour (in green) becoming a parabola and eventually a hyperbola. Importantly, the contour generator (in red) and normalised points (in blue) remain invariant, hence circles, showing that SpherO seamlessly handles all projection cases.

SpherO has three steps. First, following proposition 1, it spherically normalizes the sphere's occluding contour to form a sphero-circle. Second, it reconstructs the sphero-circle's support plane, center and radius. This induces a one-parameter sphere family. Third, it retrieves the actual sphere of prescribed radius. The letter 'O' in SpherO depicts a circle. The use of spherical normalization, following proposition 1 allows one to rid out the effect of the camera's image plane, only retaining the camera's projection center. This gives SpherO the ability to use mere plane-fitting, making it the most compelling method against the state of the art. You can read more about the details here.

Results. We evaluate the relative accuracy of the compared methods by reconstructing multiple spheres whose center distances are known. We place a target sphere with known radius R in contact with at least three auxiliary spheres whose radii R_i are also known. This yields the ground-truth distances between the target sphere's center $\mathbf{C}$ and the auxiliary spheres' centers $\mathbf{C}_i$ as $\| \mathbf{C} - \mathbf{C}_i \| = R + R_i$. Three auxiliary spheres fully constrain the target sphere's center $\mathbf{C}$, although two solutions exist. We use the compared methods to reconstruct the respective spheres' centers $\widehat{\mathbf{C}}$ and $\widehat{\mathbf{C}}_i$ from which a pairwise error is computed as $e_i = | \| \widehat{\mathbf{C}} - \widehat{\mathbf{C}}_i \| - (R + R_i) |$ for each method. Figure 6 reports the results with 3 auxiliary spheres. The relative error is computed as $\|\mathbf{e}\| = \|[e_1, e_2, e_3]^\top\|$. The target sphere's radius was $8.5\,cm$ (football) and the auxiliary spheres' radii were $3.25\,cm$ (tennis ball), $3.5\,cm$ (green ball) and $4.75\,cm$ (rainbow ball), respectively. The target sphere's center was about $40\,cm$ away from the camera. We set the RANSAC threshold $\tau_{pix} = 0.5$ pixels which was about twice of the intrinsic calibration accuracy of the camera. We repeated the experiment 1000 times with both methods. Robust-SpherO yielded $4.76\,mm$ mean error with $0.01\,mm$ std. Robust-Toth [SPE3] yielded $6.77\,mm$ mean error with $1.9\,mm$ std.

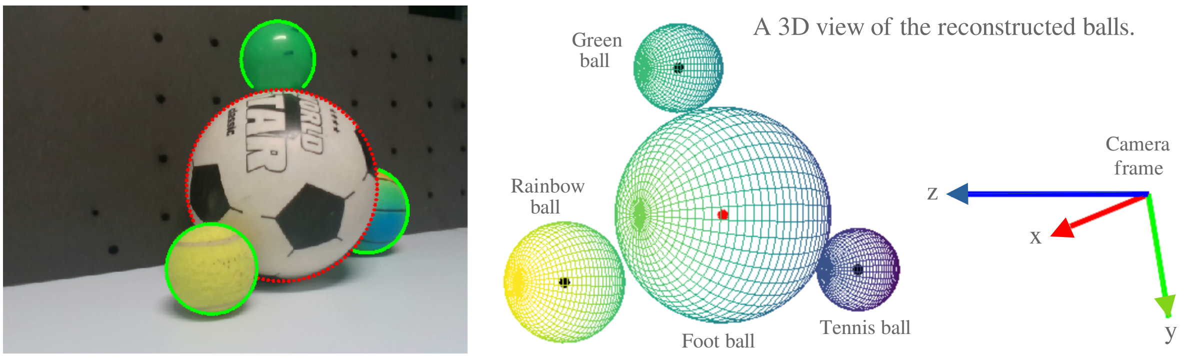

Figure 6. The input image with the spheres' occluding contours (left). A 3D view from behind the balls reconstructed by Robust-SpherO (right). The rainbow ball was mostly occluded by the football. We used 100 contour points for each ball.

References for SPE methods

[SPE1] Sun J, He H, and Zeng D. "Global calibration of multiple cameras based on sphere targets". In: *Sensors*. 2016.

[SPE2] K Shi, X Li, and H et al Xu. "Sphere localization from a minimal number of points in a single image". In: *International conference on advances in computer technology, information science and communications*. 2019.

[SPE3] Toth T and Hajder L. "A Minimal Solution for Image-Based Sphere Estimation". In: *International Journal of Computer Vision*. 2023.

[SPE4] R Halir and J Flusser. "Numerically Stable Direct Least Squares Fitting of Ellipses". In: *WSCG98*. 1998.

[SPE5] Lu C et al. "Arc-support line segments revisited: An efficient high-quality ellipse detection". In: *IEEE Transactions on Image Processing*. 2020.

[SPE6] Meng C et al. "Arc Adjacency Matrix-Based Fast Ellipse Detection". In: *IEEE Transactions on Image Processing*. 2020.

1.2 Shape perception from a single 3D image

Shape perception from a single 3D image (Sf3Di) is a practical tool for many applications. The next subsection overviews very briefly the Sf3Di problem. The subsequent subsection presents my contribution.

1.2.1 Problem overview

Problem statement. Infer the deformed 3D shape of a deformable object, given a 3D input image, the camera intrinsics, and the object's template. The template includes *(i)* a point cloud of the object's surface in its rest shape and *(ii)* a deformation law.

State-of-the-art. There exists methods on shape perception of deformable objects using 3D cameras. However, each works with either a linear or a thin-shell or a volumetric object. None of them succeeds in working with all types of objects (*i.e.*, 1D, 2D, 3D).

1.2.2 Contribution to deformable shape perception

I present `Latté` [C4]. `Latté` solves deformable shape perception problem using a 3D camera. First, `Latté`'s notable contributions with respect to the state-of-the-art are listed. Afterward, `Latté`'s methodology and results are highlighted.

Contributions. `Latté` has three notable contributions.

1. `Latté` can handle linear, thin-shell, and volumetric objects with any shape. It does not require the object's mechanical parameters. Instead, it uses geometric constraints on the object's point cloud.
2. `Latté` runs in real time without any parallelization using only the CPU. It reaches more than 30 FPS. Thus, it can be used as a deformable shape tracking solution.
3. `Latté` does not require the object to be textured. It only uses the point cloud captured by a 3D camera.

Methodology. `Latté`'s key idea is to simplify an object's shape to a lattice. `Latté` forms the lattice to manage the object's deformations. This idea has been already used in computer graphics [Sf3Di1]. The lattice simplifies complex deformations of an object's high-resolution point cloud. However, it still captures the useful deformations. These deformations are often simpler than what one would expect. Thus, the lattice can be formed with a lower-resolution point cloud. `Latté` encodes the object's shape into such lattice through the barycentric coordinates. Subsequently, `Latté` solves the deformation only for the lattice. This significantly reduces the computational load. Furthermore, most elastic objects tend to keep their local rigidity. `Latté` uses geometric As-Rigid-As-Possible (ARAP) model [Sf3Di2] to enforce this. This avoids knowing the object's mechanical parameters. `Latté`'s pipeline is presented in figure 7. It comprises 4 steps. *(i)* Capture and filtering of the point cloud. *(ii)* Rigid registration. *(iii)* Finding the corresponding points. *(iv)* Applying the deformation constraints.

The first three steps are inspired by [Sf3Di3]. `Latté` estimates the object's shape in each frame given an initial shape. Providing such an initial shape at the beginning is

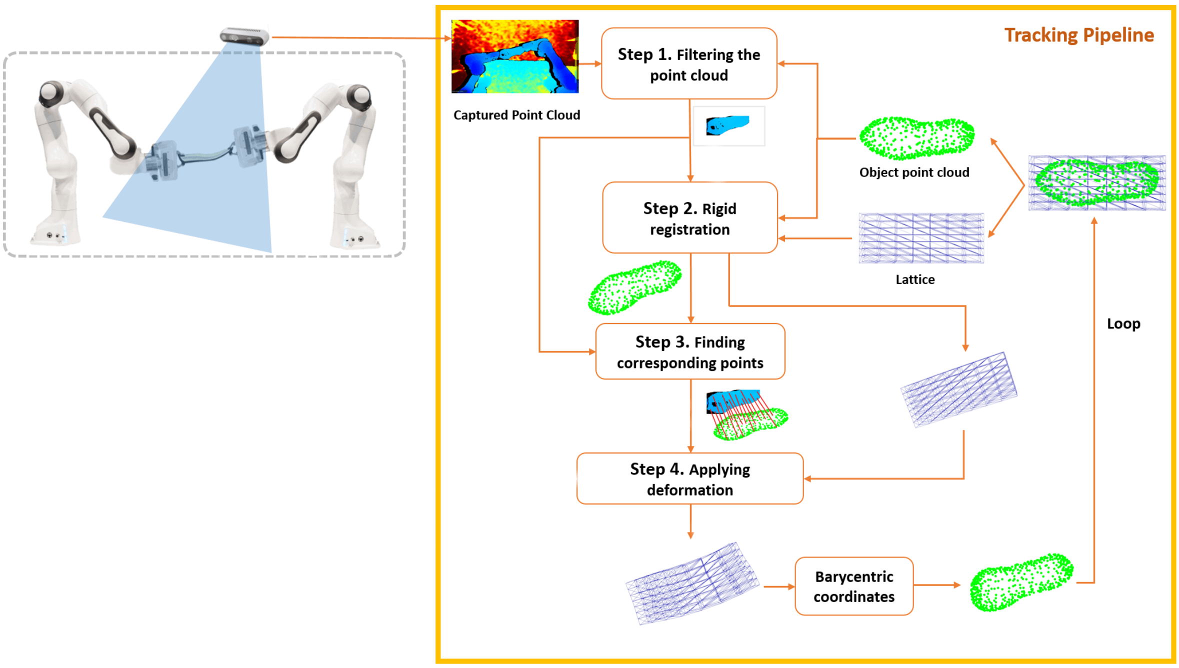

Figure 7. `Latté`'s pipeline. As inputs, it takes a captured point cloud from a 3D camera and the lattice at a given initial pose or from the previous frame in a loop. It then computes the new deformed lattice of the object shape.

common in state-of-the-art tracking methods [Sf3Di3–Sf3Di6]. `Latté` initially sets the object and lattice shapes by rigidly transforming their undeformed versions to an initial reference pose in the 3D camera's field of view.

Results. `Latté`'s performance is illustrated on a shoe sole deformation tracking. Figure 8 shows the shoe sole's lattice. Figure 9 shows the manual alignment of the shoe sole to an initial shape at a reference pose for initialization. A coarse alignment is enough to recover the shape. Figure 10 shows `Latté`'s steps on the deformed shoe sole.

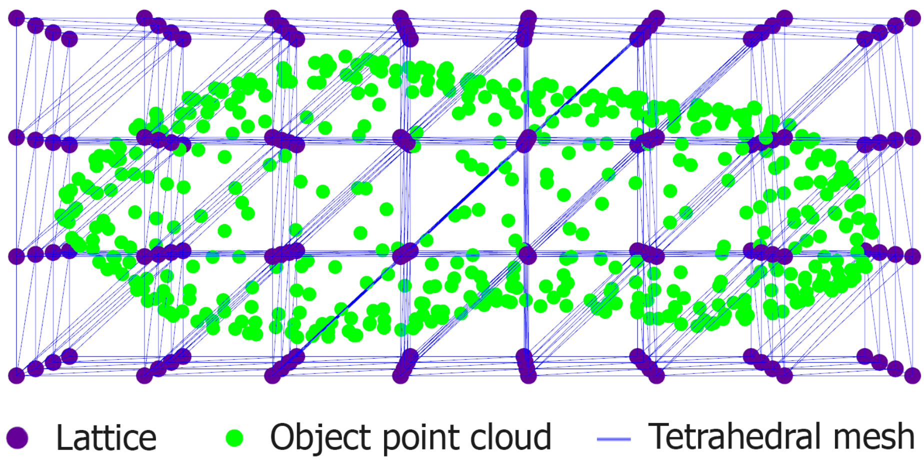

Figure 8. Lattice of the shoe sole.

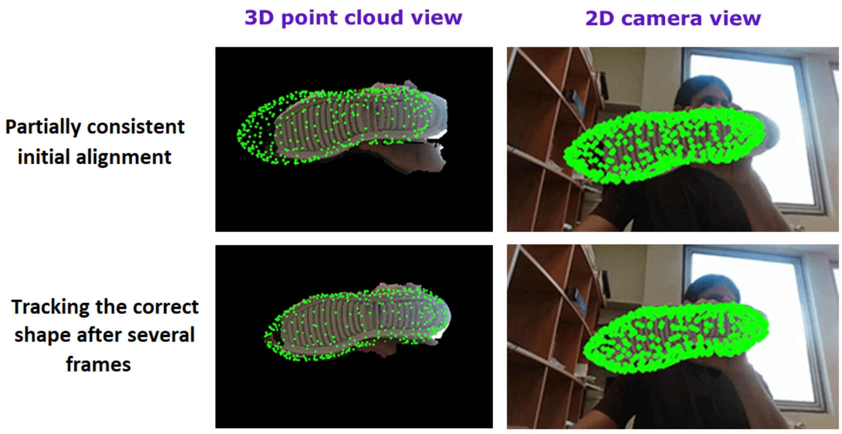

Figure 9. First row shows the manual alignment of the shoe sole to an initial shape (green) at a reference pose for initialization. A coarse alignment is enough to recover the shape. Second row shows the recovered shape.

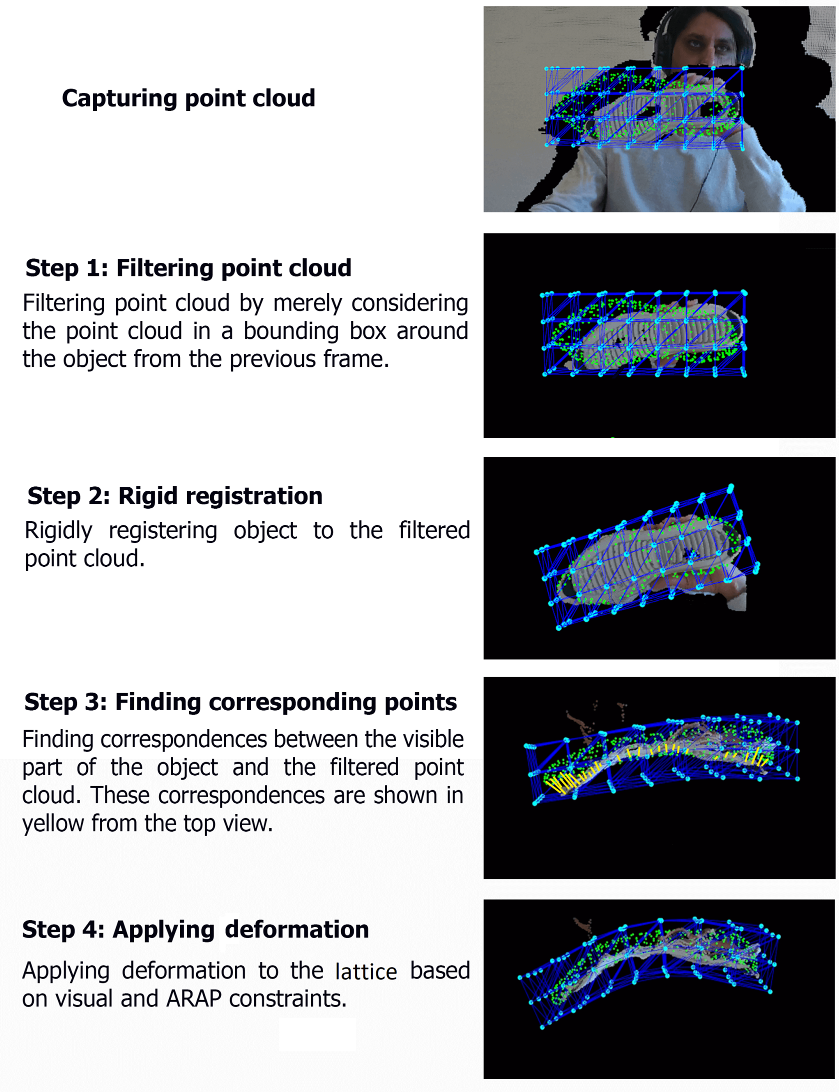

Figure 10. Output of each step in `Latté` on the deformed shoe sole. Click to watch.

References for Sf3Di methods

[Sf3Di1] Matthias Zollhöfer et al. "GPU based ARAP deformation using volumetric lattices". In: *Eurographics (Short Papers)*. 2012.

[Sf3Di2] Olga Sorkine and Marc Alexa. "As-rigid-as-possible surface modeling". In: *Eurographics Symposium on Geometry Processing*. 2007.

[Sf3Di3] Antoine Petit, Vincenzo Lippiello, and Bruno Siciliano. "Real-time tracking of 3D elastic objects with an RGB-D sensor". In: *IEEE/RSJ International Conference on Intelligent Robots and Systems*. 2015.

[Sf3Di4] Dat Tien Ngo, Jonas Östlund, and Pascal Fua. "Template-based monocular 3D shape recovery using Laplacian meshes". In: *IEEE Transactions on Pattern Analysis and Machine Intelligence*. 2015.

[Sf3Di5] Toby Collins and Adrien Bartoli. "Realtime shape-from-template: System and applications". In: *IEEE International Symposium on Mixed and Augmented Reality*. 2015.

[Sf3Di6] Teng Tang and Masayoshi Tomizuka. "Track deformable objects from point clouds with structure preserved registration". In: *The International Journal of Robotics Research*. 2022.

Chapter 2

Shape servoing of the visible

Shape servoing problem involves manipulating a deformable object towards a desired shape using robots. This is achieved through a feedback control loop that uses sensor data to understand the object's current shape and calculate the necessary robotic actions. To do so, one has to solve two important subproblems: *(P1)* real-time shape perception from sensor data, and *(P2)* shape control of infinite degrees of freedom of a deformable object with a finite number of actuation points. Solutions for *(P1)* are presented in Chapter 1. This chapter thus focuses on solutions of *(P2)*, shape control problem.

2.1 Shape servoing using 2D visual feedback

This section overviews briefly the shape servoing problem using a 2D camera (SS2D). Afterwards, it details my contributions.

2.1.1 Problem overview

Problem statement. Given a monocular 2D camera for visual feedback, control the robots to manipulate a deformable object from its current shape to a desired 3D shape.

State-of-the-art. Existing methods using 2D camera often control a few visual features in the image [SS2D1–SS2D9]. These visual features can include points, angles, distances, and object's silhouette. Set of such features represent mostly partially the object's shape. Therefore, they cannot allow one to control the object's full 3D shape. Furthermore, existing methods consider moderate deformations.

2.1.2 Contribution to shape servoing

Contributions. I present three shape servoing methods: *(i)* `IsoSS` (Isometric Shape Servoing) [C5], *(ii)* `ARAPSS` (As-Rigid-As-Possible Shape Servoing) [C6], and *(iii)* `OSS` (Optimal Shape Servoing) [C7]. These three methods use `Particle-SfT` [C2] for shape perception. They have four notable contributions.

1. Servoing the object's full shape in 3D space across large deformations. This is achieved using only a monocular 2D camera [C5–C7].
2. Use of a geometric mesh model rather than a mechanical material model. This avoids knowing the object's mechanical parameters which may not be available [C5–C7].
3. Deformation Jacobian computation uses only the object's measured current shape. Therefore, it is simpler to compute and to apply. Furthermore, it does not require initialization [C6, C7]. Conversely, existing deformation Jacobians are estimated

from sensor data. This data is collected over a time interval while the robots move. The existing deformation Jacobians also require initialization.

4. Encoding a task-focused implicit deformation behaviour directly into a shape servoing method. This avoids trajectory planning [C7]. Conversely, existing methods require replanning a trajectory each time the object's initial configuration changes.

Assumptions. The following assumptions are made.

1. The deformable object is grasped firmly throughout the task by $m \geqslant 1$ robots.
2. A fixed calibrated monocular camera observes the object.
3. A template of the object is known.
4. The object remains in quasi-static equilibrium.
5. The velocity of the end-effectors can be set exactly with no influence from the object.
6. The desired shape is reachable and the object can take the desired shape.
7. The surface of the object is visually trackable.

Isometric Shape Servoing

Methodology. `IsoSS` considers that the object deforms isometrically. Figure 1 overviews `IsoSS`. `IsoSS` has shape perception and shape control modules. Shape perception infers

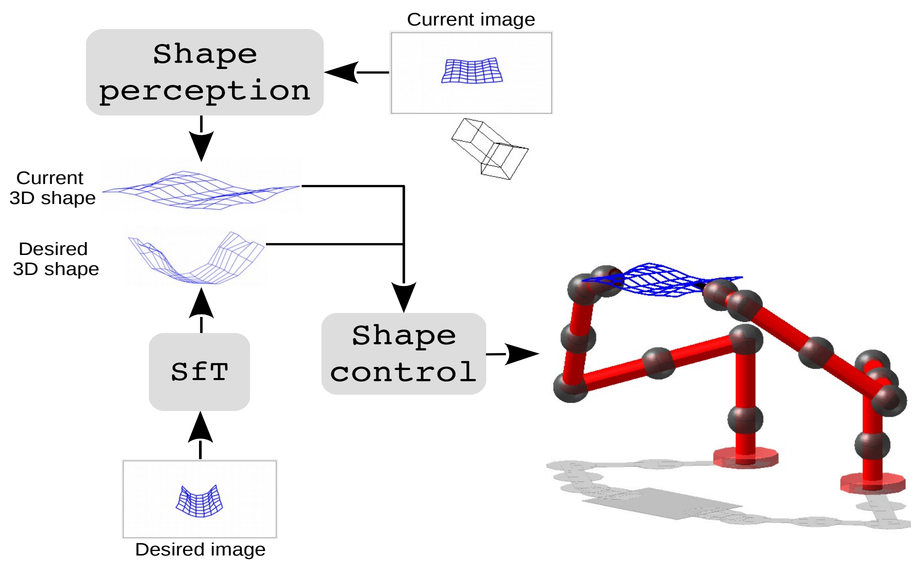

Figure 1. An overview of the isometric shape servoing method.

the object's shape using SfT. Shape control deforms the object towards the desired shape following intermediate predicted shapes generated successively through a feasible shape interpolator (FSI). FSI computes a predicted shape using the object's template, the desired shape and the current shape. Figure 2 illustrates `IsoSS`' pipeline.

Results. Figure 3 presents an experimental result for `IsoSS`.

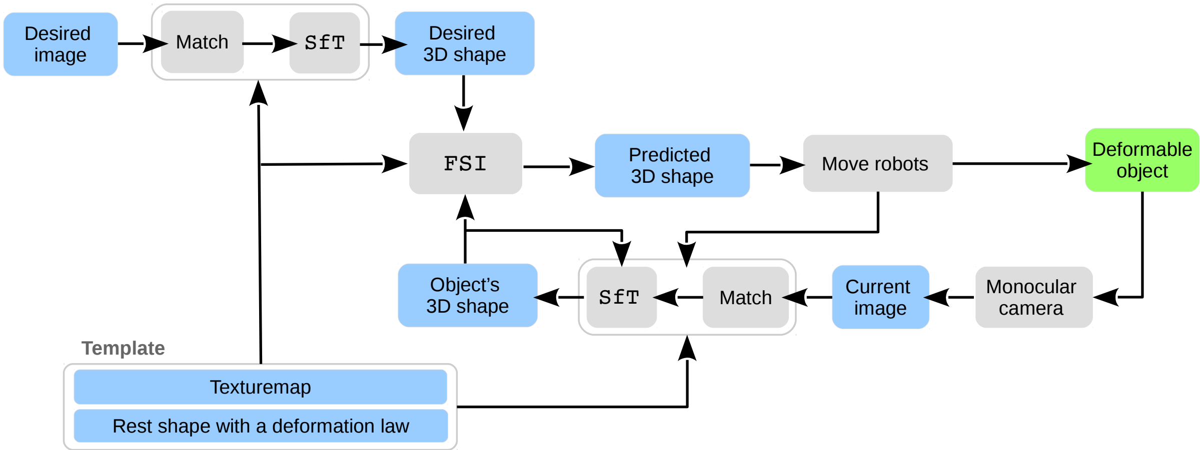

Figure 2. Pipeline of the isometric shape servoing (IsoSS) method.

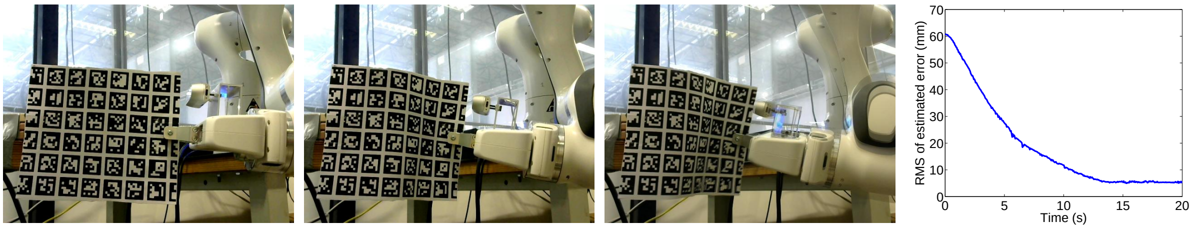

Figure 3. An experimental result for IsoSS. Left to right: the initial image, an intermediate image, and the final image (with the desired image superimposed) during servoing; RMS of the estimated shape servoing error. Click to watch.

As-Rigid-As-Possible Shape Servoing

Methodology. ARAPSS considers that the object deformation behavior can be approximated by as rigid as possible (ARAP) deformation model. It involves computing a deformation Jacobian based on the ARAP model using finite-difference estimation and simulations of the object's deformation. A proportional control law is then applied to determine the robot velocities needed to minimise the error between the object's current and desired shape, with the current shape being continuously measured through a shape tracking pipeline. Figure 4 illustrates the closed-loop control system where the measured current shape informs the Jacobian calculation and subsequent control actions to drive the object towards the desired shape.

Results. ARAPSS has been validated through various bi-arm shape servoing experiments. These experiments involved deformable objects made of different materials, including paper, rubber, and plastic. Figure 5 presents the results of seven different shape servoing tasks, showcasing the versatility of ARAPSS. These tasks involved different initial and desired shapes for objects like a Spiderman poster, a place mat, a piece of tire tread, and a shoe sole. For each task, the figure displays *(i)* images of the undeformed object, *(ii)* initial, desired (red mesh), and final (blue mesh with green nodes) shapes visualized in both camera images and RViz, and *(iii)* the evolution of the Root Mean Square (RMS) error of the shape over time. The results demonstrate that ARAPSS is able to accurately deform the objects towards their desired shapes across various materials and deformations. The error plots generally show a decrease in the RMS error over time, indicating successful

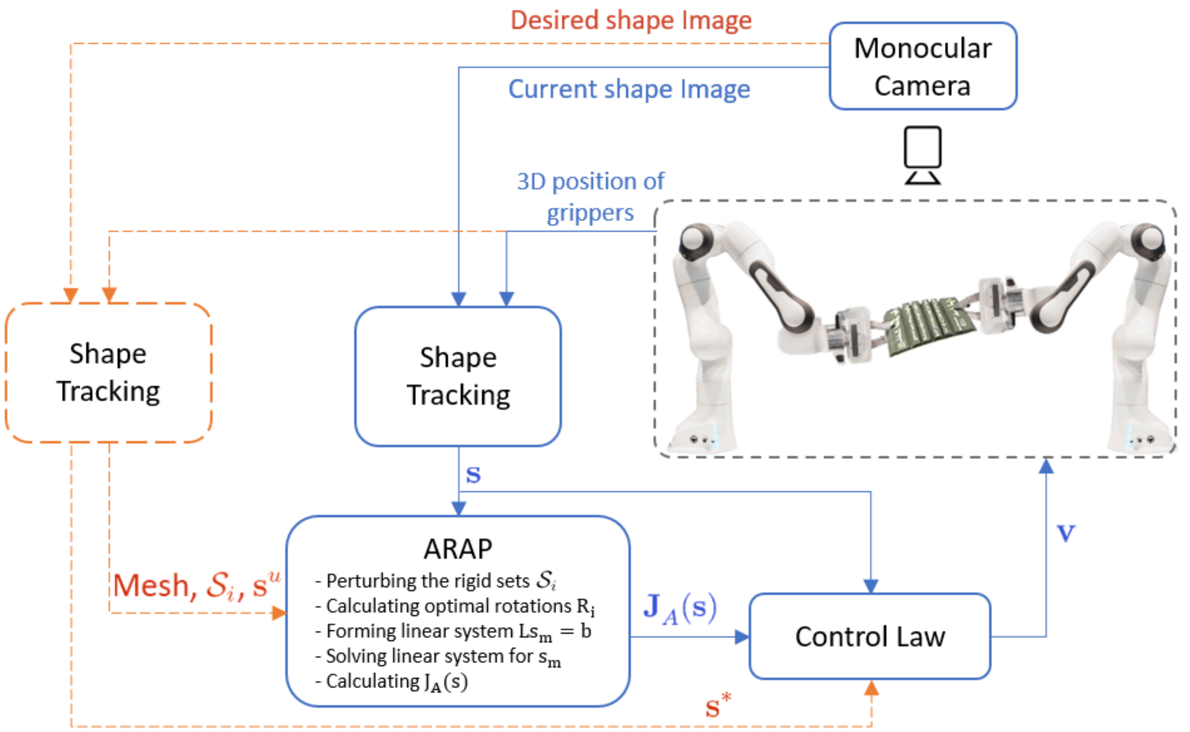

Figure 4. Pipeline of the as rigid as possible shape servoing (`ARAPSS`) method.

convergence to the desired shape. The scheme also showed robustness to minor changes in grasping conditions and uncertainties in shape tracking.

Optimal Shape Servoing

Methodology. `OSS` controls the object shape by imposing implicitly the task-focused convergence constraints. Unlike some other methods, `OSS` does not need to follow a preplanned path to get to the desired shape. Instead, it uses optimal control theory to determine the best way to deform the object to the desired shape in a fixed time. To do so, `OSS` uses shape constraints, control constraints, convergence time, and deformation prediction. The shape constraints encode how fast different parts of the object should move towards the desired shape. The control constraints decide which robot movements are most important for the task. Figure 6 shows how all these parts work together in a loop. First, the current shape of the object is perceived using SfT. This perceived shape is then compared to the desired shape to find the shape error. This error, along with a deformation model (deformation Jacobian) and the shape constraints (Q) and control constraints (R), is fed into the optimal control calculation. This calculation determines the control law for the robots. The robots then manipulate the object, and the whole process repeats. This continues until the object reaches the desired shape or the time limit is up (time horizon).

Results. `OSS` is demonstrated on a factory task. The task is to assemble rubber layers as shown in figure 7. The experimental results of figure 8 have shown that `OSS` outperforms an existing method (`ARAPSS`) in this specific task.

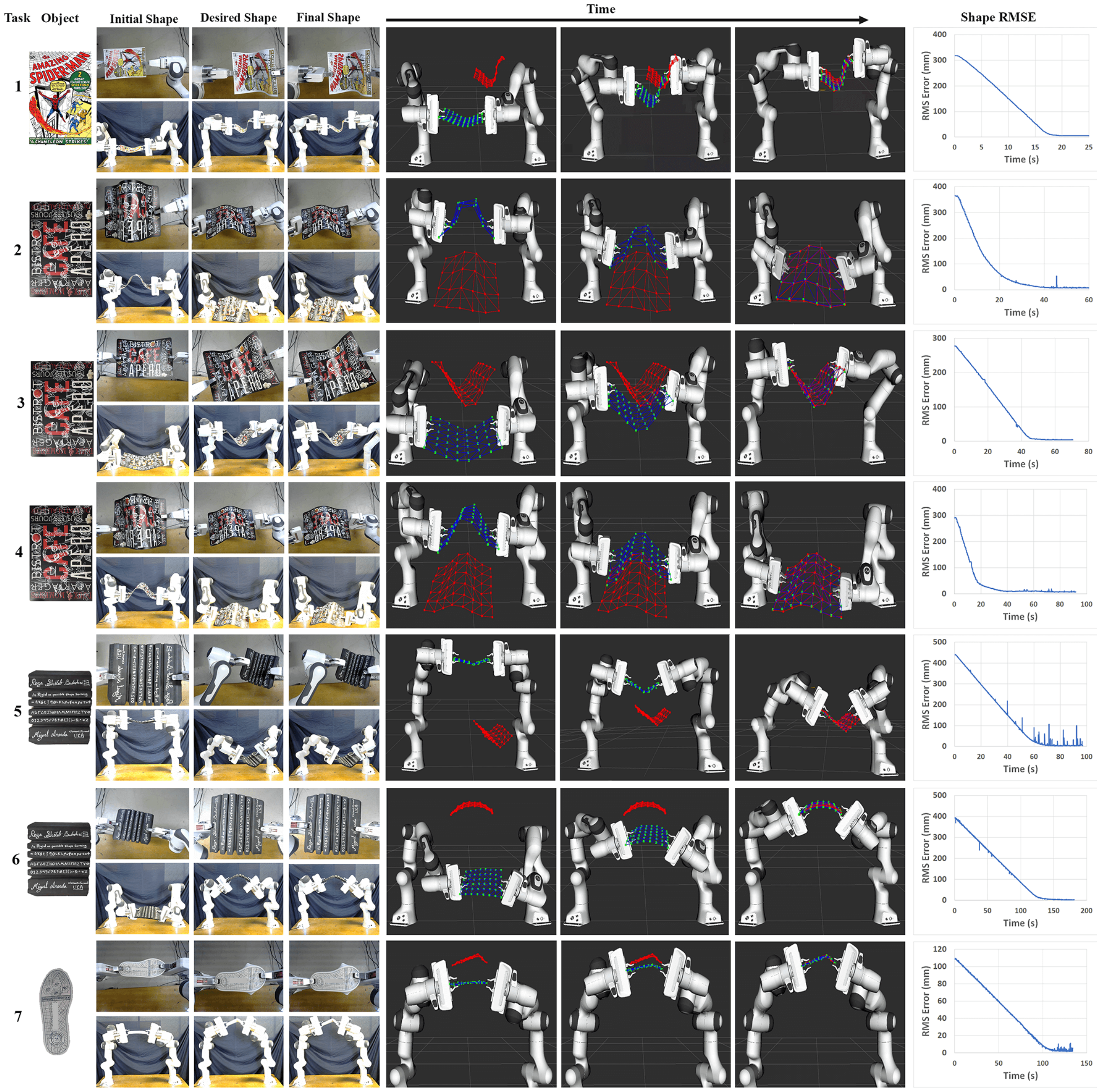

Figure 5. Experimental results for `ARAPSS`. Click to watch.

References for SS2D methods

[SS2D1] D. Navarro-Alarcon and Y.-H. Liu. "Fourier-based shape servoing: A new feedback method to actively deform soft objects into desired 2-D image contours". In: *IEEE Transactions on Robotics* 34.1 (2018).

[SS2D2] D. Navarro-Alarcon et al. "Model-free Visually Servoed Deformation Control of Elastic Objects by Robot Manipulators". In: *IEEE Transactions on Robotics* 29.6 (2013).

[SS2D3] J. Zhu et al. "Dual-arm robotic manipulation of flexible cables". In: *IEEE/RSJ International Conference on Intelligent Robots and Systems*. 2018.

[SS2D4] D. Navarro-Alarcon et al. "On the Visual Deformation Servoing of Compliant Objects: Uncalibrated Control Methods and Experiments". In: *International Journal of Robotics Research* 33.11 (2014).

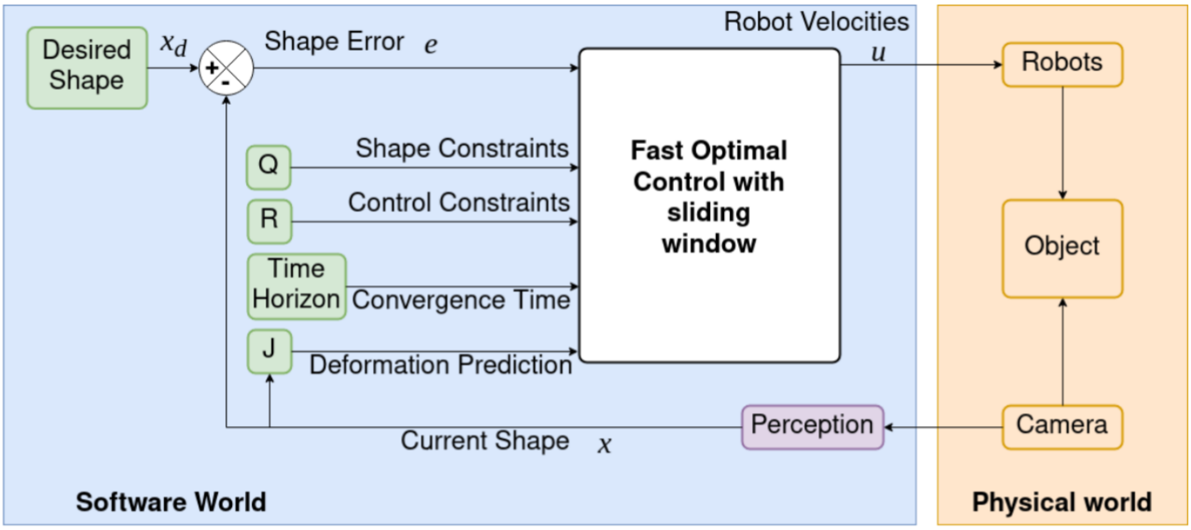

Figure 6. Pipeline of the optimal shape servoing (OSS) method.

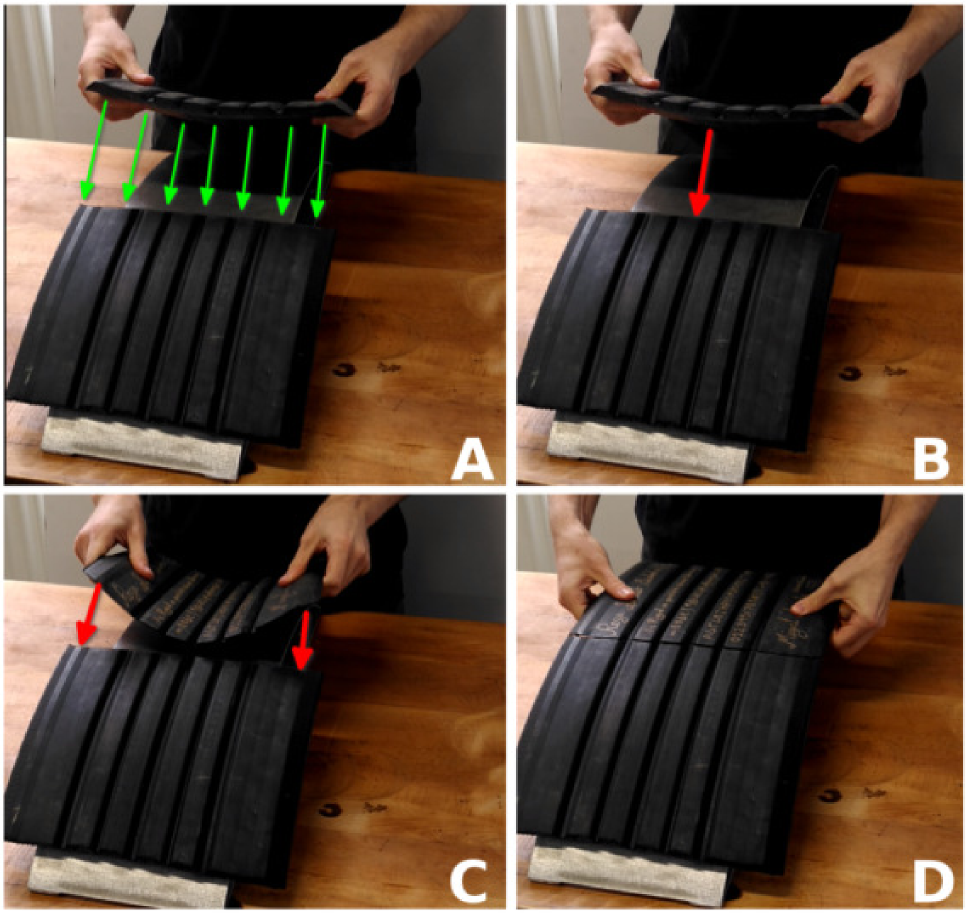

Figure 7. Manual assembly of the rubber layer as it is done in the factory.

[SS2D5] C. Shin et al. "Autonomous tissue manipulation via surgical robot using learning based model predictive control". In: *IEEE International Conference on Robotics and Automation*. 2019.

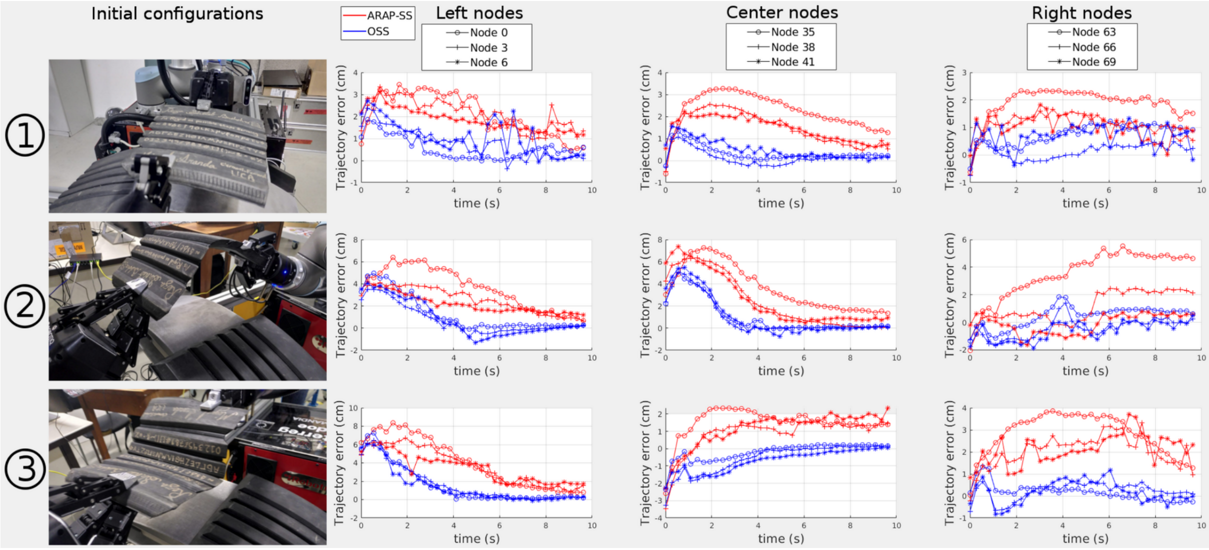

Figure 8. Nodes' trajectory errors for 3 different initial configurations. Click to watch.

[SS2D6] F. Alambeigi et al. "Autonomous data-driven manipulation of unknown anisotropic deformable tissues using unmodelled continuum manipulators". In: *IEEE Robotics and Automation Letters* 4.2 (2019).

[SS2D7] B. Jia et al. "Manipulating highly deformable materials using a visual feedback dictionary". In: *IEEE International Conference on Robotics and Automation*. 2018.

[SS2D8] A. Koessler et al. "An efficient approach to closed-loop shape control of deformable objects using finite element models". In: *IEEE International Conference on Robotics and Automation (ICRA)*. 2021.

[SS2D9] J. Zhu et al. "Vision-based manipulation of deformable and rigid objects using subspace projections of 2D contours". In: *Robotics and Autonomous Systems* 142 (2021).

2.2 Shape servoing using 3D visual feedback

This section overviews briefly the shape servoing problem using a 3D camera (SS3D). Afterwards, it details my contributions.

2.2.1 Problem overview

Problem statement. Given a 3D camera for visual feedback, control the robots to manipulate a deformable object from its current shape to a desired 3D shape.

State-of-the-art. Existing methods can be broadly grouped into *(i)* model-based shape servoing, *(ii)* model-free shape servoing, and *(iii)* learning-based shape servoing methods.

Model-based shape servoing methods use deformation models to predict how deformable objects behave. These models can be mechanical (*e.g.*, Finite Element Method (FEM) [SS3D1, SS3D2]) or geometrical (*e.g.*,, As-Rigid-As-Possible (ARAP), Position-based Dynamics (PBD), As-Similar-As-Possible (ASAP) [SS3D3]). Mechanical models can be computationally costly and depend on knowing the object's mechanical properties. While geometrical models are independent of mechanical parameters, they are often restricted to specific object forms like linear or thin-shell objects.

Model-free shape servoing methods do not require prior information about object deformation. These methods often use online sensor measurements to estimate a deformation Jacobian [SS3D4] or rely on geometric heuristics (*e.g.*, diminishing rigidity) [SS3D5]. Jacobian-based methods can be complex and sensitive to noise as they often require robot motion for calculation. Heuristic-based methods lack a proper deformation model, limiting their applicability in many real-world scenarios.

Learning-based shape servoing methods, particularly using Reinforcement Learning (RL), have gained attention [SS3D6–SS3D8]. However, these methods are typically specific to a particular object type (linear, thin-shell, or a surface of a volumetric object) and are mostly tested in simulation, often suffering from the sim-to-real gap. Training these agents for new objects requires significant amounts of data.

2.2.2 Contribution to shape servoing

I present `LattéSS` [C4]. `LattéSS` uses 3D camera for shape perception as explained in section 1.2 to solve the shape servoing problem. First, `LattéSS`' notable contributions with respect to the state-of-the-art are listed. Afterward, `LattéSS`' methodology and results are highlighted.

Contributions. `LattéSS` has three notable contributions.

1. `LattéSS` can control shapes of linear, thin-shell, and volumetric objects. It does not require the object's mechanical parameters. Instead, it uses geometric constraints on the object's point cloud.
2. `LattéSS` derives a novel analytical expression of the deformation Jacobian, which avoids numerical approximations.
3. `LattéSS` runs in real time.

Methodology. `LattéSS` controls the shape of elastic objects by using a servoing pipeline that works in conjunction with a tracking pipeline, as depicted in figure 9. The tracking pipeline continuously estimates the current object shape and the surrounding lattice using point cloud data from a 3D camera. This tracked shape, along with optimal rotations, then feeds into the servoing pipeline. Here, an analytical deformation Jacobian is calculated based on ARAP model, which represents the object's tendency to maintain local rigidity. This Jacobian is crucial for a control law that determines the velocity commands sent to the robotic grippers. By moving the parts of the lattice held by the robots, the servoing pipeline aims to deform the current lattice shape towards a desired lattice shape, which in turn guides the object's current shape towards a desired object shape due to the geometrical constraints between them. This creates a feedback loop where the tracked shape informs the control actions to achieve the desired deformation.

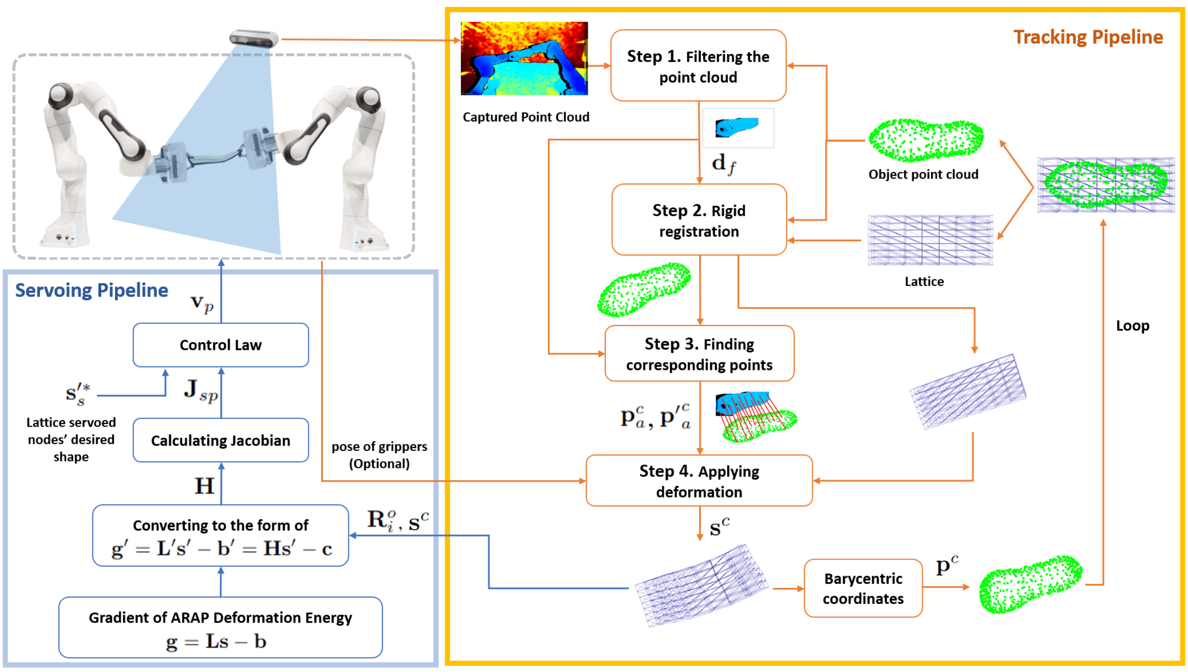

Figure 9. Pipeline of the lattice based shape servoing (`LattéSS`) method.

Results. Figure 10 illustrates two experiments conducted with a volumetric object: a bulky shoe sole. The figure displays the current shape (green) and the desired shape (red) of the shoe sole for each task, along with a graph showing the Root Mean Square Error (RMSE) of the servoing error over time.

The top row shows a task (Task 3.1) of full shape servoing of the shoe sole. The goal in this experiment was to drive the shoe sole from its initial configuration to a desired shape that exhibits a large deformation. The visual representation likely shows a significant change in the overall form of the shoe sole between the initial (green) and desired (red) states. The corresponding RMSE graph on the right would demonstrate how the servoing error decreases over time as the control law attempts to match the current shape to the desired shape.

The bottom row shows a task (Task 3.2) similar to Task 3.1 but with a key difference in the definition of the desired shape. In this task, the desired shape is defined with a severe rotation with respect to the initial shape. The figure visually highlights this, likely showing the shoe sole in the desired (red) configuration being significantly rotated and potentially flipped compared to its initial (green) orientation. The caption explicitly mentions that the visible part of the shoe sole in the desired shape is considerably different from the visible part in the initial shape. This demonstrates the approach's capability to handle large rigid body motions in addition to non-rigid deformations. The RMSE graph for this task would again illustrate the convergence of the servoing error as `LattéSS` attempts to achieve the significantly rotated and potentially deformed target shape.

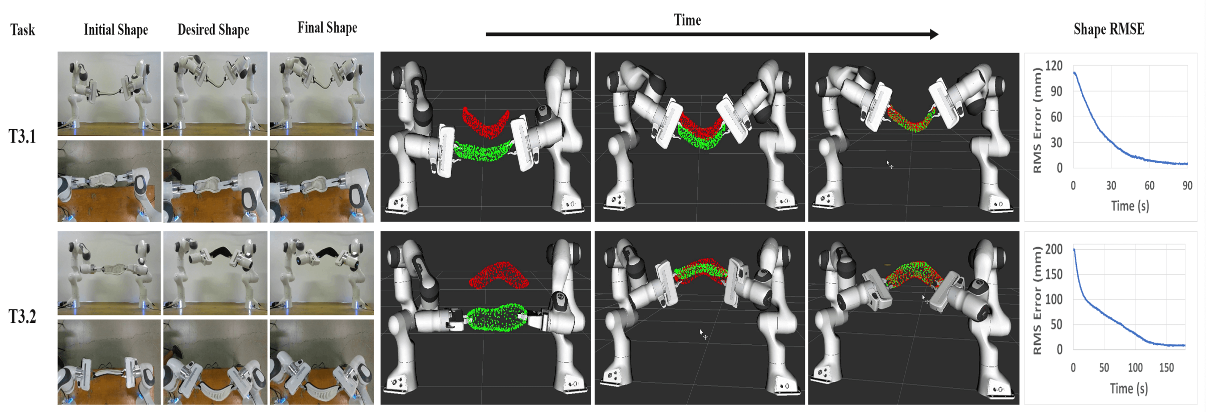

Figure 10. Experimental results of `LattéSS` with the shoe sole. Click to watch.

References for SS3D methods

[SS3D1] Fouad Makiyeh et al. "Shape Servoing of a Soft Object Using Fourier Series and a Physics-Based Model". In: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2023.

[SS3D2] A. Koessler et al. "An efficient approach to closed-loop shape control of deformable objects using finite element models". In: *IEEE International Conference on Robotics and Automation (ICRA)*. 2021.

[SS3D3] O. Aghajanzadeh et al. "An offline geometric model for controlling the shape of elastic linear objects". In: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2022.

[SS3D4] D. Navarro-Alarcón and Y.-H. Liu. "On the visual deformation servoing of compliant objects: un-calibrated control methods and experiments". In: *International Journal of Robotics Research* (2014).

[SS3D5] D. Berenson. "Manipulation of deformable objects without modeling and simulating deformation". In: *IEEE/RSJ International Conference on Intelligent Robots and Systems*. 2013.

[SS3D6] R. Jangir, G. Alenya, and C. Torras. "Dynamic cloth manipulation with deep reinforcement learning". In: *IEEE International Conference on Robotics and Automation (ICRA)*. 2020.

[SS3D7] R. Laezza and Y. Karayiannidis. "Learning shape control of elastoplastic deformable linear objects". In: *IEEE International Conference on Robotics and Automation (ICRA)*. 2021.

[SS3D8] M. H. Daniel Zakaria et al. “Robotic control of the deformation of soft linear objects using deep reinforcement learning”. In: *IEEE International Conference on Automation Science (CASE)*. 2022.

Part II – The Hidden

If an object cannot be observed by a camera but other available sensors, then it is considered hidden.

Objectives

I set two objectives.

Objective-1. Shape perception of a hidden object from a single 2D image.

Objective-2. Depth perception of a hidden object by means of an augmented 2D image.

Outline

Part II explores my works addressing these objectives in the medical context. Specifically, two medical projects `Hepataug` and `IMMORTALLS` are targeting the above objectives. The projects help locate tumors in laparoscopic liver surgery.

Chapter 1 focuses on problems related to **Objective-1**. Section 1.1 presents 3D-to-2D deformable registration problem as a possible solution. It then highlights my contributions [C1–C4]. Section 1.2 proposes 3D-to-2D rigid registration as another possible solution. It subsequently demonstrates my contributions [C5, C6].

Chapter 2 focuses on **Objective-2**. Section 2.1 explains the occluded object visualization problem in Augmented Reality. Section 2.2 details my contribution [C7]

The last chapter puts forward **shape servoing of a hidden object** as a research problem. If solved, it might yield use cases in minimally invasive surgeries.

Contributions

[C1] B. Koo et al. "Deformable Registration of a Preoperative 3D Liver Volume to a Laparoscopy Image using Contour and Shading Cues". In: *International Conference on Medical Image Computing and Computer Assisted Intervention*. 2017.

[C2] E. Ozgur et al. "Preoperative Liver Registration for Augmented Monocular Laparoscopy using Backward-Forward Biomechanical Simulation". In: *International Journal of Computer Assisted Radiology and Surgery*. 2018.

[C3] Y. Espinel et al. "Combining Visual Cues with Interactions for 3D-2D Registration in Liver Laparoscopy". In: *Annals of Biomedical Engineering*. 2020.

[C4] B. Le Roy et al. "A Case Series Study of Augmented Reality in Liver Laparoscopic Resection with a Deformable Preoperative Model". In: *Surgical Endoscopy*. 2020.

[C5] M.M. Kalantari et al. "Markerless Ultrasound Probe Pose Estimation in Mini-Invasive Surgery". In: *IEEE International Conference on Robotics and Automation*. 2024.

[C6] M. M. Kalantari et al. "LARLUS: Laparoscopic Augmented Reality from Laparoscopic Ultrasound". In: *International Journal of Computer Assisted Radiology and Surgery*. 2024.

[C7] E. Ozgur, A. Lafont, and A. Bartoli. "Visualizing In-Organ Tumors in Augmented Monocular Laparoscopy". In: *IEEE International Symposium on Mixed and Augmented Reality*. 2017.

Chapter 1

Shape perception of the hidden

This chapter focuses on shape perception of the hidden (SPH) from a single 2D image.

1.1 3D-to-2D deformable registration

1.1.1 Problem overview

Hepataug project. The project's objective is to develop and use `Hepataug`, a software for Augmented Reality (AR) guidance in monocular Laparoscopic Liver Resection (LLR). `Hepataug` enhances the surgeon's ability to visualise and localise internal hidden liver anatomy, such as tumors and vessels, by creating a virtual liver transparency during laparoscopic surgery. `Hepataug` creates the virtual liver transparency by overlaying a preoperative 3D model onto the laparoscopic images.

Achieving AR in laparoscopy requires registration of a preoperative organ volume to an intraoperative laparoscopic image. This is an extremely challenging objective for the liver. This is because the preoperative volume is textureless, and the liver is deformed and only partially visible in the laparoscopic image.

Problem statement. Given the preoperative 3D liver model, segmented from a volumetric CT or MRI, register the preoperative 3D liver model to a 2D intraoperative laparoscopic image by taking into account the liver's deformation. The problem is known as 3D to 2D multimodal deformable registration. See figure 1.

State-of-the-art. The current state of the art in preoperative 3D to intraoperative 2D deformable multimodal liver registration for monocular laparoscopy presents several approaches, each with its own limitations. Traditionally, a common method involves manually overlaying a rigid preoperative 3D model onto the 2D laparoscopic image [SPH1]. However, this approach suffers from limited accuracy because it fails to account for the significant deformation the liver undergoes during surgery due to factors like pneumoperitoneum, gravity, and manipulation.

To address deformation, other semi-automatic methods have been developed that utilise visual cues extracted from the laparoscopy image, such as the liver's silhouette and anatomical landmarks like the falciform ligament and inferior ridge. These methods often employ biomechanical models to guide the deformation of the preoperative 3D model to fit these visual cues, sometimes using an Iterative Closest Point (ICP)-like procedure [SPH2]. Nevertheless, a key challenge with these methods is the poor registration of occluded or hidden parts of the liver, as the visual cues only provide information about the visible surface. Furthermore, they typically struggle with irregular changes in shape, such as those caused by surgical cutting.

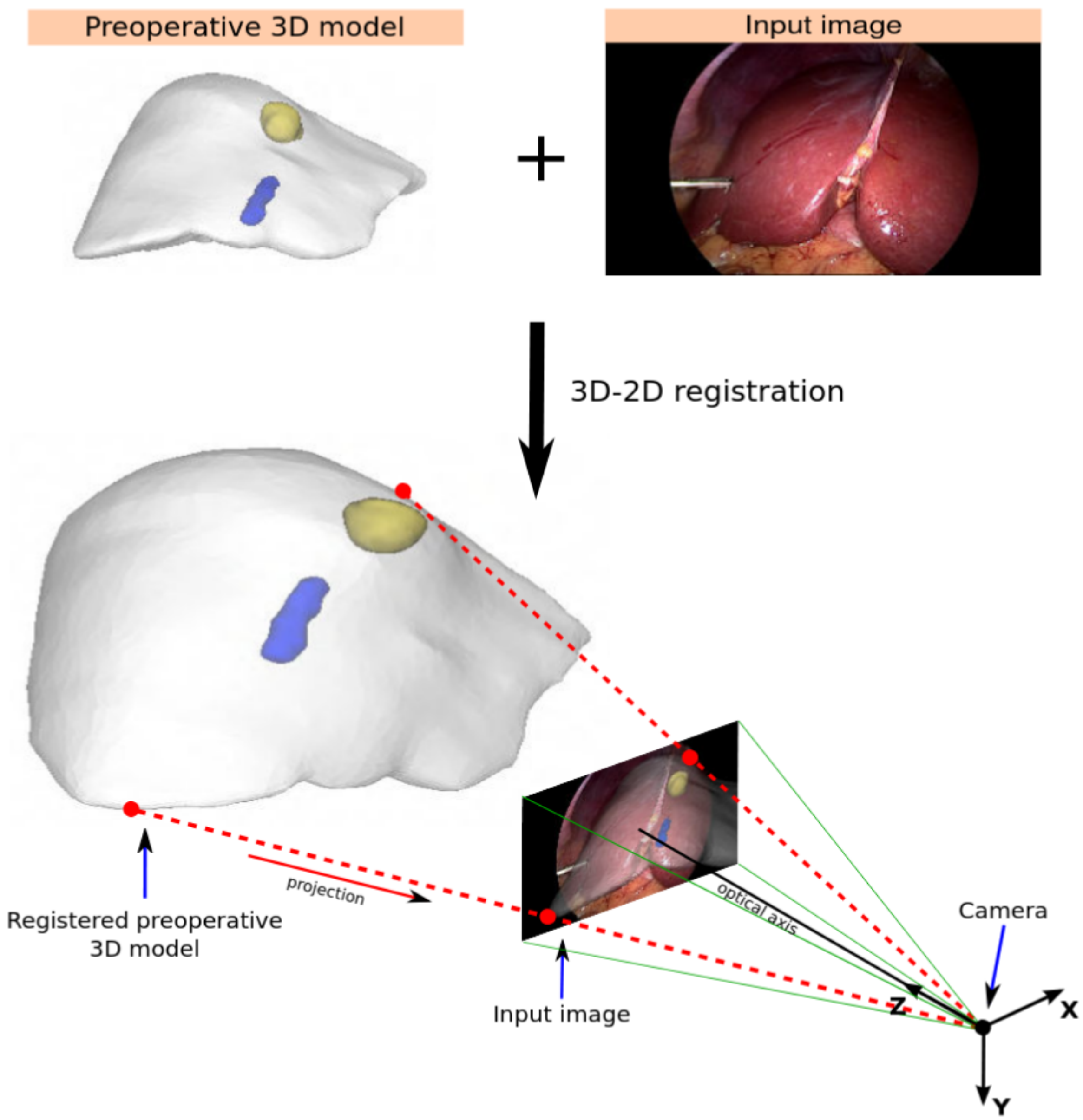

Figure 1. Registration of a preoperative 3D liver model with a laparoscopic image. The preoperative 3D model is extracted from CT and the camera represents the laparoscope. The preoperative 3D model comprises the liver volume whose external surface is in gray and parts of its inner anatomy, namely a subsurface tumor in yellow and vein in blue.

Some research explores the use of stereo-laparoscopy to reconstruct the 3D shape of the visible liver intraoperatively, which can then be used to constrain registration [SPH3]. However, these methods often depend on non-standard laparoscopy equipment or special hardware that is not usually available. Additionally, with the exception of some, they often do not adequately address the registration of the liver's hidden parts.

More advanced methods attempt to incorporate environment priors, such as the effects of pneumoperitoneum pressure and gravity, into the registration process [SPH4]. However, effectively exploiting these priors in vivo is challenging due to unknown boundary conditions related to the surrounding organs [SPH5]. From an optimisation perspective, many existing methods attempt to solve for both the deformation and the pose (position and orientation) of the liver simultaneously in a single step, which often leads to non-convex problems that are difficult to solve without a good initial alignment [SPH6].

Finally, achieving full automation of these registration methods in standard monocular laparoscopy remains difficult due to the need for manual initialisation and the complexities of automatically detecting and recognising anatomical landmarks and contours in the laparoscopic images [SPH7].

In summary, while significant progress has been made, the state of the art still presents limitations in achieving accurate and robust deformable registration of a preoperative 3D liver model to a 2D laparoscopic image in the standard operating room setting, particularly concerning the registration of hidden structures and the robustness to liver deformation.

1.1.2 Contribution to 3D-to-2D deformable registration

Contributions. `Hepataug` has three notable contributions.

1. It proposed the first 3D-to-2D multimodal deformable registration algorithm in liver laparoscopy [C1]. (`RegVisC`)
2. It proposed backward-forward biomechanical simulation to consider environmental factors like gravity and pneumoperitoneum to predict liver deformation before registration [C2]. (`RegVisCEP`)
3. It introduced a registration method combining visual cues and user interaction with a biomechanical model for more accurate liver registration [C3]. (`RegVisCUI`)

1.1.2.1 Registration using Visual Cues

Methodology. The preoperative liver model is a volumetric shape with a tetrahedral topology, and its deformation uses the Neo-Hookean elastic model. The visual cues from a monocular laparoscopic image are the *(i)* contour cues and *(ii)* shading cues.

Contour cues include the liver's silhouette and curvilinear anatomical landmarks such as ridge contours and the falciform ligament. The user marks visible contour segments in the image, and may also mark corresponding points on anatomical landmarks if available. Due to the lack of direct point correspondences, an Iterative Closest Point (ICP) mechanism is embedded within the algorithm to handle these contour cues. Shading cues use the Lambertian model without ambient light, assuming the laparoscope is the sole light source with light fall-off following the inverse-square distance law.

The registration uses a convergent alternating projections algorithm. This algorithm iteratively cycles through projection mappings for the deformation model and the visual constraints (contour and shading). The algorithm consists of four main stages: marking, rough automatic initialisation, contour-based refinement, and contour-and-shading-based refinement. Shading is introduced after an initial convergence using contour constraints due to its local validity. You can read more about the details here.

Quantitative results with semi-synthetic data. Patients' liver models are deformed to different shapes used as ground-truths and their synthetic images are rendered. Each column in figure 2 shows one registration using `RegVisC`. In figure 2, the first row shows the ground-truth images with marked contours (yellow for silhouette, red for ridges, blue for ligament) inside the field of view rectangle (cyan). The second and third rows show the colour mapped volumetric registration errors (mm) from a different viewpoint. We observe that the preoperative model's inner structures are registered within 1 cm error. This corresponds to about 4% registration error regarding the greatest transverse diameter of the liver.

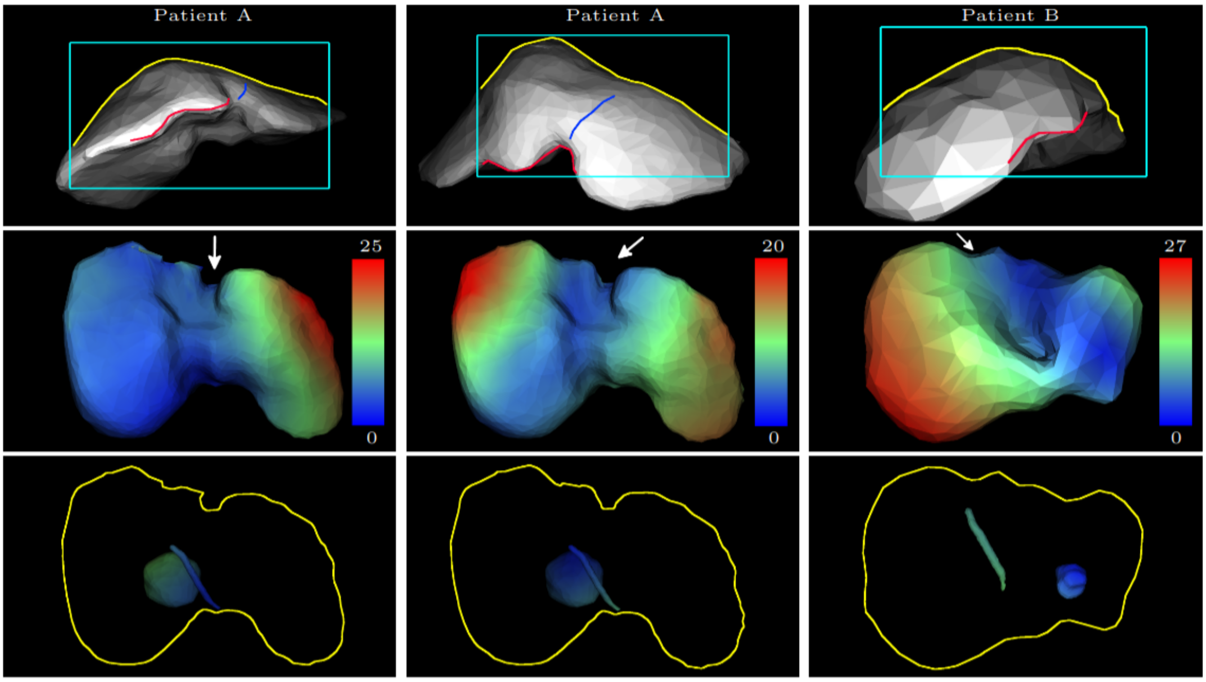

Figure 2. Quantitative registration experiments with the preoperative liver model. White arrows show the laparoscope's look at directions.

Qualitative results with real data. Figure 3 shows the in-vivo liver registration experiments. Each column shows a different registration. The second row shows the registered liver models on the input images. The third row shows the state-of-the-art manual rigid registration results. We observe that the manual rigid registration cannot align the models as good as `RegVisC`. Click to watch how `RegVisC` functions on an image from laparoscopic liver surgery.

1.1.2.2 Registration using Visual Cues and Environment Priors

Methodology. A semi-automatic solution is proposed for the initial liver registration problem. Its inputs are the liver's preoperative 3D model and a single laparoscopy image. It thus complies well with monocular laparoscopy and exploits both the environment priors (gravity and pneumoperitoneum) and visual cues. We formulate the deformable registration problem as two consecutive subproblems, solving for *(i)* deformation and *(ii)* pose. Both subproblems are well-posed and well-constrained. We solve subproblem *(i)* by simulating the biomechanical deformations between the preoperative and intraoperative states to predict the liver's unknown initial intraoperative shape. This requires us to simulate first the gravityless shape from the gravity-loaded preoperative 3D model in supine orientation and then the shape under pneumoperitoneum pressure and gravity-load in a reverse trendelenburg orientation. We then solve subproblem *(ii)* by registering rigidly the simulated shape to the input laparoscopy image using contour cues. These visual

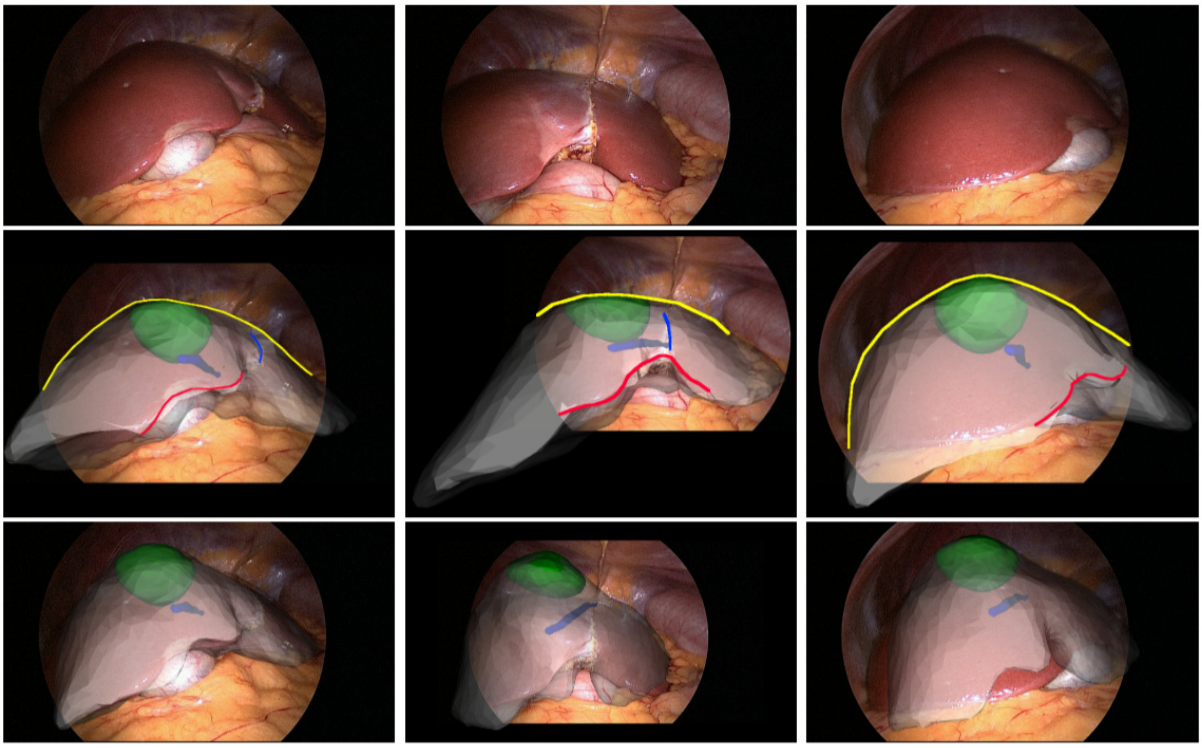

Figure 3. Qualitative in-vivo registration experiments. The augmented images show the registered tumors and veins seen through the translucent liver models. The second row is for `RegVisC` and the third row for the rigid manual registrations.

cues combine the liver's silhouette and a few curvilinear anatomical landmarks. They are marked on the input laparoscopy image by the user. Their corresponding location on the preoperative 3D model is then assigned automatically. The pipeline of the registration solution, `RegVisCEP`, using visual cues and environment priors is shown in figure 4. You can read more about the details here. The proposed registration solution using visual cues and environment priors is compared with deformable registration [C1] and manual rigid registration, referred to as deformable and rigid, respectively.

Quantitative phantom experiments. We simulated the deformation induced by the pneumoperitoneum pressure on a patient's gravityless preoperative 3D liver model. We then 3D printed out the mold of the deformed 3D liver model to eventually create its silicone phantom. We placed the phantom in a different pose compared to preoperative CT scanning, in order to obtain a realistic gravity-induced deformation, similarly to the reverse trendelenburg position. The resulting ground-truth shape was then acquired by structured-light 3D scanning using a turntable (DAVID SLS- 3 3D Scanner). The ground-truth shape thus encodes both gravity and pneumoperitoneum deformation. Figure 5 shows the registration results of the compared methods. The ground-truth shape's top part has less visibility in the camera and the bottom part was mostly occluded during scanning. Therefore, it has two holes, one on the top and one at the bottom.

Qualitative results for laparosurgery images. Figure 6 shows the segmented model from the CT volume, an input image and a control image of a patient for qualitative comparison of the registrations. The control image shows the liver with the same deformation

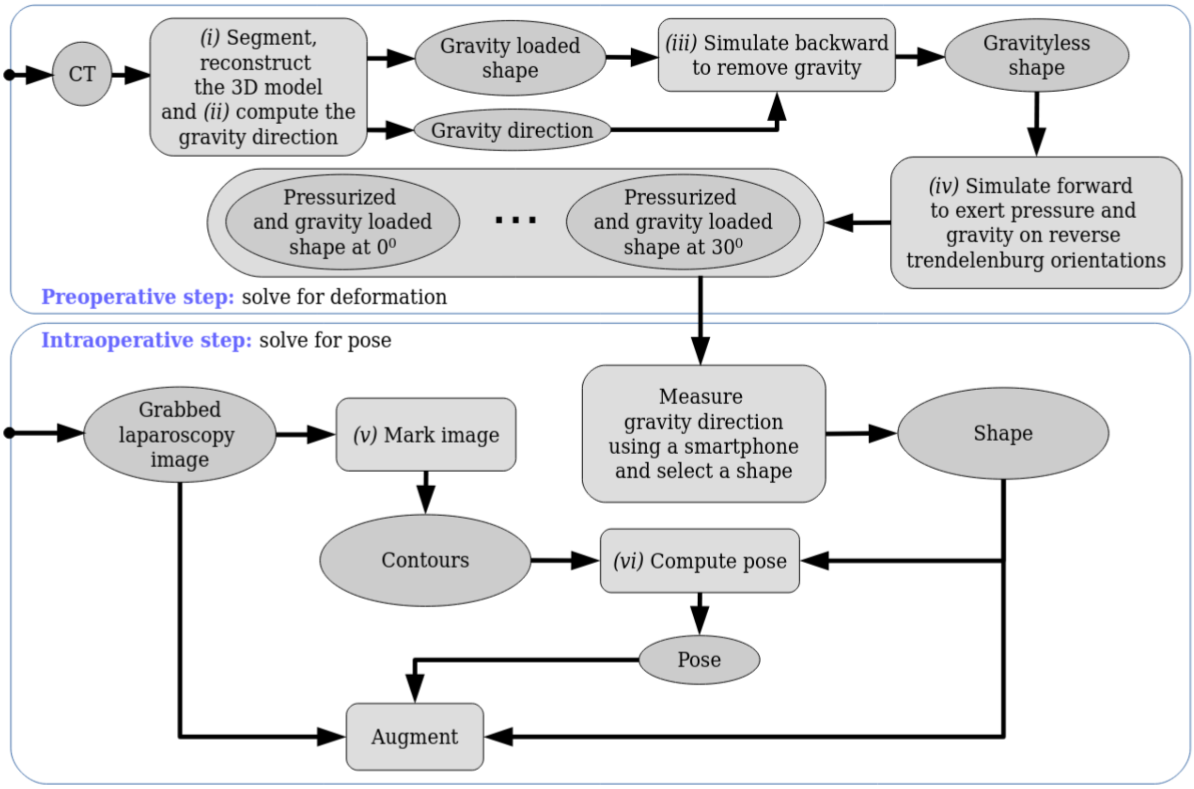

Figure 4. Pipeline of `RegVisCEP` using visual cues and environment priors.

but from a different viewpoint. The left-most half of the control image reveals a part of the liver which is unseen in the input image. In the first row of figure 7, we observe that the proposed registration and deformable match the observation well. On the other hand, in the second row, we observe that the proposed registration covers substantially better the deformation of the unseen part. This is because deformable optimizes the deformation using only the input image, which contains no information about the unseen part and therefore constrains the shape weakly and locally. In contrast, proposed registration optimizes the deformation using the strong environment priors (pneumoperitoneum pressure and gravity) which constrain the shape globally. In this case, rigid was significantly worse.

1.1.2.3 Registration using Visual Cues and User Interaction

Methodology. `RegVisCUI` is an hybrid registration approach. The key idea is that the manual rigid and deformable semi-automatic approaches are highly complementary. The hybrid approach extends and draws on the strengths of both by combining user interaction with visual cues and a biomechanical model. In other words, both the machine and the user perception are valuable and should be taken into account via the visual cues and interaction respectively. In the presence of both user interaction and visual cues, the hybrid approach bundles all constraints in a single registration. In the absence of user interaction, it behaves similarly to the existing deformable semi-automatic approaches. In the absence of visual cues, it allows the user to edit the registration under guidance of the biomechanical model. This is a significant improvement compared to the existing manual rigid approach as it allows the user to fully express their expertise in anatomy,

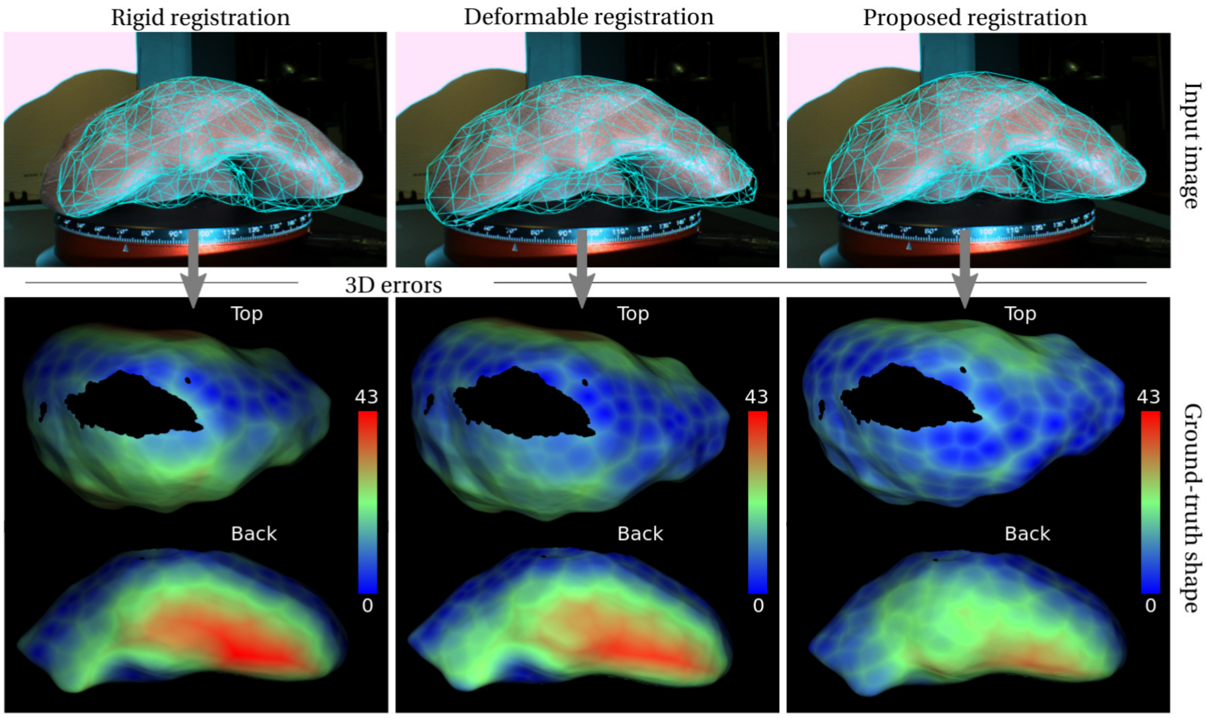

Figure 5. The first row shows the registration for the compared methods. The second row shows the colormapped 3D errors (mm) from different viewpoints on the ground-truth shape measured with respect to the registered shapes in terms of the closest point distances.

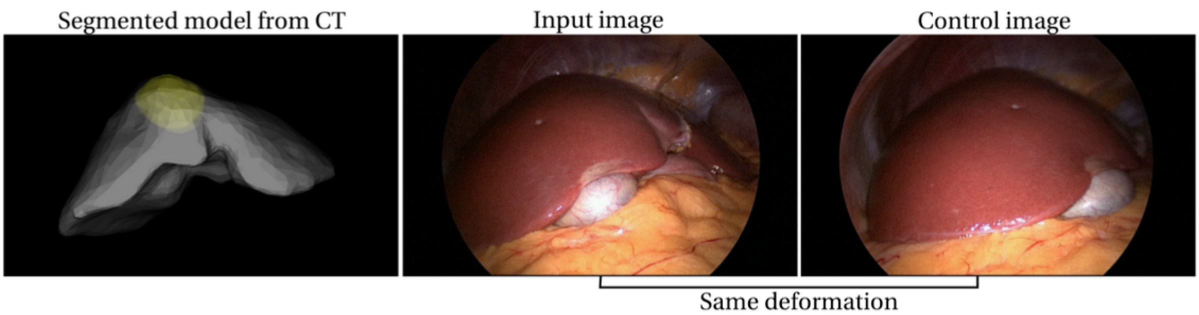

Figure 6. The segmented liver model with the tumor (yellow) from the CT volume, an input image and a control image for qualitative comparison of the registrations. The control image shows the liver with the same deformation from a different viewpoint.

prior experience and spatial understanding of the case at hand to the system. `RegVisCUI` implements this idea following the cage-based paradigm from the field of shape editing. The cage is a set of handle points enclosing the organ. Dragging these handle points interactively deforms the model. A tactile graphical user interface is designed and implemented as in figure 8. Shape editing is a widely studied problem. The cage-based paradigm is well-adapted to registration owing to its flexibility. You can read more about the details here.

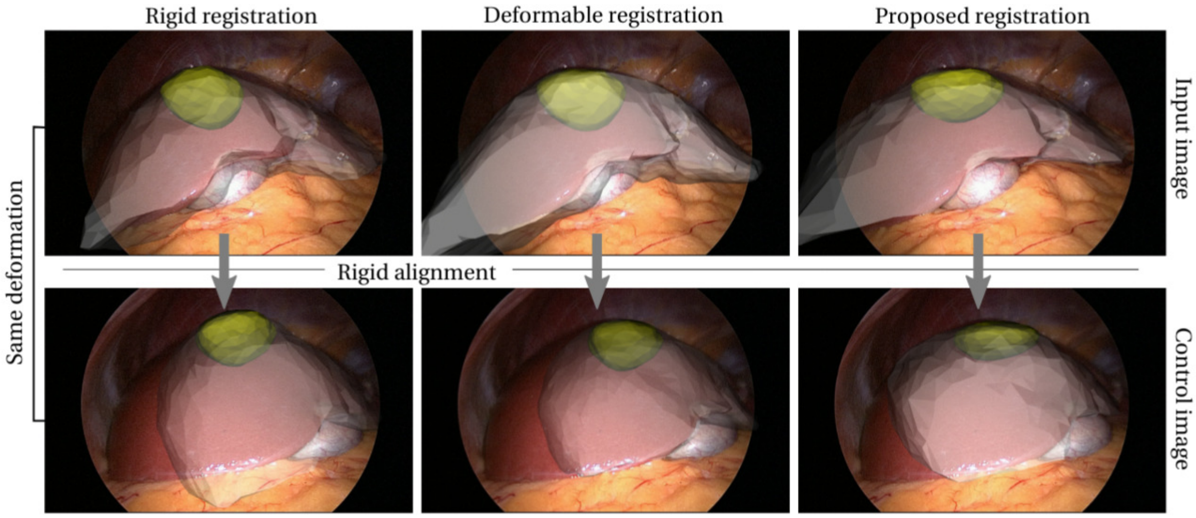

Figure 7. Qualitative comparison of in-vivo registrations for the images seen in figure 6. On the input image, both deformable and the proposed registration look convincing. In the second row, we check the quality of the registered shapes by rigidly aligning them to the control image.

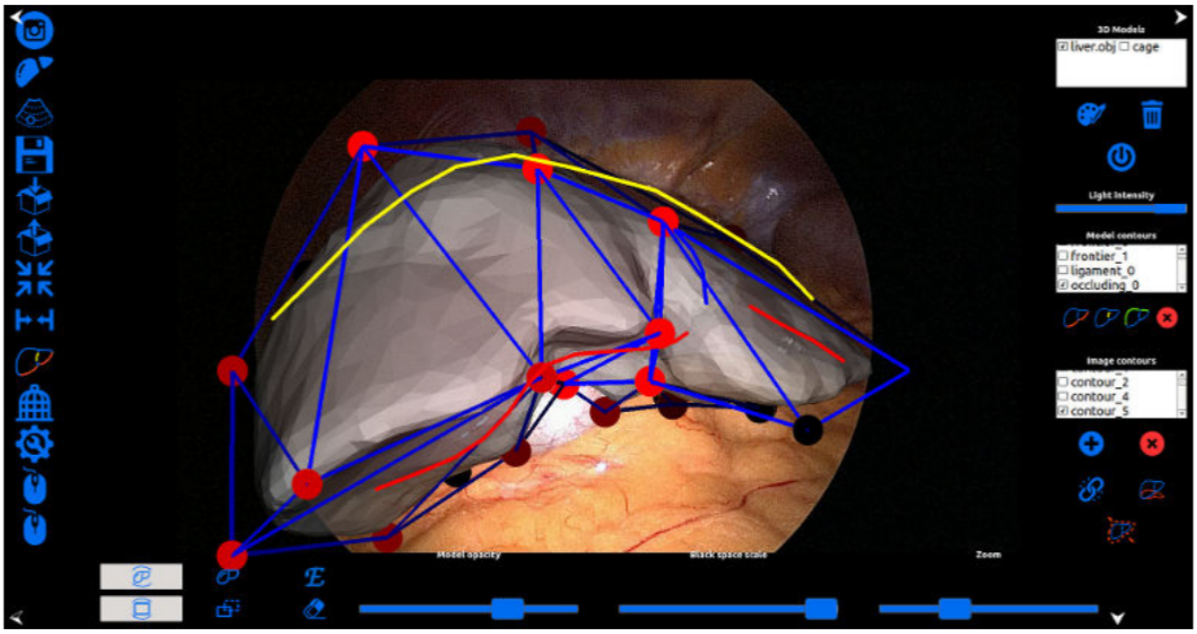

Figure 8. Proposed tactile Graphical User Interface for RegVisCUI.

Results. Qualitative registration results delivered by the state-of-the-art methods and the proposed RegVisCUI are shown in figure 9. Top right result is from semi-automatic deformable registration method [C1] based on visual cues (contour constraints in yellow, blue and red). The visual cues are sparse and do not convey enough information to unambiguously constrain registration outside the field of view. Bottom left result is the manual rigid registration method. It is too restrictive to accurately model the liver deformation. Bottom right result shows the proposed hybrid method, combining visual cues with a biomechanical model through cage-based tactile interaction. The cage's control points (red dots) are used to edit the deformable registration while also optimizing simultaneously and automatically for the constraints of the visual cues.

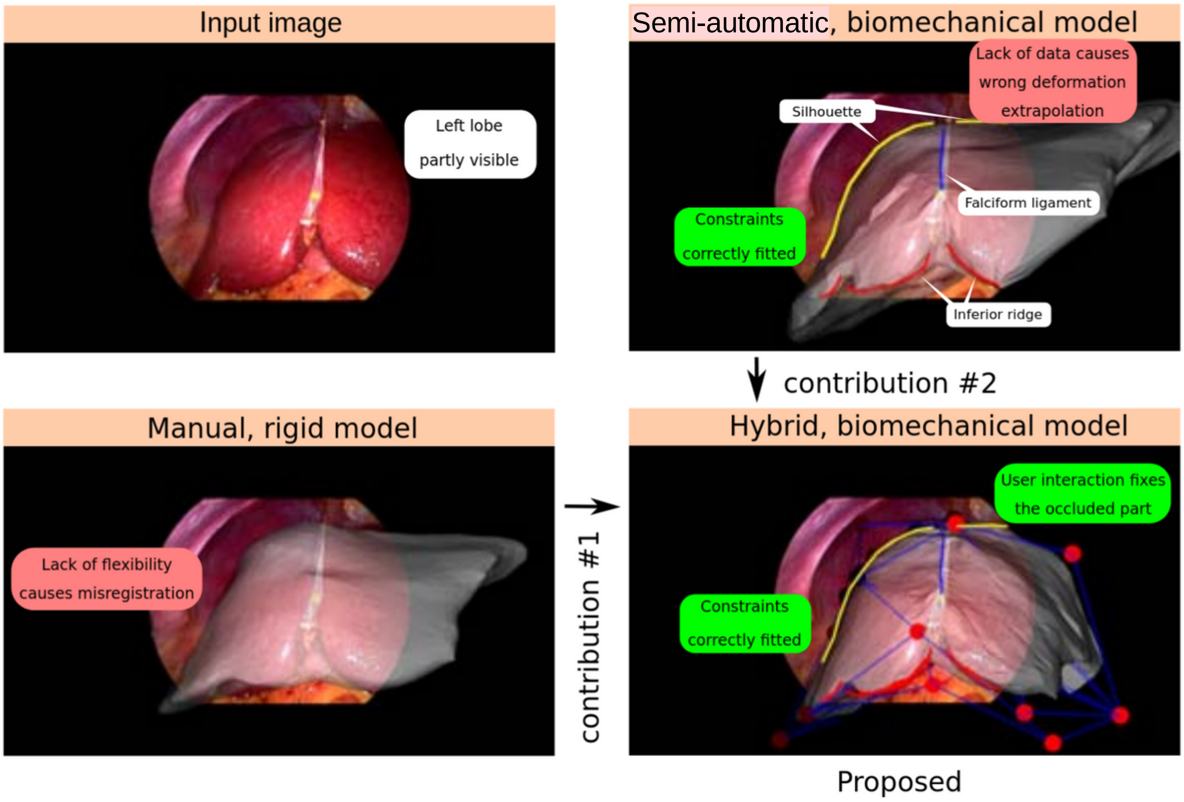

Figure 9. A comparison of the registrations with `RegVisCUI`.

References for SPH methods

[SPH1] S. Nicolau et al. "Augmented reality in laparoscopic surgical oncology". In: *Surgical Oncology* (2011).

[SPH2] N. Haouchine et al. "Using contours as boundary conditions for elastic registration during minimally invasive hepatic surgery". In: *IROS*. 2016.

[SPH3] Stoyanov D et al. "Real-time stereo reconstruction in robotically assisted minimally invasive surgery". In: *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*. 2010.

[SPH4] S. Bernhardt et al. "Using shading to register an intraoperative CT scan to a laparoscopic image". In: *Workshop CARE at MICCAI*. 2015.

[SPH5] J. Bosman et al. "The role of ligaments: patient-specific or scenario-specific?" In: *ISBMS*. 2014.

[SPH6] S. F. Johnsen et al. "Database-based estimation of liver deformation under pneumoperitoneum for surgical image-guidance and simulation". In: *MICCAI*. 2015.

[SPH7] Y. Adagolodjo et al. "Silhouette-based pose estimation for deformable organs application to surgical augmented reality". In: *IROS*. 2017.

1.2 3D-to-2D rigid registration

1.2.1 Problem overview

IMMORTALLS project. The project's objective is to enable automatic and real-time augmented reality guidance in liver laparoscopy prior to tumor resection. The project relies on intraoperative laparoscopic ultrasound (LUS) imaging data and preoperative imaging data to reveal the tumor and to track the tumor with a robotically-controlled LUS probe on the liver surface while the liver deforms and moves. In the following, I present the progress made on the registration problem related to `IMMORTALLS` project.

Problem statement. Given the preoperative 3D tumor model, segmented from a volumetric CT or MRI, register the preoperative 3D tumor model to a laparoscopic image using 2D intraoperative LUS imaging. This 3D to 2D multimodal registration problem can be divided into two subproblems: *(i)* LUS pose estimation in the laparoscope's camera coordinate frame and *(ii)* preoperative 3D tumor model to 2D intraoperative LUS image registration. See figure 10. In the following subsections, subproblem *(i)* is addressed.

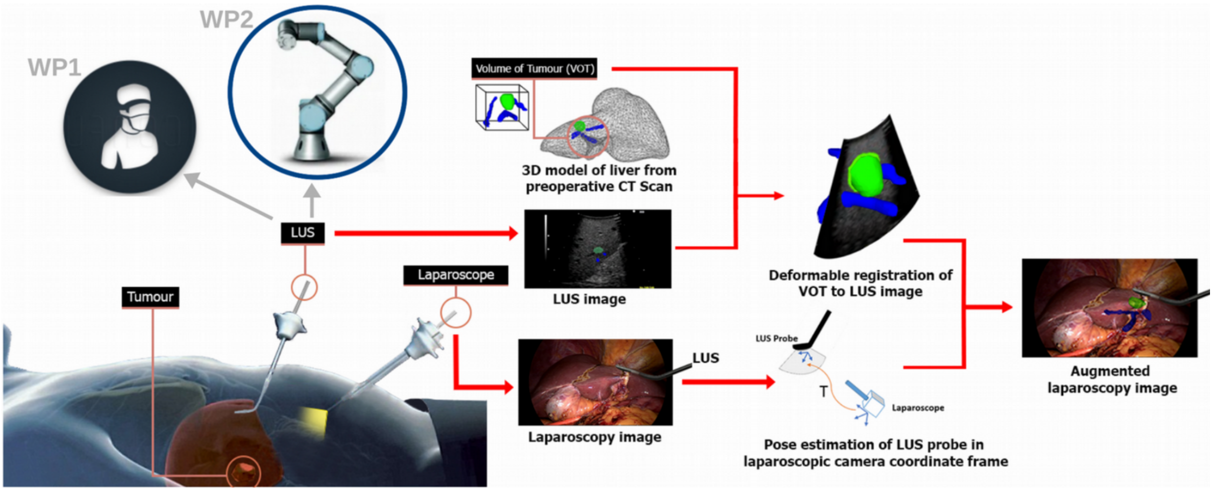

Figure 10. Overview of IMMORTALLS project.

1.2.2 LUS pose estimation subproblem

Challenges. The LUS pose problem is challenging, owing to the operating room (OR) conditions, with four main reasons. First, the laparoscope is the only sensor and it provides monocular RGB images. Second, the LUS probes are usually textureless, shiny, and partially occluded. This makes it difficult to find features in the laparoscopic images that strongly constrain the pose. Third, an approach using markers, optical or electromagnetic tracking would require one to introduce new elements or sensors in the abdominal cavity, which is highly impractical and hardly feasible for obvious sterilization and safety reasons. They would also increase the OR clutter and the risk of occlusions by the staff and other devices. Fourth, surgical robots with cable-driven actuation do not allow one to estimate LUS pose reliably via their internal encoders. This is because of the inherent inaccuracy of the readings and the error accumulation of encoder values over time.

Furthermore, most existing commercial LUS probes have cylindrical shaft and head shapes and hemispherical tips. These include Fujifilm L44LA, Siemens Acuson P300 LP323, Philips L10-4lap, SonoScape LAP7, and Aloka UST-5418. One such LUS probe can be modeled mathematically as a straight homogeneous spherical cylinder (SHSC) segment, which is also known as a spherocylinder. See figure 11. The LUS pose can be

Figure 11. Fujifilm's L44LA LUS probe.

defined as the position of the tip's hemispherical center and as the direction vector of the head's cylinder axis. It is obvious that, owing to the symmetry of the spherocylinder, a 1 dof rotational ambiguity exists on the pose, as an unrecoverable angle about the cylinder axis.

Requirements. We set four requirements for a practical LUS pose estimation method. It should be free of *(i)* markers, *(ii)* extra sensors, *(iii)* initialization, and should *(iv)* run in real-time in any OR.

State-of-the-art. As for the vast majority of pose estimation problems, existing LUS pose estimation methods can be broadly split into two categories: *(C1)* methods requiring new elements or sensors to be integrated into the OR, and *(C2)* methods using only the standard laparoscopic monocular RGB images.

Methods in category *C1* use either an off-the-shelf tracking device requiring special equipment such as electromagnetic trackers and optical motion capture systems [LUS0], [LUS0], or marker-based vision techniques [LUS0]. Methods in this category thus fail requirements *(i)* or *(ii)*; yet, they form the state-of-the-art in LUS pose estimation.

On the other hand, methods in category *C2* are promising, as they generally fulfill the four requirements for a practical solution by construction. Yet, this category is still void of existing methods for LUS pose.

1.2.2.1 Contributions

I present `LUP`, a single shot 5 dof Laparoscopic Ultrasound Pose estimation method, and its improved version `LARLUS` working with sequence of images. `LUP` has three notable contributions.

1. `LUP` respects all the four requirements mentioned above.
2. `LUP` has a well-engineered algorithmic pipeline, including image segmentation and robust model estimation.

3. LUP proposes a novel minimal formulation for the reconstruction of a spherocylinder from its occluding contour.

LARLUS has three notable contributions.

1. LARLUS improves LUP in terms of run time and robustness.
2. LARLUS provides 6 dof pose estimation whenever the LUS shaft is visible.
3. LARLUS propose a geometry-specific pose filtering method on the pose estimates by taking into account the LUS probe's geometry. This substantially reduces pose flickering across successive frames.

LUP

Methodology. LUP has two main steps. *(i)* LUS probe segmentation and contour extraction, and *(ii)* pose estimation from unclassified contour points. Both steps represent challenging problems. LUP's steps are illustrated in figure 12.

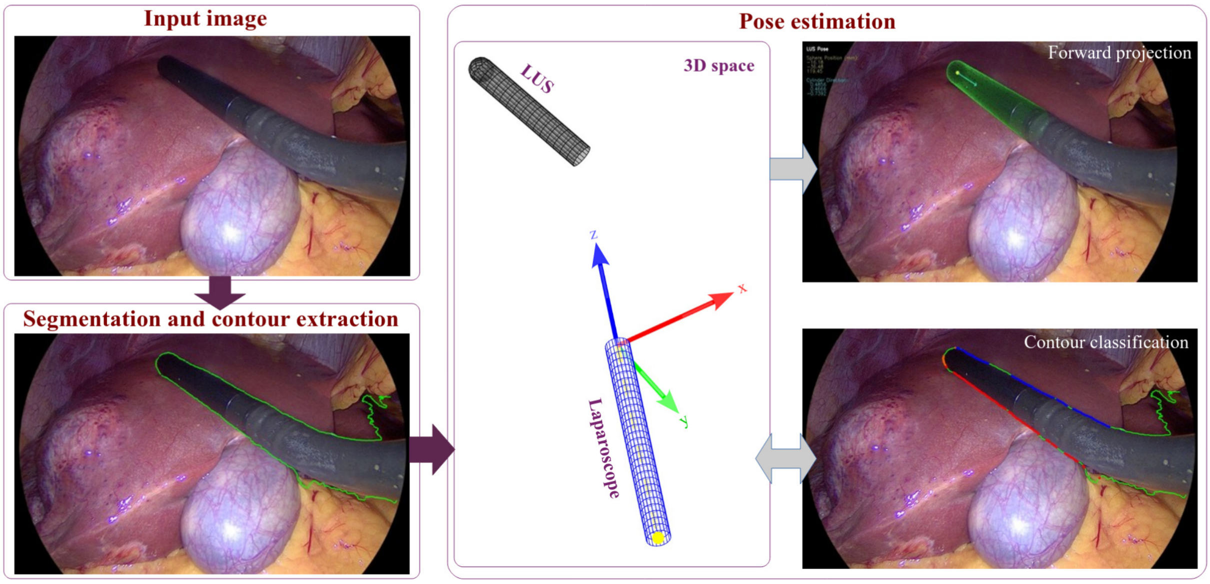

Figure 12. A single shot 5 dof LUS pose estimation with LUP on a laparoscopic image.

Step *(i)*, LUS probe segmentation and contour extraction, uses deep learning for semantic segmentation. Training used liver surgery images. Experts manually marked the LUS probes. Contours of the probe were then extracted.

Step *(ii)*, pose estimation from unclassified contour points, uses robust estimation with minimal solutions. Explicitly, the minimal solutions are for a line and a sphere. Sequential RANSACs are employed with the minimal solutions to build the LUS pose (*i.e.*, spherocylinder pose). First, it uses pairs of points to fit lines for cylinder edges and builds the cylinder with the best consensus. Second, it uses two points and two lines to fit spheres and selects the sphere of best consensus. Third, it assembles sphere and cylinder to form the LUS pose. You can read more about the details here.

Qualitative results with clinical data. `LUP` is applied on laparoscopic images from 5 different liver surgeries. Figure 13 shows these qualitative results. It is observed that the projected LUS head model from the estimated LUS poses yields visually perfect alignments with the input images. In these experiments, `LUP` runs at 10 fps (65 ms for image segmentation and contour extraction, and 35 ms for pose estimation). This is fast enough considering that the LUS probe moves slowly in search of the tumor.

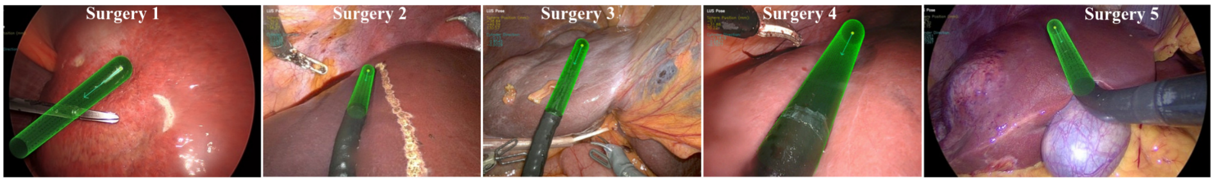

Figure 13. Forward projections of the LUS head model onto the five different surgery images from their estimated poses by `LUP`. Click to watch how `LUP` functions on these laparoscopic liver surgery videos.

LARLUS

Methodology. `LARLUS` relies on two components. *(i)* The first component is a single-shot pose estimation `LUP*`. It is an improved version of `LUP`. It estimates 6 dof LUS poses faster and more robustly. *(ii)* The second component is the geometry-specific LUS pose filtering. You can read more about the method here.

6 dof LUS pose estimation. We can only estimate 5 DoF poses from the LUS probe's head's spherocylinder shape. The 6th dof cannot be observed directly because of the rotational symmetry around the LUS head's axis. However, most LUS probes have 4-way (*i.e.*, left, right, up and down) articulation for the transducer head. We make the observation that the shaft axis and the transducer head axis of the LUS probe form a plane Π_L, when only an up-or-down displacement is applied on the 4-way articulation. In such a LUS probe posture, by construction, the head transducer's imaging plane $\Pi_{\underline{L}}$ overlaps the plane Π_L. See figure 14. It follows that, given the 5 dof LUS pose by `LUP` and the LUS shaft's cylinder axis direction in the laparoscope coordinate frame, we can estimate the pose of the LUS imaging plane $\Pi_{\underline{L}}$. Subsequently, we can build the 6 dof LUS pose. Any of the following scenarios allows us to recover the LUS shaft's cylinder axis direction. First scenario is when the LUS shaft is also visible in the laparoscopic image. Second scenario is when both the LUS probe and the laparoscope are controlled by robots providing their relative pose.

Geometry-specific filtering. We study the LUS probe's geometry to define the most suitable variables for filtering. This is for two reasons. First, we want to filter only the most noise-sensitive parts of the pose estimates. Second, we want to avoid over-filtering on the pose estimates because of the possible couplings between the variables. Consequently, we ended up forming the quadruplets $\{\,\alpha, \beta, \theta, ||\mathbf{t}||\,\}$ shown in figure 14. We consider that they are well decoupled from each other. The variable $||\mathbf{t}||$ is an absolute value and is estimated at each single frame. The variables $\{\,\alpha, \beta, \theta\,\}$ are incremental values and are estimated as differences from two consecutive frames. $\{\alpha, \beta, \theta, \}$ are expressed within the coordinate system of the subsequent frame. We filtered each variable using a 1D Unscented Kalman Filter (UKF) with a constant velocity motion model and white noise for the measurements and the motion model.

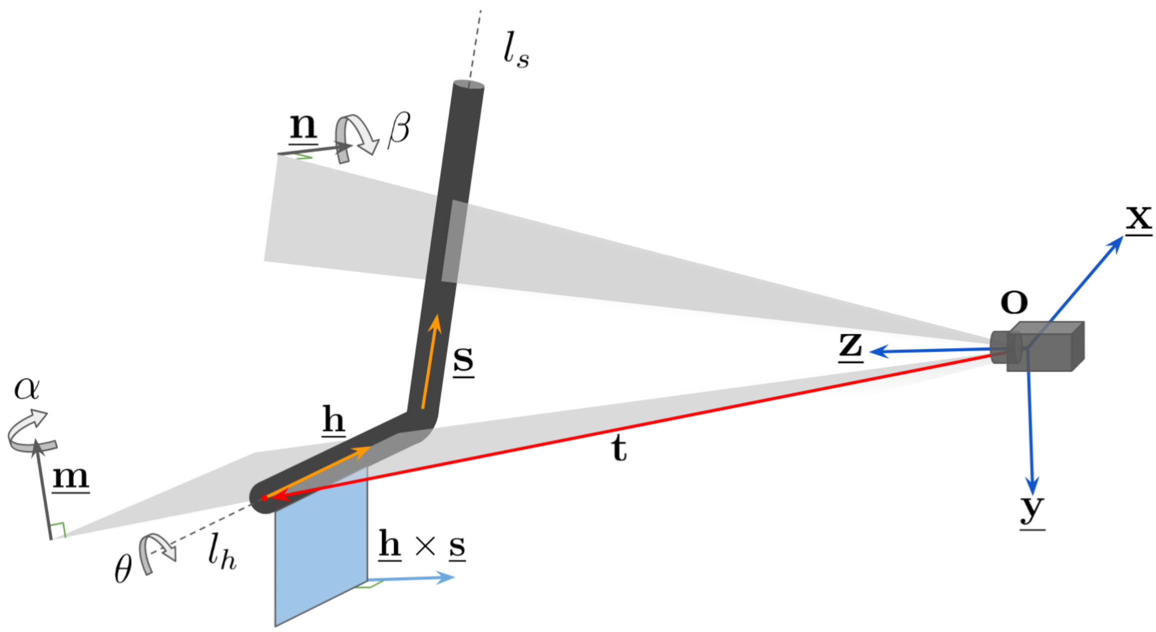

Figure 14. LUS probe geometry and modeling. $\{\underline{\mathbf{x}}, \underline{\mathbf{y}}, \underline{\mathbf{z}}\}$ denote the axes and $\mathbf{o}$ the origin of the camera coordinate frame, $\mathbf{t}$ denotes the tip position (*i.e.*, hemisphere center) and the unit vector $\underline{\mathbf{h}}$ represents the LUS probe's head axis direction. Similarly, the unit vector $\underline{\mathbf{s}}$ represents the LUS probe's shaft axis direction. The vector $\underline{\mathbf{h}} \times \underline{\mathbf{s}}$ forms the normal of the LUS image plane (*i.e.*, the blue plane) when the head's lateral movement lever is left constant. The vector $\underline{\mathbf{m}}$ represents the normal of the interpretation plane defined by head axis line l_h and $\mathbf{o}$. The vector $\underline{\mathbf{n}}$ represents the normal of the interpretation plane defined by shaft axis line l_s and $\mathbf{o}$. The rotation angle α around $\underline{\mathbf{m}}$ quantifies the depth variation of the LUS probe's head axis direction $\underline{\mathbf{h}}$ on its interpretation plane. Similarly, the rotation angle β around $\underline{\mathbf{n}}$ quantifies the depth variation of the LUS probe's shaft axis direction $\underline{\mathbf{s}}$ on its interpretation plane. Finally, the rotation angle θ quantifies the LUS probe's head rotation around its axis direction $\underline{\mathbf{h}}$.

Results. Figure 15 shows results of 6 dof LUS pose estimation when the LUS shaft is visible on a clinical data. The upper row shows the LUS imaging planes without filtering. The lower row shows the LUS imaging planes with filtering. In each frame, the green contour denotes a current configuration at an instant k, while the white contour denotes the previous configuration at the instant $k-1$. One can observe that, without filtering, even a slight LUS probe movement might flicker the LUS imaging plane considerably because of sensitivity to noise. An instance of localized hidden tumors through `LARLUS` is shown in figure 16.

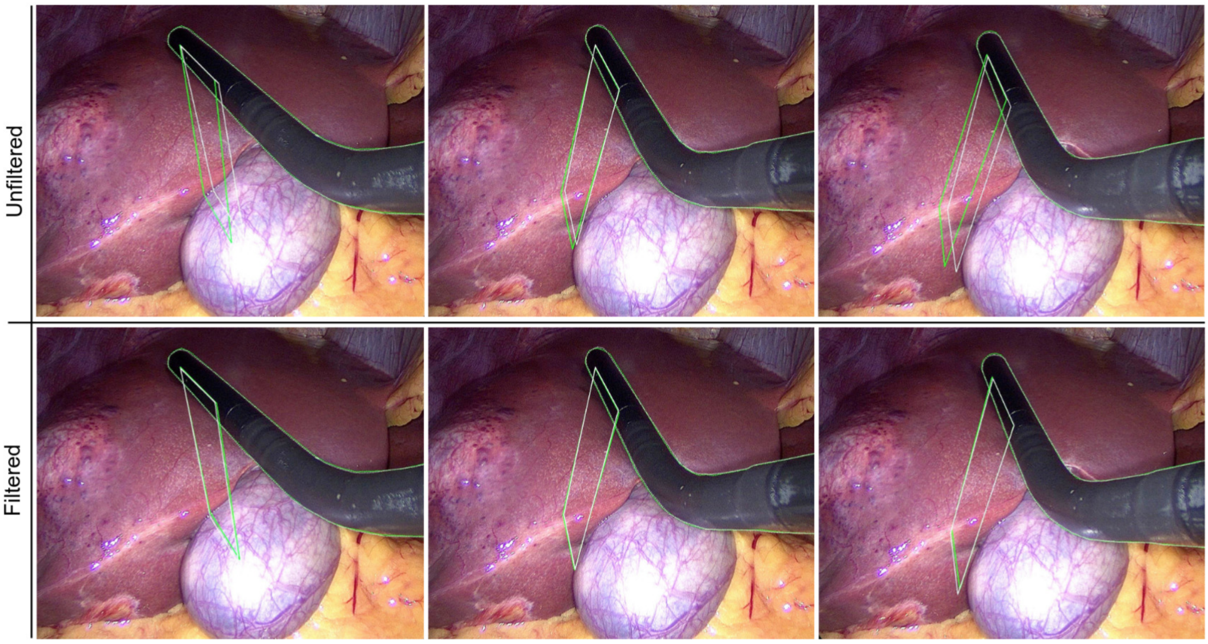

Figure 15. `LARLUS` using `LUP`* for 6 dof pose estimations whenever the LUS shaft is visible. Upper row shows unfiltered 6 dof poses. Lower row shows filtered 6 dof poses. Click to watch how `LARLUS` functions on this laparoscopic liver surgery video.

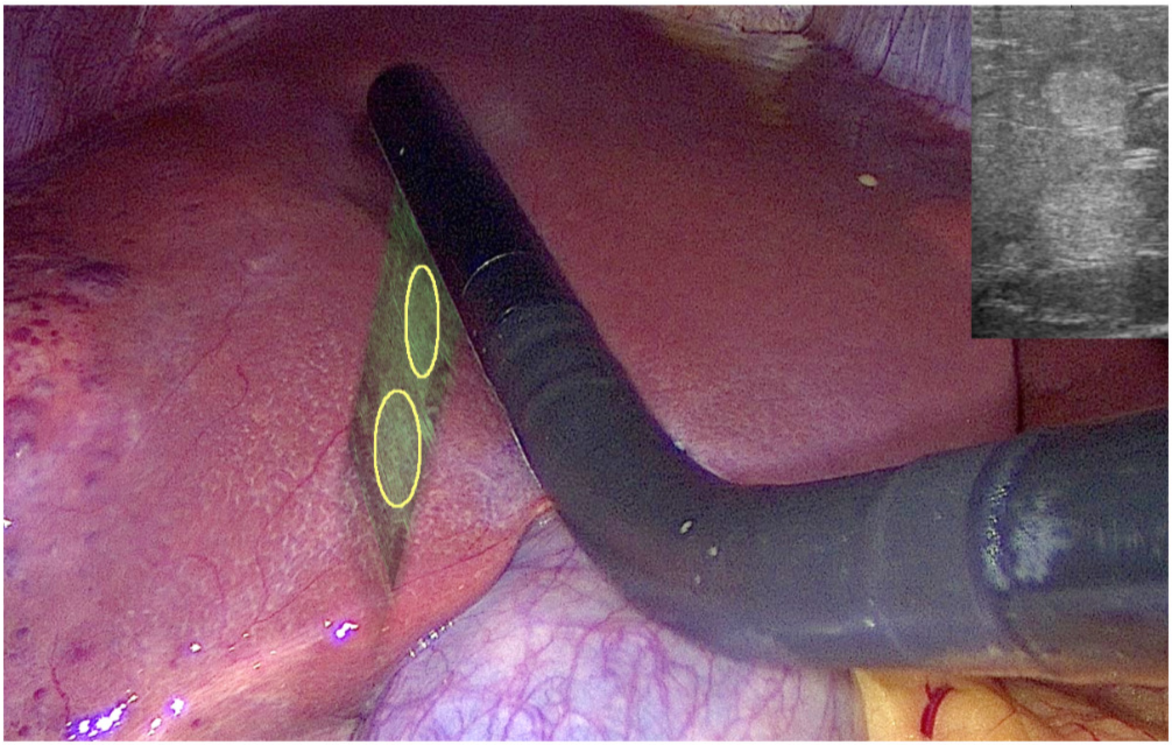

Figure 16. An instance of `LARLUS` output. The LUS image is augmented on the laparoscopic image with the bounding ellipses of hidden tumors. The original LUS image is also overlaid on the top-right corner.

Chapter 2

Augmentation of the hidden

2.1 Problem overview

Augmented Reality (AR) plays an increasingly important role in medicine. It may become especially important in laparoscopy, which still challenges surgeons in some procedures, such as the tumorectomies. Tumorectomy is the surgical removal of a tumor with a minimal amount of healthy tissue from the host organ. Laparoscopic tumorectomy may be difficult when the tumor is hidden in the organ, because there is currently no means to localize it. In particular, there is no tactile feedback. AR can help surgeons by augmenting the tumor, available from preoperative CT or MR volumes, on the live laparoscopy video. It can be implemented without additional hardware. In order to achieve this, one must solve two difficult problems: *(i)* registration between the preoperative volume and the laparoscope's image, and *(ii)* occluded object visualization. The medical AR literature mostly concentrates on the registration problem.

Problem statement. Given a registration solution, propose a visualization method for the hidden anatomy in monocular laparoscopy such that the visualization conveys a correct depth perception to a surgeon for a hidden part of the anatomy. This may help a surgeon to plan an optimal resection path towards a tumor and reduce damage in the healthy tissue.

State-of-the-art. It is unnatural for the human eye to see through an opaque surface. Thus the visualization problem of the occluded objects forms a challenging task. Most of the literature in depth perception for occluded objects (*a.k.a.*, X-Ray AR) focused on mobile hand-held displays and head-mounted displays (HMD) [AUG1–AUG7]. These platforms allow one to use multiple depth cues (static/dynamic, monocular/stereo) to counter-balance the depth misperception of occluded object visualization. In monocular laparoscopy however, one must use monocular X-Ray AR. Occlusion is then known to be the most powerful depth cue [AUG8]. In this case, occluded object visualization becomes self-conflicting, because the occluded tumor is rendered onto its occluding organ, and thus occludes the occluding surface. Consequently, the surgeon's visual perception stimulates the occlusion message: "the rendered tumor occludes the organ". This makes the surgeon see the rendered tumor as if floating on top of its occluding surface [AUG2]. To fix this misperception, a virtual window was introduced on the occluding surface [AUG2]. This fakes the surgeon's visual perception by giving the impression of seeing the other side of the occluding surface. However, it was reported that the rendered tumor continues to float even under motion [AUG9], until some occlusion also happens in the virtual window [AUG3].

In medical AR, few methods have so far been proposed for the visualization of the hidden anatomical structures [AUG5, AUG10, AUG11]. None of the existing methods form a satisfying solution to visualize the hidden anatomy in monocular laparoscopy.

2.2 Contribution to depth perception

Contributions. I propose `OTViS` (occluded tumor visualization) method. It exploits the strongest depth cues of monocular laparoscopy: occlusion and relative size. `OTViS` contributes to hidden tumor depth perception by combining the following features.

1. It introduces auxiliary orthographic tumor silhouettes on the front and back surfaces of the organ, creating two distinct depth planes and a perceivable metric space for the tumor. The use of orthographic projection ensures the silhouettes are the exact size of the tumor, enabling the strong relative size depth cue.
2. A virtual transparent dark screen, in the form of the front silhouette, lowers the luminance contrast, making the tumor appear deeper and reducing the illusion of it floating on the organ's surface.
3. Rendering the tumor as a transparent surface generates spontaneous occlusions by overlaying the organ's surface context features, strengthening the perception of the tumor being inside the organ.

Methodology. `OTViS`' methodology is illustrated in figure 1 on uterus with its tumor. The silhouettes are obtained by orthographic projection onto the front and back planes. The planes' orientations are defined from the sight-line passing through the tumor's center of mass. The planes are located at the intersection points of the organ's surface with this sight-line. You can read more about the method here.

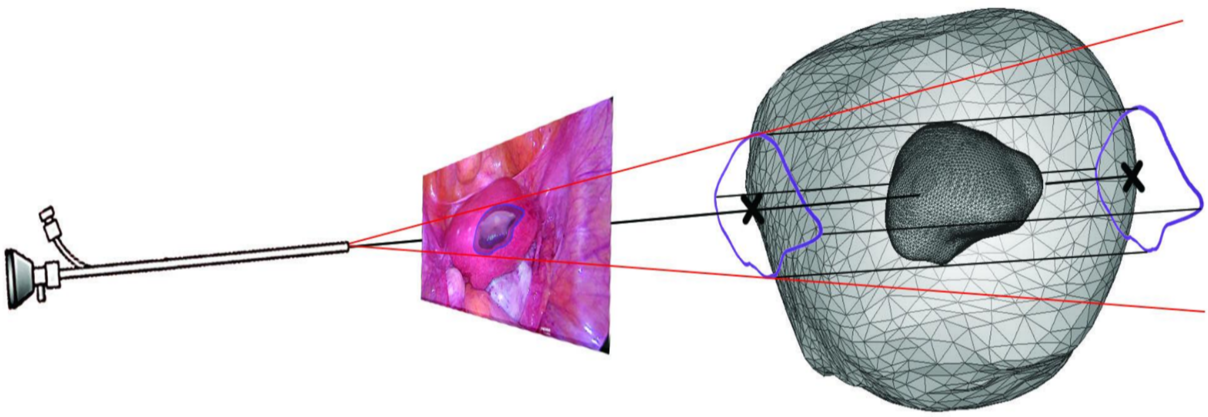

Figure 1. Illustration of `OTViS` on uterus with its tumor.

Results. We conducted a user study to compare `OTViS` against the conventional transparent overlay visualization. We used a patient's uterus for the tumor augmentation. We selected four different laparoscopy images of the uterus for the user study. We used four uterine tumors with different sizes and locations for each image. This thus yielded 16 different cases to evaluate. Figure 2 shows visualization examples. The results of the study have shown that OTViS almost doubled the quality of depth perception as compared with the conventional transparent overlay.

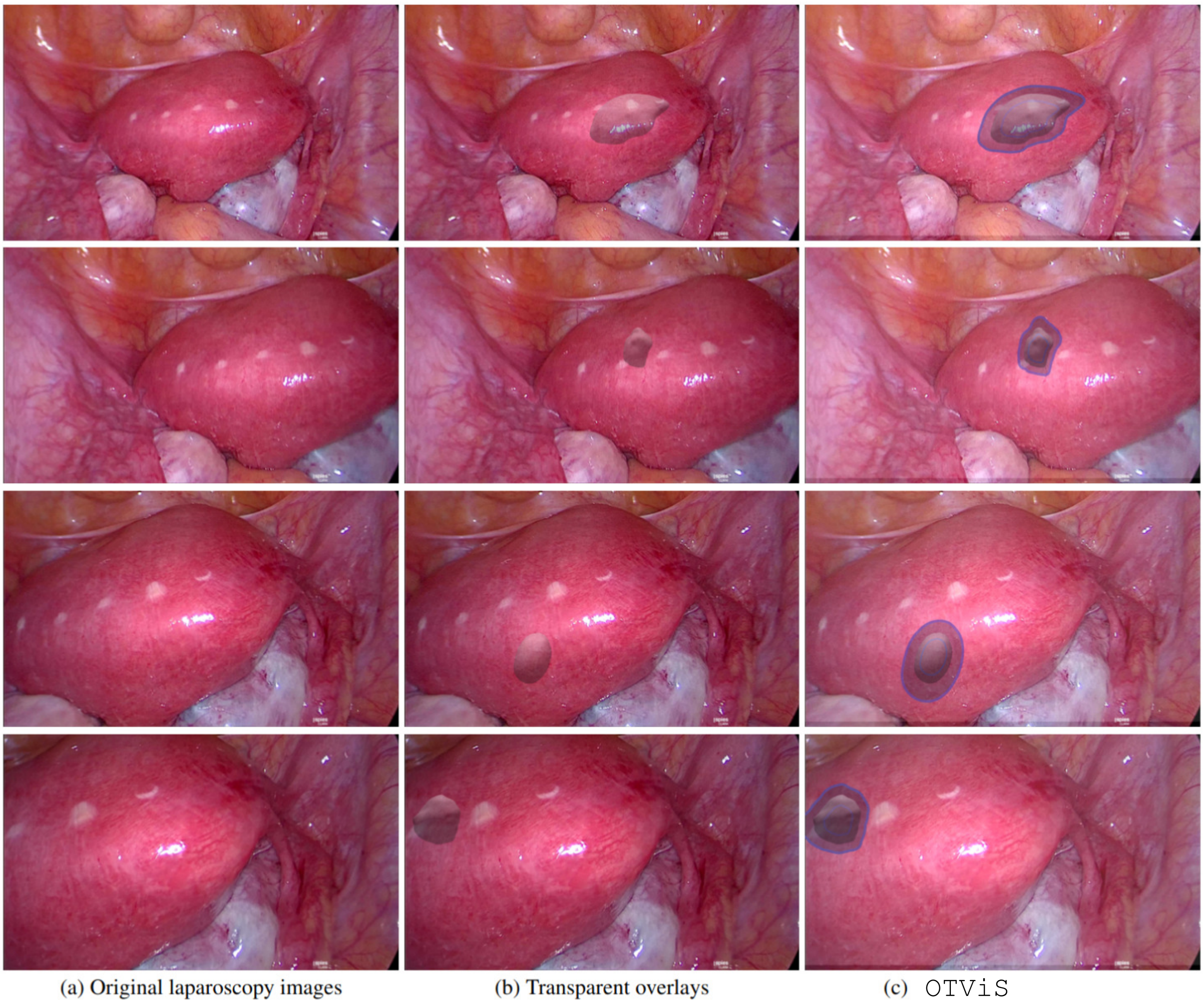

Figure 2. Augmentation examples of in-organ uterus tumors. Click to watch OTViS on this laparoscopic uterus surgery video.

References for AUG methods

[AUG1] M. Bajura, H. Fuchs, and R. Ohbuchi. "Merging virtual objects with the real world: Seeing ultrasound imagery within the patient". In: *International Conference on Computer Graphics and Interactive Techniques*. 1992.

[AUG2] C. Bichlmeier et al. "Improving depth perception in medical AR: A virtual vision panel to the inside of the patient". In: *Bildverarbeitung für die Medizin*. 2007.

[AUG3] H. Fuchs et al. "Augmented reality visualization for laparoscopic surgery". In: *Medical Image Computing and Computer Assisted Intervention*. 1998.

[AUG4] M. Kyto et al. "Improving relative depth judgments in augmented reality with auxiliary augmentations". In: *ACM Transactions on Applied Perception* (2013).

[AUG5] T. Sielhorst et al. "Depth perception – a major issue in medical AR: evaluation study by twenty surgeons". In: *Medical Image Computing and Computer Assisted Intervention*. 2006.

[AUG6] M. Berning et al. "A study of depth perception in hand-held augmented reality using autostereoscopic displays". In: *International Symposium on Mixed and Augmented Reality*. 2014.

[AUG7] A. Dey, A. Cunningham, and C. Sandor. "Evaluating depth perception of photorealistic mixed reality visualizations for occluded objects in outdoor environments". In: *IEEE Symposium on 3D User Interfaces*. 2010.

[AUG8] J. Cutting and P. Vishton. "Perceiving layout and knowing distances: The integration, relative potency, and contextual use of different in-formation about depth". In: *Perception of Space and Motion*. 1995.

[AUG9] C. Furmanski, R. Azuma, and M. Daily. "Augmented-reality visualization guided by cognition: Perceptual heuristics for combining visible and obscured information". In: *International Symposium on Mixed and Augmented Reality*. 2002.

[AUG10] R. Wang et al. "Visualization techniques for augmented reality in endoscopic surgery". In: *Medical Imaging and Augmented Reality*. 2016.

[AUG11] M. Kersten-Oertel et al. "Augmented reality in neurovascular surgery: feasibility and first uses in the operating room". In: *International Journal of Computer Assisted Radiology and Surgery* (2015).

Shape servoing of the hidden

I present below a research problem proposal that I would like to address in the future.

Problem statement. Given a 2D camera for visual feedback, control the robots to manipulate a deformable object so that a hidden object inside it deforms from its current shape to a desired shape.

State-of-the-art. None.

An application. Shape control of the hidden anatomy inside an organ for precise and safe tumorectomy in computer-assisted minimally invasive robotic surgery.

Part III – The Invisible

If an object cannot be observed by the available sensors, then it is considered invisible.

Objective

I set the following objective.

Objective. Shape perception and augmentation of an invisible object.

Outline

Part III presents my research project proposal `FulllyALIVE`. It considers the above objective in the context of a medical application founded on computer vision, robotics, augmented reality, medical imaging and liver laparoscopy fields. I submitted `FulllyALIVE` for a European Research Council (ERC) consolidator grant (CoG) in 2024. `FulllyALIVE` succeeded the 1st step, and was interviewed in the 2nd step, and was graded as B. I shall dedicate myself to solving the fundamental research problems of this proposal.

FullyALIVE

`FullyALIVE` is the acronym for "Toward Full-time Augmented-reality Guidance in Liver Endoscopy".

Proposal's context, motivation and objectives

Context. Liver cancer is a leading cause of cancer death worldwide. An estimated 830,000 people around the world died from the disease in 2020. Liver resection is considered as one of the most effective treatments. In this respect, laparoscopic liver resection (LLR) comes up by reducing substantially patient trauma compared to open liver resection. The patient recovers faster which in return reduces healthcare costs.

Challenges of LLR. However the use of LLR remains limited. This is because of three challenges. First, controlling intraoperative bleeding using laparoscopic instruments requires advanced technical skills. Second, the surgeon cannot manually palpate the liver and thus cannot locate the tumors and their resection margins easily. Consequently this raises a risk of inadequate resection on the patient's liver such as the removal of too much healthy tissue and the leaving of some cancerogenous tissue behind. Third, laparoscopic ultrasonography (LUS), the only tool for intraoperative subsurface imaging which allows real-time tumor localization, has a long learning curve. This is because its design consists of a small transducer with a small field of view attached to the end of a long shaft with a pivoting mechanism.

Existing Solutions. In order to ease LLR, augmented reality (AR) guidance solutions were proposed. AR allows a surgeon to see the subsurface tumors and veins by overlaying preoperative imaging (e.g., CT, MRI) or intraoperative imaging (e.g., LUS) data on a live laparoscopic (LAP) video. The AR guidance solutions can be categorized broadly into two groups: (G1) solutions based on preoperative imaging data [RP1–RP3] and (G2) solutions based on intraoperative LUS imaging data [RP4, RP5].

G1 solutions predict the location of the tumor by overlaying the preoperative data onto the laparoscopy image. G1 solutions require the whole liver to be visible as much as possible in the laparoscopy image to make a reliable prediction. However, the liver is usually very partially visible (i.e., about 30% or less) in laparoscopic images. G1 solutions are useful to guide surgeons at the very beginning of the laparoscopic surgery prior to tumor resection. G1 methods are neither automatic (except [RP6, RP7]) nor real-time (except [RP8]).

G2 solutions simply overlay the LUS images onto the laparoscopy images. G2 solutions require the LUS probe's pose to be known in the laparoscope's coordinate frame. The LUS probes are usually textureless, shiny, and partially occluded. Therefore, G2 solutions use either markers or optical or electromagnetic trackers to estimate LUS pose [RP4, RP5]. This requires one to introduce new elements or sensors in the abdominal cavity of the patient, which is highly impractical and hardly feasible for obvious sterilization and safety reasons. This also increases the operating room (OR) clutter and the risk of occlusions by the staff and other devices. A recent work [RP9] proposed a markerless and trackerless LUS pose estimation solution. G2 solutions can operate locally and they are not affected by the restricted liver visibility. G2 solutions might suffer in revealing the tumor in some specific cases. For instance, if a tumor is isoechoic, then it appears very similar to the echotexture of the surrounding liver parenchyma; or if a tumor is small and a chemotherapy is applied, then the tumor might get smaller or disappear completely.

G1 and G2 solutions are fundamentally different and complete each other. They form together an AR guidance solution especially prior to the tumor resection. Figure 1 shows results from G1 and G2 solutions.

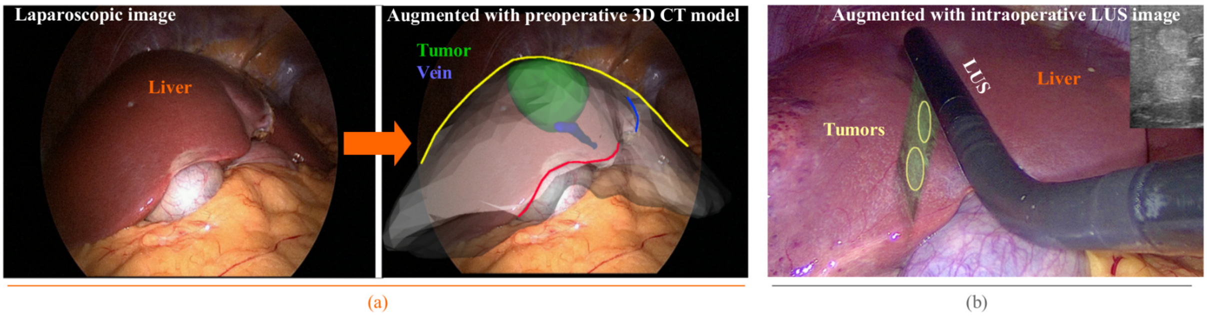

Figure 1. (a) a G1 solution from [RP3]. (b) a G2 solution from [RP9].

G1 solutions cannot provide an AR guidance during the resection. This is because a very small portion of the liver is visible and when the liver gets cut, its 3D model topology becomes wrong. G2 solutions provide barely a satisfactory AR guidance during resection. This is because of three reasons. First, the acoustic reflection artifacts increase substantially during the tumor resection. This usually obscures the tumor in LUS images in most of the viewing configurations of the LUS probe. Second, the LUS probe is retrieved away during resection, otherwise it obstructs the resection. Third, even if the LUS probe is not retrieved away, an electrocautery resection instrument blinds its LUS imaging.

Figure 2 illustrates on a timeline where and how long the existing AR guidance solutions can be applied during a laparoscopic liver surgery.

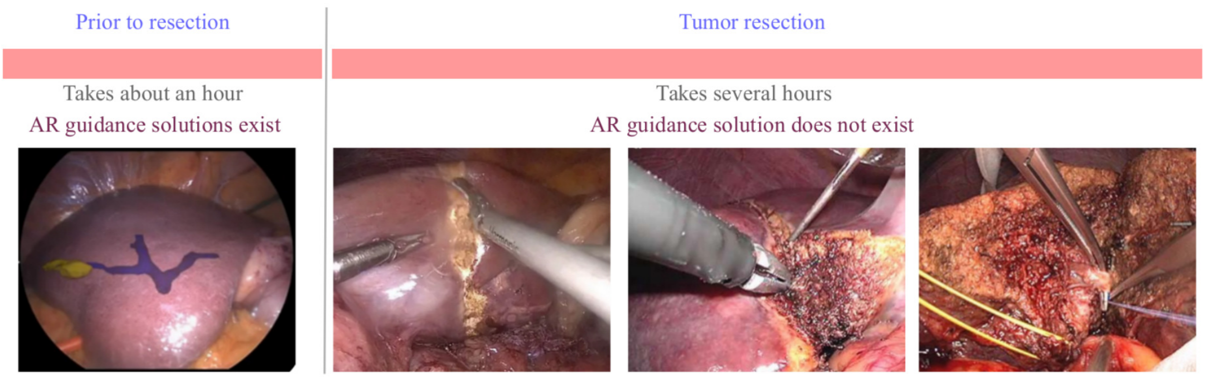

Figure 2. AR guidance during laparoscopic liver surgery.

Why does AR guidance matter during tumor resection? Prior to resection, a surgeon usually makes traces for the resection plan on the liver surface region which occludes the tumor. The traces are made using an electrocautery instrument guided by either LUS or an AR solution. Electrocautery yields crude traces. Once the guidance cannot be sustained anymore, only these traces can help the surgeon form mental resection planes for the remaining several hours of the surgery. These mental resection planes can easily deviate from the initial planned resection planes during the resection. This is because of multiple factors. First, the liver deforms and the tumor shifts whenever a resection instrument interacts with the liver. Second, the more resection progresses, the more liver's shape and appearance change compared to the moment where mental resection planes are formed in the surgeon's mind. Third, due to the first two factors, it is probable that a very slight angle deviation on liver surface can occur for a following resection plane. The deeper the tumor, the more this angle deviation amplifies the distance of the resection plane to the tumor. A deviated resection plane might move towards critical structures and even miss to include the tumor zone. A possible AR guidance solution during tumor resection has potential to reduce such risks. It can provide *(i)* visual guidance for the alignment of the resection instrument to a resection plane; *(ii)* real-time feedback on deviation from a planned resection plane; *(iii)* warnings when instruments are moving toward critical structures; and *(iv)* real-time updates on tumor resection margins.

Objective. The proposal's objective is to enable automatic and real-time AR guidance during the tumor resection period of the laparoscopic liver surgery by tracking the tumor and nearby vessels and augmenting them with a resection plan in the laparoscopic images.

Ambition. The proposal's first ambition is to achieve the objective in the de-facto LLR OR setup. That is to say, a 2D monocular laparoscope and a LUS probe are the only sensors available. The proposal's solution shall not use any markers nor trackers. Subsequently, the proposal's AR guidance solution has potential to be deployed to ORs worldwide. The second ambition is to put the proposal's AR guidance solution into practice in the collaborator hospital CHU Clermont-Ferrand.

Proposed Solution. Proposal will first form a set of keyframe AR solutions (KARS) prior to resection. Then, a favorable keyframe AR solution will be registered to a laparoscopic image showing a resection scene by taking into account topology, deformation and appearance changes. KARS set will be updated with the new registrations to ease future registrations. The laparoscope's motion will also be controlled by a robotic arm to optimize the surgical gestures and the AR guidance.

Hypothesis. The occluding liver surface of the tumor is observable in the laparoscopy video. The use of robotic control for the laparoscope makes this hypothesis even more viable than for a hand-held laparoscope.

Expected Result. The proposal will simplify LLR. It will shorten hospital stays, improve surgical safety and accuracy, and contribute to an overall better quality of life and reduction of healthcare costs.

State-of-the-art

A practical AR guidance pipeline for LLR should typically solve three problems. *(p1)* Registration prior to tumor resection: 3D-to-2D multimodal deformable registration. *(p2)* Registration during tumor resection: 3D-to-2D multimodal, deformable, topology-and-appearance varying registration. *(p3)* Augmented reality: Visualization of an occluded tumor and a laparoscopic resection guidance model. As it is explained in the previous section, all the state-of-the-art LLR AR guidance solutions address problem *p1*. Therefore, here only the related work on problems *p2* and *p3* is reviewed.

3D-to-2D multimodal, deformable, topology-and-appearance varying registration

There exists no work which registers a 3D model (i.e., a combination of preoperative CT/MRI imaging data and intraoperative LUS and laparoscopic imaging data) to a 2D laparoscopic image under deformation, topology and appearance changes in LLR.

It has been reported so far two registration methodologies related to this problem. The first registration methodology uses a geometric criterion to handle cuts already happened on an organ along a fixed depth direction [RP10]. Although it has promising results in controlled experiments, it has two limitations hindering its application in the clinical setting of an LLR. The first limitation is that, to detect cuts, it requires the deformable registration is solved perfectly from the point features, which is very unlikely in practice. Furthermore since the model topology is wrong before the registration and solving for the deformation is a difficult problem, then we cannot expect a highly accurate registration. Subsequently, even a temporary and mild noise in the registration will cause spurious volume deletion with no possibility of later recovery. The second limitation is that it cannot update the model topology while a cut is in progress, which may take long minutes. It cannot handle either a partial cut or a cut with curved depth. The second registration methodology uses image-based detection to handle cuts [RP11]. Although the registration methodology is validated on clinical data, it is tailored for the laparoscopic uterus surgery. The uterus is smaller and has mostly rigid behaviour compared to the liver. Since the uterus is smaller, it is also usually fully visible even during resection. Therefore the second registration methodology neither forms a solution for LLR.

On visualization of an occluded tumor and a laparoscopic resection guidance model

Occluded tumor visualization. Although there exists several depth cues (e.g., motion, height, relative size, aerial perspective, occlusion, etc.) to exploit for transmitting the depth of an occluded object in an augmentation, the most widely used augmentation method in medical AR is the transparent overlay visualization of the subsurface tumor onto the occluding visible surface. On the other hand, the occlusion is known to be the most powerful depth cue for the human visual perception [RP12]. This makes the transparent overlay visualization self-conflicting, because the occluded tumor is now rendered onto its occluding organ, and thus occludes the occluding surface. Consequently, the surgeon's visual perception stimulates the occlusion message: "the rendered tumor occludes the organ". This makes the surgeon see the rendered tumor as if floating on top of its occluding surface [RP13]. To fix this misperception different augmentation methods are proposed such as the virtual window [RP13]. In [RP14], the authors proposed an augmentation of the virtual windows in the form of tumor's silhouette in order to also transfer the depth information naturally to the surgeon by forming a ratio-scaled metric space. Occluded object visualization still forms an open problem in AR and it is more difficult to solve in the de-facto LLR OR setup since we are constrained to use only a few monocular depth cues (i.e., relative size, occlusion and motion).

Laparoscopic resection model visualization. LLR's one of the objectives is to minimize the removal of the healthy liver tissue to the least possible while removing all the cancerogeneous cells. A properly designed laparoscopic tumor resection guidance model would lead the surgeon to resect accurately the tumor with the defined safe margins. It has been reported so far only one such resection guidance model, also known as, "Tool-port Projection" [RP15]. It is a promising guidance model. However, it guides the surgeon simply by showing a straight line. The straight line gives an idea of where the tumor is, but it is not the right path. A right path should avoid critical structures (e.g., veins, arteries). One can pre-establish such a resection path on the preoperative CT model which avoids critical structures. Once the preoperative CT model with the resection path is registered to the laparoscopic image, then the question is how to show effectively this resection path to the surgeon.

Contribution to the state-of-the-art

There exists no AR guidance solution during the resection period of LLR. `FullyALIVE` will break this limit by building an automatic and real-time AR guidance solution for the resection period of LLR. `FullyALIVE` will contribute to the state-of-the-art by proposing the following four novelties.

1. `FullyALIVE` shall solve a multimodal registration problem with deformation, topological and appearance changes between a 3D model (i.e., a combination of preoperative CT/MRI and intraoperative LUS and laparoscopic images) and a 2D laparoscopic image.
2. Ground-truth registrations are not available in most medical applications. This is also true for LLR. `FullyALIVE` thus shall assess the registration uncertainty of the subsurface structures (e.g., tumor, veins). This is crucial for the surgeon's decision-making.
3. `FullyALIVE` shall study how to convey correctly both the tumor's depth with its uncertainty and a resection path to the surgeon via an intuitive and distractionless AR visualization.
4. `FullyALIVE` shall provide a practical robotic control system of the laparoscope's field-of-view to optimize both the surgical gestures and AR guidance.

Methodology

`FullyALIVE` is located at the intersection of the medical, computer vision, augmented reality and robotics fields. `FullyALIVE` has three work-objectives (WO) on the core scientific problems. WO1 will address 3D-to-2D deformable multimodal topology-and-appearance varying registrations. WO2 will address AR guidance visualization. WO1 and WO2 form already a stand-alone AR guidance solution for de-facto OR setup of liver laparoscopy. WO3 proposes to adapt the AR guidance solution formed by WO1 and WO2 for a minimally-invasive surgery (MIS) OR where a robotic arm is available or for a de-facto LLR OR equipped with a robotic arm. More explicitly, WO3 will explore robotic motion control of the laparoscope to improve AR guidance by tracking stably the tumor resection zone. WO3 will be designed to perform its task independently from the WO1's solution. By doing so, WO1 and WO3 will have capacity to mitigate each other's failures and to make their respective solutions more robust when they will be coupled.

WO1. 3D-to-2D Multimodal, Deformable, Topology and Appearance Varying Registrations

Goals. First, WO1 shall (Task 1.1) formulate and solve the registration problem between a 3D liver model and a 2D laparoscopic image showing a liver resection. Then, WO1 shall (Task 1.2) estimate and evaluate the registration uncertainty. Finally, WO1 shall (Task 1.3) validate the registration solution clinically and improve it along the validation period.

Challenges. Solving an intraoperative registration problem in the de-facto LLR OR setup is extremely difficult. This is because a 2D monocular laparoscope and a LUS are the only sensors available. Subsequently, the most important challenges to confront are as follows.

1. *(c1) Deformation.* A 3D liver model can only be formed prior to resection using either preoperative CT data or intraoperative LUS imaging data or both. This implies that the registration of this prior 3D model to a 2D laparoscopic image grabbed further in time during resection should inevitably compensate for the deformation. This is because the liver moves by the breathing motion, is deformed by the abdominal insufflation and is compressed by the resection instruments compared to the prior 3D model.
2. *(c2) Topological changes.* During resection the liver gets cut. The prior 3D model's topology thus becomes wrong. Consequently, the registration method should also be able to adapt the 3D model's topology.
3. *(c3) Appearance changes ≡ correspondence problem.* The 3D liver model can always be textured with the laparoscopic camera (LAP) images prior to resection using existing G1 and G2 solutions. However, the liver's appearance changes drastically during the resection in LAP images. Deformation and the cuts on the liver are the main causes of its appearance change.
4. *(c4) Restricted liver visibility ≡ correspondence problem.* During resection the hand-held laparoscope zooms in the resection zone. This reduces the laparoscope's field-of-view (FoV). A great portion of the liver's visible surface goes out of FoV. Furthermore, the laparoscope's reduced FoV is visually cluttered during resection by the surgical instruments, blood, smoke, specularities, and blurring/defocusing effects due to the movements of the zoomed-in hand-held laparoscope. These restrict the liver's visibility substantially. Consequently, corresponding primitives among LAP images become sparse and very difficult to match because of challenges *c3* and *c4*.

5. *(c5) Registration uncertainty evaluation.* Estimating registration uncertainty in medical images provides valuable information for clinical decision-making. However the ground-truth registrations are not available for LLR. It is thus challenging to evaluate the quality and validate the accuracy of an uncertainty estimation.

Work programme. WO1's goals are split into 3 tasks. Each task is explained one by one in detail below.

Task 1.1. Registration problem formulation, solution development and evaluation

Goal. First, to formulate and solve the multimodal, deformable, topology-and-appearance varying registration problem between a 3D liver model and a 2D laparoscopic image showing a liver resection. Second, to evaluate the accuracy of the proposed solution.

Work plan. I would like to enrich an input 2D laparoscopic image showing a liver resection scene with the preoperative 3D CT imaging data and the inraoperative 2D LUS imaging data. I consider that the length of the resection only grows in time (which is always true) and that the resection is visible entirely from time to time during the resection period. These assumptions can allow detection of the resection in an input laparoscopic image and update of the 3D liver model when the length of the detected resection has grown. I conceive the following scenario to achieve the task goal. I split the scenario into 8 steps. Step 1.1.1: I build intraoperatively a set of timestamped keyframe deformable registration solutions prior to the resection. The keyframe laparoscopic images are taken from different laparoscope poses. Some keyframe laparoscopic images show the liver under various local deformations around the tumor location. These local deformations are generated by the instruments and/or the LUS probe while the LUS probe is visible in the images and can observe the tumor. This step helps alleviate challenge *c1*. Step 1.1.2: I find the best matching keyframe laparoscopic image (showing the liver prior to the resection) to an input laparoscopic image (showing the liver during resection). Step 1.1.3: I perform deformable image registration between the best matched keyframe laparoscopic image and the input laparoscopic image. Step 1.1.4: I detect the resection region boundary in the input laparoscopic image. Step 1.1.5: I transfer the detected resection region boundary to the best matched keyframe image (step 1.1.2) using the known deformable image registration (step 1.1.3). The transferred resection region boundary should collapse to a curve segment on the keyframe laparoscopic image since the liver was still intact. Step 1.1.6: I backproject the keyframe resection curve segment to its registered intact 3D liver model surface prior to resection. The 3D liver model is then virtually cut according to this resection curve segment. This step helps alleviate challenge *c2*. Step 1.1.7: I register the updated 3D liver model (step 1.1.6) to the input laparoscopic image under the guidance of the known deformable image registration (step 1.1.3). Step 1.1.8: I include this registration solution (step 1.1.7) as a new element into the set of the timestamped keyframe registration solutions. I also remove the oldest timestamped keyframe registration solution from the set. This step helps alleviate challenges *c1*, *c3* and *c4* by applying progressive updates on the set. I finally loop back to the step 1.1.2. One might also conceive a different methodology to solve the registration problem (e.g.,, if possible, an end-to-end unsupervised deep learning approach). I recall that this task still forms an open problem.

Inputs. A multimodal 3D liver model (i.e., a registered combination of preoperative CT/MRI imaging data and intraoperative LUS and laparoscopic imaging data), a 2D laparoscopic image showing a liver resection scene, the laparoscope's camera intrinsic calibration matrix.

Output. The registered multimodal deformed 3D liver model with the updated topology and appearance in the laparoscope's camera coordinate frame. The registration method.

Methods. I will explore the state-of-the-art 3D-to-2D deformable registration methods to initialize the set of deformable registration solutions prior to the resection. I will explore deep learning methods to solve the image matching, resection detection, and deformable image registration problems. I will next re-explore, once the 3D liver's model topology is updated, the state-of-the-art 3D-to-2D deformable registration methods prior to the resection to solve the final registration step.

Task 1.2. Registration uncertainty estimation and evaluation

Goal. To estimate and evaluate the registration uncertainty.

Work plan. To reach the task goal, I will follow 2 steps. Step 1.2.1: *Uncertainty estimation*. The registration method conceived in Task 1.1 is composed of multiple steps. I will develop uncertainty estimations of these steps. I will then form the resultant uncertainty estimation of the registration method from the uncertainty estimations of its steps. Step 1.2.2: *Uncertainty estimation evaluation*. I will have possibility to evaluate the registration uncertainty estimations of the tumor prior to the resection by comparing with the registered LUS tumor profiles (e.g., see figure 1.b). Once the tumor resection starts, LUS tumor profiles may not be observed anymore. Evaluating uncertainty estimation when the ground-truth registrations are not available is another open problem.

Inputs. The registration method developed in Task 1.1.

Output. A registration uncertainty estimation method, an evaluation method.

Methods. I will explore both conventional and deep learning based state-of-the-art uncertainty estimation methods. I will explore also the uncertainty metrics to assess the quality of an uncertainty estimation method when the ground-truth is not available.

Task 1.3. Clinical validation and improvements

Goal. To validate the registration solution clinically and improve it.

Work plan. To reach the task goal, I will follow 3 steps. Step 1.3.1: Validation on clinical datasets. I will validate the registration software (i.e., registration method with an AR visualization) on the collected clinical datasets from LLR surgeries. Step 1.3.2: Clinical protocol. I will follow a protocol which will be defined by the hospital and the surgeons so as to bring a personal computer (PC) installed with the registration software and one of the team members into the operating room. Step 1.3.3: Tests in the operating room. PC with registration software will grab the laparoscopy and LUS images, and stream automatic and real-time augmentations on its own screen. By doing so, I will not modify the existing setup of the operating room neither the surgery. I will take feedback from the surgeon about the augmentation of the tumor and take notes about the failure cases so that these can be integrated as improvements.

Inputs. An implementation of the registration method with its uncertainty estimation from Tasks 1.1 and 1.2, and an AR visualization from Task 2.2, the clinical datasets, an LLR surgery (i.e., operating room, patient, surgeon) with a clinical protocol to follow, patient's preoperative 3D liver model from CT images, the laparoscope's camera intrinsic calibration matrix.

Output. A registration method ready to be integrated in an AR guidance software for LLR use cases.

Methods. I will conduct tests first on the clinical datasets and afterward in the operating rooms, and improve the registration method.

WO2. Subsurface Structures and Resection Guidance Model Visualizations

Goals. First, WO2 shall (Task 2.1) explore the monocular depth visual cues in the context of laparoscopic liver surgery. Then, WO2 shall (Task 2.2) study the subsurface structures visualizations (i.e., occluded object visualization problem) such that their depth and uncertainty information can be conveyed to the surgeon precisely. Later, WO2 shall (Task 2.2) also study the visualizations for a laparoscopic resection guidance model so that it is correct, intuitive and distractionless. Finally, WO2 shall (Task 2.3) evaluate the visualizations clinically and improve them along the validation period.

Challenges. Solving AR visualizations in the de-facto LLR OR setup also imposes new challenges. The most important ones to confront are as follows.

1. *(c1) Occluded tumor visualization problem.* I recall that the state-of-the-art subsurface tumor augmentation method in medical AR is the transparent overlay visualization. This visualization is self-conflicting, because the occluded tumor occludes its occluding surface. Consequently, the surgeon's visual perception stimulates the occlusion message: "the rendered tumor occludes the organ". This makes the surgeon see the rendered tumor as if floating on top of the organ surface. Occluded object visualization still forms an open problem in AR and it is more difficult to solve in a de-facto LLR OR since we are constrained to use only a few monocular depth cues (i.e., relative size, occlusion and motion) on a flat 2D screen.
2. *(c2) Laparoscopic tumor resection guidance model visualization.* I recall that the state-of-the-art laparoscopic resection guidance model requires the resection instrument's incision point on the patient's abdominal wall to be known precisely in the laparoscope's camera coordinate frame. *(c2a)* Tracking the 3D coordinates of the resection instrument's incision point is a difficult problem to solve in the de-facto LLR OR setup where a hand-held 2D monocular laparoscope moves in an unpredictable manner. Consider it solved, then the resection path visualization problem

has to be confronted. The state-of-the-art visualization shows a straight line, which gives an idea of where the tumor is, but it is not the right path. A right path should avoid critical structures (e.g., veins, arteries). One can pre-establish such a right path on the preoperative CT model which will subsequently be registered to the laparoscopic image by WO1's solution. *(c2b)* Then the question is how to visualize effectively this resection path to the surgeon.

Work programme. WO2's goals are split into 3 tasks. Each task is explained one by one in detail below.

Task 2.1. Analysis and evaluation of monocular depth visual cues in liver laparoscopy

Goal. To explore the monocular depth visual cues in the context of laparoscopic liver surgery.

Work plan. Considering that WO1's solution is achieved in real-time, I will have the opportunity to exploit all the monocular depth cues, e.g., motion parallax. I will follow 2 steps. Step 2.1.1: Study the contributions of all the relevant monocular depth cues (occlusion, motion parallax, familiar size, relative size, shading and lighting, texture gradient, blur, accommodation) in visualization of the subsurface tumor and veins on a flat screen in conveying the right depth perception. Step 2.1.2: Perform user studies with surgeons and inexperienced subjects to evaluate each depth cue.

Inputs. Subsurface tumor and veins' 3D models, LLR surgery videos, the laparoscope's intrinsic camera matrix, surgeons and inexperienced subjects for answering the questions in the user studies.

Output. A set of prominent monocular depth cues to exploit in LLR AR guidance visualization.

Methods. I will explore the rendering and overlaying methods which allow mixing virtual and real objects on images. I will design thoughtful visualizations for the user studies. I will use statistical analysis to evaluate the results.

Task 2.2. Visualization study on the subsurface structures and a resection guidance model

Goal. To study the subsurface structures visualizations such that their depth and uncertainty information can be conveyed to the surgeon accurately and intuitively. To also study the visualizations for a laparoscopic tumor resection guidance model so that it is intuitive and distractionless.

Work plan. To reach the task goal, I will follow 4 steps. Step 2.2.1: Segment the non-patient pixels in the laparoscopy image (e.g., surgical instruments). Step 2.2.2: Implement a visualization of the subsurface tumor and veins with their uncertainties using the most efficient depth cues in the laparoscopy image only on the patient-pixels. Step 2.2.3: Compute the resection instrument's 3D incision point on the abdominal wall and then implement a visualization of the optimal resection path guidance. Step 2.2.4: Design preoperative training sessions for the surgeons on depth perception and resection path perception in an LLR context. The objectives are to make surgeons (i) learn to see the right depth and uncertainty for the proposed visualization; (ii) understand the resection path guidance better. The surgeon's perceptual cognition will be trained with real contextual objects such as the patient's liver phantom, the tumor phantom, surgical instruments, etc. of an LLR case and with the relevant AR visualizations. The sessions will be repeated as m-minutes per day during the last n-days before surgery. This is expected to improve the surgeon's perception capabilities during the intraoperative visualizations beyond what can be done only with the monocular depth cues.

Inputs. WO1's registration solution with its uncertainty and pre-planned resection path, the laparoscopic image, the laparoscope's intrinsic camera matrix.

Output. An effective AR visualization method for the subsurface tumor and veins, and the resection path guidance on the laparoscopy image with the monocular depth cues and uncertainty estimations.

Methods. I will explore the following methods. (i) Realistic perspective projection methods. These methods take into account the display field of view (DFOV) and the geometric field of view (GFOV), i.e., the view frustum. A perspective projection with matching DFOV and GFOV provides a geometrically correct image on a screen, as if the user was viewing a virtual window through the frame of the screen. It is frequently assumed that, in such a setup, observers perceive the metric properties of a virtual scene almost in the same way as they would perceive the properties of a corresponding real-world scene. (ii) Rendering and overlaying methods. These methods allow mixing virtual and real objects on images. (iii) X-ray visualization methods in AR, e.g., alpha-blending, ghosting, virtual windows, etc. These methods can help alleviate challenge *c1*. (iv) Animated artistic augmentation methods. These methods can allow to amplify depth cues, and thus help improve depth perception. (v) Deep-learning methods. These methods can help to segment non-patient pixels in the laparoscopy images. (vi) Colour codes and shape aura representation techniques (e.g., multiple nested isosurfaces around the shape with decreasing opacity) for the uncertainty visualization. (vii) WO1's registration solution with a hand-held laparoscope or robotically-controlled laparoscope in order to localize the resection instrument's 3D incision point (challenge *c2a*).

Task 2.3. Clinical validation and improvements

Goal. To evaluate the visualizations clinically and improve them.

Work plan. To reach the task goal, I will follow 3 steps. Step 2.3.1: Validation on clinical datasets. I will validate the AR guidance software (i.e., the implementation of the effective AR visualization method from Task 2.2 and the registration method with its uncertainty estimation from Tasks 1.1 and 1.2) on the collected clinical datasets from LLR surgeries. Step 2.3.2: Clinical protocol. I will follow a protocol which will be defined by the hospital and the surgeons so as to bring a personal computer (PC) installed with the AR guidance software into the operating room. Step 2.3.3: Tests in the operating room. PC with AR guidance software will grab the laparoscopy and LUS images, and stream automatic and real-time augmentations on its own screen. By doing so, I will not modify the existing setup of the operating room neither the surgery. I will take feedback from the surgeon about the AR visualizations and take notes about the failure cases so that these can be integrated as improvements.

Inputs. The AR guidance software (i.e., the implementation of the effective AR visualization method from Task 2.2 and the registration method with its uncertainty estimation from Tasks 1.1 and 1.2), the clinical datasets, an LLR surgery (i.e., operating room, patient, surgeon) with a clinical protocol to follow, patient's preoperative 3D liver model from CT images, patient's pre-planned resection path, the laparoscope's camera intrinsic calibration matrix.

Output. An AR guidance software ready for LLR use cases.

Methods. I will conduct tests first on the clinical datasets and afterward in the operating rooms, and improve the registration method.

WO3. Visual Servoing of the Laparoscope with Constraints Satisfaction

Goals. First, WO3 shall (Task 3.1) analyze and model the tasks (e.g., keeping the resection zone visible), hard constraints (e.g., remote-center-of-motion), and soft constraints (e.g., minimizing the surgeon's eye-hand misorientation by controlling the laparoscope's roll angle) for a task-driven laparoscope field-of-view (FoV) control in LLR surgery. Then, WO3 shall (Task 3.2) formulate a provably stable visual servoing control law of the laparoscope for a task-driven FoV control and constraints satisfaction. Finally, WO3 shall (Task 3.3) validate the motion control of the laparascope on laparoscopic phantom and animal tumor resection models.

Challenges. Visual servoing of the laparoscope will be challenged strongly during LLR and since it forms a stand-alone control solution independently from WO1. The important challenges to confront are as follows.

1. *(c1) Task instantiations.* LLR surgical workflow requires many tasks to be performed prior to resection (e.g., ligament dissection, tracing the tumor resection zone on liver surface) and during resection. Usually, a patient's abdominal cavity has never been seen before the surgery starts. Every patient is different. Thus, a task should be able to be recognized from a first-time seen surgical images. Subsequently, instantiating a task for a task-driven FoV control becomes challenging.
2. *(c2) FoV and Constraints satisfaction.* The remote-center-of-motion (RCM) forms a hard (i.e., highest priority) constraint for MIS. Its objective is to avoid hurting the patient by restricting the lateral movements of the laparoscope at its incision point on the abdominal wall. RCM is a well-studied constraint. However, it will hinder the task-driven FoV control and soft constraints' satisfaction. This is because RCM allows the laparoscope to move only with 4 degrees of freedom (dof) while LLR scene can deform and move in 6 dof.

3. *(c3) Safety.* Satisfying RCM under possible collisions between the laparoscope's robot and the surgeon or OR staff is a crucial challenge which must be addressed.
4. *(c4) Occlusions.* It is inevitable that the liver's resection zone will be cluttered by the blood, smoke and surgical instruments. These will occlude the laparoscope's FoV.

Work programme. WO3's goals are split into 3 tasks. Each task is explained one by one in detail below.

Task 3.1. Analysis and modeling of the laparoscope's tasks and constraints

Goal. To analyze and model the tasks, hard constraints, and soft constraints for a task-driven laparoscope field-of-view (FoV) control in LLR surgery.

Work plan. I will set a minimum number of tasks and constraints but enough to provide a practical solution to optimize surgical gestures and AR guidance. To do so, I will follow 4 steps. Step 3.1.1: Tasks and FoV formulations. Autonomous laparoscope FoV control should be triggered either for an inference from the laparoscopic image or for a prediction in the laparoscopic image. I will analyze and model two inference tasks: "LUS scans the liver" and "instrument cuts the liver" for FoV formulation. I will analyze and model one prediction task: "laparoscope sees the tumor resection zone" for FoV formulation. I will make a prediction on where the tumor resection zone can be in the laparoscopic image (e.g., whole liver) and refine the prediction (e.g., right liver lobe) while the surgery progresses. Integration of this prediction task into the FoV control can help AR guidance software (WO1+WO2) reveal the tumor. Or the other way around, if I fail to predict the resection zone in WO3 then the AR guidance software (WO1+WO2) can help predict it. Step 3.1.2: Multi-task FoV formulation. I will model a "union" operator for different FoVs formulated from multiple concurrent tasks to form one FoV per laparoscopic image. Step 3.1.3: Constraints and laparoscope motion. I will analyze and model two constraints of the laparoscope motion: "collision-resistant programmable Remote-Center-of-Motion" (safe RCM) constraint and "Eye-Hand Misorientation" (EHM) constraint. Safe RCM is hard constraint and must be satisfied. EHM is a soft constraint and should be satisfied as much as possible. RCM constraint will control 2 dofs of the laparoscope motion to stop its lateral movements about the incision point. Safe RCM will be analyzed and modelled using a redundant collaborative robot arm (i.e., a robot with force sensors at each joint and has at least 7 dofs) and a human-robot interaction technique. This will ensure the safety of both the staff in the OR if they collide with the robot and the patient by respecting the RCM at the incision point. EHM problem happens when the laparoscope's line of sight diverges from the surgeon's intended line of sight as (s)he observes directly the abdomen. EHM is also amplified when the laparoscope moves under an RCM constraint. EHM forms an open problem. In modeling, EHM constraint will control 1 dof (i.e., roll angle) of the laparoscope motion to optimze surgeon's gestures. Autonomous FoV will control the remaining 3 dofs of the laparoscope motion to focus the laparoscope on a desired location. I will also analyze and model three more constraints on the laparoscope motion: (i) minimal motion constraint (i.e., minimal number of displacements and minimum total displacement), (ii) smooth motion constraint and (iii) responsiveness constraint (i.e., fast convergence). These additional three constraints help decrease surgery time and surgeon's visual fatigue, and improve surgical gestures.

Inputs. A redundant collaborative robot arm and its geometric parameters, calibrated laparoscope, the clinical datasets, tumor's location in the preoperative 3D liver model (e.g., segment 8).

Output. A set of modelled tasks and constraints for optimal FoV control.

Methods. I will explore the following methods. (i) deep learning methods for context-aware task inference or prediction using action-triplets <instrument, verb, target> [RP16]. (ii) deep learning methods for semantic segmentation and recognition for predicting resection zone in laparoscopic images. (iii) Existing RCM and safe RCM formulations. (iv) Eye-hand coordination formulations. (v) Human-robot interaction techniques.

Task 3.2. Provably stable visual servoing control law formulation with constraints satisfaction

Goal. To formulate a provably stable visual servoing control law of the laparoscope for a task-driven FoV control and constraints satisfaction.

Work plan. Again to be able to provide a practical solution for laparoscope FoV control in ORs, I will formulate and implement two complementary control schemes. To do so, I will follow 2 steps. Step 3.2.1: Autonomous laparoscope FoV control. I will use the models of the tasks and the constraints developed in Task 3.1 to put them into a unified optimization problem for the task-driven laparoscope FoV control. I will design a provably stable visual servoing control law to solve the unified optimization problem. I will implement the provably stable autonomous visual servoing scheme. Step 3.2.2: Collaborative laparoscope FoV control. I will formulate and implement a collaborative control scheme between the laparoscope-holding robot and an assistant surgeon. The collaborative control scheme will allow an assistant surgeon help the laparoscope-holding robot focus better to a desired FoV. The laparoscope-holding robot will always respect the RCM and safety constraints, while it is being moved by the assistant surgeon. While the assistant surgeon moves or holds the robot, for instance a small red dot will be shown on a corner of the screen indicating the robot is in the collaborative mode. Once the assistant surgeon leaves the robot, the collaborative mode will remain active if no task is recognized in the scene and the robot will not move. Otherwise, if a task is recognized and the assistant surgeon does not hold or move the robot, then the robot will resume the autonomous laparoscope FoV control. This time, for instance, a small green dot will be shown on a corner of the screen to indicate that the robot is in the autonomous mode.

Inputs. The tasks and constraints that are modelled in Task 3.1, a redundant collaborative robot arm's geometric parameters, the laparoscope's intrinsic camera matrix.

Output. Control scheme software for autonomous and collaborative laparoscope FoV control.

Methods. I will explore the following methods. (i) Image-based and position-based eye-in-hand visual servoing schemes. (ii) Lyapunov stability theorem. (iii) Robust control theories. (iv) Collaborative control schemes (e.g., Cartesian compliance control method). (v) Null-space projection-based hierarchical multi-task methods. These methods allow a stack of tasks to be servoed and the constraints to be satisfied in a non-conflicting and convergent way. (vi) Model predictive control methods. These methods allow to take into account explicitly the constraints both on the manipulated (i.e., control inputs of the robot) and controlled variables. They also avoid excessive variations on the control inputs of the robot. Therefore they can allow us to ensure safe upper boundedness of the robot velocities. (vii) Fuzzy approximation methods. These methods can compensate efficiently for the unknown time-varying periodic disturbances, such as the breathing moving the abdomen rhythmically.

Task 3.3. Laparoscope's motion validation on laparoscopic phantom and animal resection models

Goal. To validate experimentally the motion control of the laparascope on laparoscopic phantom and animal tumor resection models.

Work plan. I will follow 2 steps. Step 3.3.1: Experiments on phantom resection models. I will test robotically-controlled laparoscope FoV control on phantom resection models. I will take notes about the failure cases. Then I will propose solutions to overcome these failure cases and integrate them as improvements to the control scheme software. Step 3.3.2: Experiments on animal resection models. I will follow a protocol which will be defined by the hospital and the surgeons so as to bring the robotic control system (robot + controller computer) of the laparoscope into the laparoscopy training room of trainee surgeons. Controller computer will grab the laparoscopic images and servo the laparoscope-holder-robot to control the laparoscope's FoV on animal resection models. Once more, I will take notes about the failure cases. Then I will propose solutions to overcome these failure cases and integrate them as improvements to the control scheme software.

Inputs. Calibrated laparoscope, 7 dof collaborative robot arm, control scheme software for autonomous and collaborative laparoscope FoV control, phantom set up, animal, trainee surgeon, trainee assistant, protocol.

Output. The robot's joints or end-effector velocities to control FoV while satisfying constraints.

Methods. I will conduct the phantom experiments in our laboratory and the animal model experiments in the laparoscopy training room of trainee surgeons in the "Centre International de Chirurgie Endoscopique (CICE)" (www.cice.fr) under the respect of ethical constraints.

Positioning with respect to the state-of-the-art. In the state-of-the-art, robot-assisted laparoscope FoV control with a programmable RCM constraint and an eye-hand misorientation constraint form a difficult threefold problem. This is because its parts are still open research problems. Furthermore, only a unified solution of the problem's parts would be realistic to put into practice in the laparoscopic ORs to optimize the surgical scene visualization and gestures. Only few recent works have addressed this threefold problem [RP17, RP18] but only with a simple tool tracking scenario. WO3 shall address this threefold problem with a safe programmable RCM to enhance safety with respect to the patient and OR staff. Second, it shall follow a task-driven FoV control approach (i.e., recognize a context-aware task from the surgical workflow, then activate a relevant FoV control) while ensuring minimal smooth and responsive laparoscope motion for the surgical site visualization all along the surgery. In case if the surgeon is not satisfied, WO3 shall also provide a collaborative mode where the surgeon can continue to operate as in the conventional laparoscopic surgery by instructing an assistant surgeon to focus the laparoscope's FoV on a desired location.

References for Research Project Proposal

[RP1] Y. Espinel et al. "Combining Visual Cues with Interactions for 3D-2D Registration in Liver Laparoscopy". In: *Annals of Biomedical Engineering* (2020).

[RP2] B. Le Roy et al. "Augmented Reality Guidance in Laparoscopic Hepatectomy with Deformable Semi-automatic Computed Tomography Alignment". In: *Journal of Visceral Surgery* (2019).

[RP3] B. Koo et al. "Deformable Registration of a Preoperative 3D Liver Volume to a Laparoscopy Image using Contour and Shading Cues". In: *MICCAI - International Conference on Medical Image Computing and Computer Assisted Intervention*. 2017.

[RP4] N. Rabbani et al. "A Methodology and Clinical Dataset with Ground-truth to Evaluate Registration Accuracy Quantitatively in Computer-assisted Laparoscopic Liver Resection". In: *Computer Methods in Biomechanics and Biomedical Engineering: Imaging and Visualization* special issue (2021).

[RP5] X. Liu, W. Plishker, and R. Shekhar. "Hybrid Electromagnetic-ArUco Tracking of Laparoscopic Ultrasound Transducer in Laparoscopic Video". In: *Journal of Medical Imaging* (2021).

[RP6] B. Koo et al. "Automatic, Global Registration in Laparoscopic Liver Surgery". In: *International Journal of Computer Assisted Radiology and Surgery* 17 (2022), pp. 167–176.

[RP7] M. Labrunie et al. "Automatic Preoperative 3D Model Deformable Registration in Laparoscopic Liver Resection". In: *International Journal of Computer Assisted Radiology and Surgery* 17.8 (2022), pp. 1429–1436.

[RP8] M. Labrunie et al. "Automatic 3D/2D Deformable Registration in Minimally Invasive Liver Resection using a Mesh Recovery Network". In: *MIDL - Medical Imaging with Deep Learning*. 2023.

[RP9] M M Kalantari et al. *Markerless Ultrasound Probe Pose Estimation in Mini-Invasive Surgery*. submitted to ICRA - IEEE International Conference on Robotics and Automation. 2024.

[RP10] C J Paulus et al. "Handling Topological Changes during Elastic Registration". In: *International Journal of Computer Assisted Radiology and Surgery* 12.3 (2017), pp. 461–470.

[RP11] T. François et al. "Image-based Incision Detection for Topological Intraoperative 3D Model Update in Augmented Reality Assisted Laparoscopic Surgery". In: *MICCAI - International Conference on Medical Image Computing and Computer Assisted Intervention*. 2021.

[RP12] J. Cutting and P. Vishton. "Perceiving layout and knowing distances: The integration, relative potency, and contextual use of different information about depth". In: *Perception of Space and Motion*. 1995.

[RP13] M. Bajura, H. Fuchs, and R. Ohbuchi. "Merging virtual objects with the real world: Seeing ultrasound imagery within the patient". In: *International Conf. on Computer Graphics and Interactive Techniques*. 1992.

[RP14] E. Ozgur, A. Lafont, and A. Bartoli. "Visualizing In-Organ Tumors in Augmented Monocular Laparoscopy". In: *IEEE International Symposium on Mixed and Augmented Reality, ISMAR*. 2017.

[RP15] P. Chauvet et al. "Augmented Reality in a Tumor Resection Model". In: *Surgical Endoscopy* 32.3 (2018), pp. 1192–1201.

[RP16] Chinedu Innocent Nwoye. "Deep learning methods for the detection and recognition of surgical tools and activities in laparoscopic videos". PhD thesis. PhD Thesis, 2021.

[RP17] B. Li et al. "Data-driven Holistic Framework for Automated Laparoscope Optimal View Control with Learning-based Depth Perception". In: *IEEE International Conference on Robotics and Automation (ICRA)*. 2021.

[RP18] S. Yang et al. "Autonomous Laparoscope Control for Minimally Invasive Surgery With Intuition and RCM Constraints". In: *IEEE Robotics and Automation Letters* 7.3 (2022), pp. 7934–7941.

Conclusion

Presented research work. The presented research work demonstrates a coherent progression through the challenges of understanding and interacting with objects based on their observability.

Part I, "The Visible" lays the foundational work on shape perception and servoing of deformable objects using both 2D and 3D visual feedback. This addresses fundamental problems in computer vision and vision-based control for robotic manipulation.

Part II, "The Hidden" builds upon these foundations by tackling the more complex scenario of shape perception and augmentation of hidden objects, primarily in the context of medical applications like laparoscopic liver surgery. This work explores techniques like 3D-to-2D registration and novel visualization methods to enhance a surgeon's perception of subsurface anatomy.

Finally, Part III, "The Invisible" proposes future research that aims to push the boundaries further by addressing the even more challenging problem of shape perception and augmentation during tumor resection when the target is essentially invisible to standard OR sensors, outlining a project that integrates advanced registration, visualization, and robotic control.

This structured approach addresses a spectrum of observability, advancing from direct visual interaction to inference and augmentation of the unobservable.

Unpresented research work. Beyond the research work presented in this manuscript, I have research works which explore other significant areas within computer vision and robotics. To name a few, these include investigations into uncalibrated visual servoing for tasks such as object pushing (e.g., page 77, [12]), then kinematic modelling and control of robot arms using unit dual quaternions (e.g., page 77, [17]), and contributions to the field of dexterous hand synthesis and grasping (e.g., page 77, [14]).

Research perspectives. I have one short-term research perspective and another long-term research perspective.

The short-term research perspective shall address the problem of shape servoing of hidden objects, which aims controlling the deformation of a hidden object inside another using external manipulation and visual feedback.

The long-term research perspective shall address `FullyALIVE` research project proposal, which aims to develop automatic and real-time augmented reality guidance during laparoscopic liver tumor resection by tracking invisible tumors and utilising robotic control of the laparoscope.

Journal Articles

[1] K. Hanifati, M. Alkhatib, **E. Ozgur**, E. Buc, B. Le Roy, H. Rante, Y. Mezouar, and A. Bartoli. "Hidden Tumor Visualization in Augmented Monocular Liver Laparoscopy". In: *IET Healthcare Technology Letters, special issue: selected papers from AE-CAI - Joint AE-CAI — CARE — OR 2.0 Workshop at MICCAI* (2025).

[2] **E. Ozgur**, M. Alkhatib, Y. Mezouar, and A. Bartoli. "Reconstructing a Sphere and the Camera Focal Length from a Single View by Fitting Planes". In: *International Journal of Computer Vision (IJCV), special issue: selected papers from BMVC 2024* (accepted September 2025).

[3] M. M. Kalantari, **E. Ozgur**, M. Alkhatib, E. Buc, B. Le Roy, R. Modrzejewski, Y. Mezouar, and A. Bartoli. "LARLUS: Laparoscopic Augmented Reality from Laparoscopic Ultrasound". In: *International Journal of Computer Assisted Radiology and Surgery (IJCARS-IPCAI)* (2024).

[4] M. Shetab-Bushehri, M. Aranda, Y. Mezouar, and **E. Ozgur**. "Lattice-Based Shape Tracking and Servoing of Elastic Objects". In: *IEEE Transactions on Robotics (TRO)* 40 (2024), pp. 364–381.

[5] M. Shetab-Bushehri, M. Aranda, **E. Ozgur**, Y. Mezouar, and A. Bartoli. "ROBUSfT: Robust Real-Time Shape-from-Template, a C++ Library". In: *Image and Vision Computing* 141 (2024).

[6] H. Abaunza, R. Chandra, **E. Ozgur**, J.A. Corrales, and Y. Mezouar. "Kinematic screws and dual quaternion based motion controllers". In: *Control Engineering Practice* (2022).

[7] M. Shetab-Bushehri, M. Aranda, Y. Mezouar, and **E. Ozgur**. "As-Rigid-As-Possible Shape Servoing". In: *IEEE Robotics and Automation Letters, (RAL)* (2022).

[8] B. Le Roy, M. Abdallah, Y. Espinel, L. Calvet, B. Pereira, **E. Ozgur**, D. Pezet, E. Buc, and A. Bartoli. "A Case Series Study of Augmented Reality in Liver Laparoscopic Resection with a Deformable Preoperative Model". In: *Surgical Endoscopy* (2020).

[9] Y. Espinel, **E. Ozgur**, B. Le Roy L. Calvet, E. Buc, and A. Bartoli. "Combining Visual Cues with Interactions for 3D-2D Registration in Liver Laparoscopy". In: *Annals of Biomedical Engineering* 48 (2020), pp. 1712–1727.

[11] **E. Ozgur**, B. Koo, B. Le Roy, E. Buc, and A. Bartoli. "Preoperative Liver Registration for Augmented Monocular Laparoscopy using Backward-Forward Biomechanical Simulation". In: *International Journal of Computer Assisted Radiology and Surgery (IJCARS)* (2018).

[12] G. Lopez-Nicolas, **E. Ozgur**, and Y. Mezouar. "Parking Objects by Pushing using Uncalibrated Visual Servoing". In: *Autonomous Robots* (2018).

[13] P. Phutane, E. Buc, K. Poirot, **E. Ozgur**, D. Pezet, A. Bartoli, and B. Le Roy. "Preliminary Trial of Augmented Reality Performed on a Laparoscopic Left Hepatectomy". In: *Surgical Endoscopy* (2017).

[14] **E. Ozgur**, G. Gogu, and Y. Mezouar. "A Study on Dexterous Grasps via Parallel Manipulation Analogy". In: *Journal of Intelligent and Robotic Systems* 87:1 (2017), pp. 3–14.

[15] **E. Ozgur** and A. Bartoli. "Particle-SfT: a Provably-Convergent, Fast Shape-from-Template Algorithm". In: *International Journal of Computer Vision (IJCV)* 123:2 (2017), pp. 184–205.

[16] S. Briot, V. Rosenzveig, P. Martinet, **E. Ozgur**, and N. Bouton. "Minimal Representation for the Control of Parallel Robots via Leg Observation Considering a Hidden Robot Model". In: *Mechanism and Machine Theory* 106 (2016), pp. 115–147.

[17] **E. Ozgur** and Y. Mezouar. “Kinematic Modeling and Control of a Robot Arm using Unit Dual Quaternions”. In: *Robotics and Autonomous Systems* 77 (2016), pp. 66–73.

[18] **E. Ozgur**, N. Andreff, and P. Martinet. “Linear Dynamic Modeling of Parallel Kinematic Manipulators from Observable Kinematic Elements”. In: *Mechanism and Machine Theory* 69 (2013), pp. 73–89.

[19] M. Unel, O. Soldea, **E. Ozgur**, and A. Bassa. “3D Object Recognition using Invariants of 2D Projection Curves”. In: *Pattern Analysis and Applications* 13-4 (2010), pp. 451–468.

International Conference Papers

[20] A. Samai, J. Thomas, M. Alkhatib, **E. Ozgur**, and Y. Mezouar. “Dual Quaternion based Compliant Movement Primitives for Deformable Object Manipulation”. In: *IEEE International Conference on Robotics and Automation (ICRA)* (2026).

[21] A. Khalife, M. Alkhatib, **E. Ozgur**, E. Buc, B. Le Roy, Y. Mezouar, and A. Bartoli. “Radiomics LUS Transformer for Liver Tumour Segmentation”. In: *IEEE International Symposium on Biomedical Imaging (ISBI)* (2026).

[22] M. M. Kalantari, **E. Ozgur**, M. Alkhatib, N. Rabbani, Y. Espinel, R. Modrzejewski, B. Le Roy, E. Buc, Y. Mezouar, and A. Bartoli. “Stronger Together: Registering Preoperative Imagery, LUS, and MIS Liver Images”. In: *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)* (2025).

[23] **E. Ozgur**, M. Alkhatib, Y. Mezouar, and A. Bartoli. “Reconstructing Spheres by Fitting Planes”. In: *British Machine Vision Conference (BMVC)* (2024).

[24] M.M. Kalantari, **E. Ozgur**, M. Alkhatib, E. Buc, B. Le Roy, R. Modrzejewski, Y. Mezouar, and A. Bartoli. “Markerless Ultrasound Probe Pose Estimation in Mini-Invasive Surgery”. In: *IEEE International Conference on Robotics and Automation (ICRA)* (2024).

[25] R. Laezza, M. Shetab-Bushehri, **E. Ozgur**, Y. Mezouar, and Y. Karayiannidis. “Offline Reinforcement Learning for Shape Control of Deformable Linear Objects from Limited Real Data”. In: *3rd Workshop on Representing and Manipulating Deformable Objects - IEEE International Conference on Robotics and Automation (ICRA)* (2023).

[26] V. Giraud, M. Padrin, M. Shetab-Bushehri, C. Bouzgarrou, Y. Mezouar, and **E. Ozgur**. “Optimal Shape Servoing with Task-focused Convergence Constraints”. In: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (2022).

[27] M. Aranda, J. A. Corrales, Y. Mezouar, A. Bartoli, and **E. Ozgur**. “Monocular Visual Shape Tracking and Servoing for Isometrically Deforming Objects”. In: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (2020).

[28] Y. Espinel, **E. Ozgur**, L. Calvet, B. Le Roy, E. Buc, and A. Bartoli. “Combining Visual Cues and Interactions for 3D-2D Registration in Liver Laparoscopy”. In: *International Conference on Information Processing in Computer-Assisted Interventions (IPCAI)* (2019).

[29] **E. Ozgur**, A. Lafont, and A. Bartoli. “Visualizing In-Organ Tumors in Augmented Monocular Laparoscopy”. In: *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)* (2019).

[30] B. Koo, **E. Ozgur**, B. Le Roy, E. Buc, and A. Bartoli. “Deformable Registration of a Preoperative 3D Liver Volume to a Laparoscopy Image using Contour and Shading Cues”. In: *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)* (2017).

[31] A. Bartoli and **E. Ozgur**. “A Perspective on Non-Isometric Shape-from-Template”. In: *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)* (2016).

[32] G. Lopez-Nicolas, **E. Ozgur**, and Y. Mezouar. “Image-Based Control of Two Mobile Robots for Object Pushing”. In: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (2015).

[33] **E. Ozgur**, G. Gogu, and Y. Mezouar. “Structural Synthesis of Dexterous Hands”. In: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (2014).

[34] **E. Ozgur**, R. Dahmouche, N. Andreff, and P. Martinet. “A Vision-Based Generic Dynamic Model of PKMs and its Experimental Validation on the Quattro Parallel Robot”. In: *IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM)* (2014).

[35] **E. Ozgur**, Y. Mezouar, and G. Gogu. “Control-Based Synthesis and Tracking of Grasping Points”. In: *European Control Conference (ECC)* (2014).

[36] V. Rosenzveig, S. Briot, P. Martinet, **E. Ozgur**, and N. Bouton. “A Method for Simplifying the Analysis of Leg-Based Visual Servoing of Parallel Robots”. In: *IEEE International Conference on Robotics and Automation (ICRA)* (2014).

[37] **E. Ozgur**, R. Dahmouche, N. Andreff, and P. Martinet. “High Speed Parallel Kinematic Manipulator State Estimation from Legs Observation”. In: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (2013).

[38] **E. Ozgur**, N. Bouton, N. Andreff, and P. Martinet. “Dynamic Control of the Quattro Robot by the Leg Edgels”. In: *IEEE International Conference on Robotics and Automation (ICRA)* (2011).

[39] **E. Ozgur**, N. Andreff, and P. Martinet. “Vector-Based Dynamic Modeling and Control of the Quattro Parallel Robot by means of Leg Orientations”. In: *IEEE International Conference on Robotics and Automation (ICRA)* (2010).

[40] **E. Ozgur**, N. Andreff, and P. Martinet. “On the Adequation of Dynamic Modeling and Control of Parallel Kinematic Manipulators”. In: *International Conference on Multibody System Dynamics (IMSD)* (2010).

[41] **E. Ozgur**, M. Unel, H. Erdogan, and A. Ercil. “Evolving Implicit Polynomial Interfaces”. In: *British Machine Vision Conference (BMVC)* (2008).

[42] **E. Ozgur**, B. Yilmaz, H. Karabalkan, H. Erdogan, and M. Unel. “Lip Segmentation Using Adaptive Color Space Training”. In: *International Conference on Auditory-Visual Speech Processing (AVSP)* (2008).

[43] H. Bilen, M. A. Hocaoglu, **E. Ozgur**, and M. Unel. “A Comparative Study of Conventional Visual Servoing Schemes in Microsystem Applications”. In: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (2007).

[44] **E. Ozgur** and M. Unel. “Positioning and Trajectory Following Tasks in Microsystems Using Model Free Visual Servoing”. In: *IEEE International Conference on Industrial Electronics (IECON)* (2007).

[45] **E. Ozgur** and M. Unel. “Image Based Visual Servoing Using Bitangent Points Applied to Planar Shape Alignment”. In: *IASTED International Conference on Robotics and Applications (RA)* (2007).

[46] A. Malima, **E. Ozgur**, and M. Cetin. “A Fast Algorithm for Vision-Based Hand Gesture Recognition for Robot Control”. In: *IEEE Conference on Signal Processing and Communications Applications (SIU)* (2006).